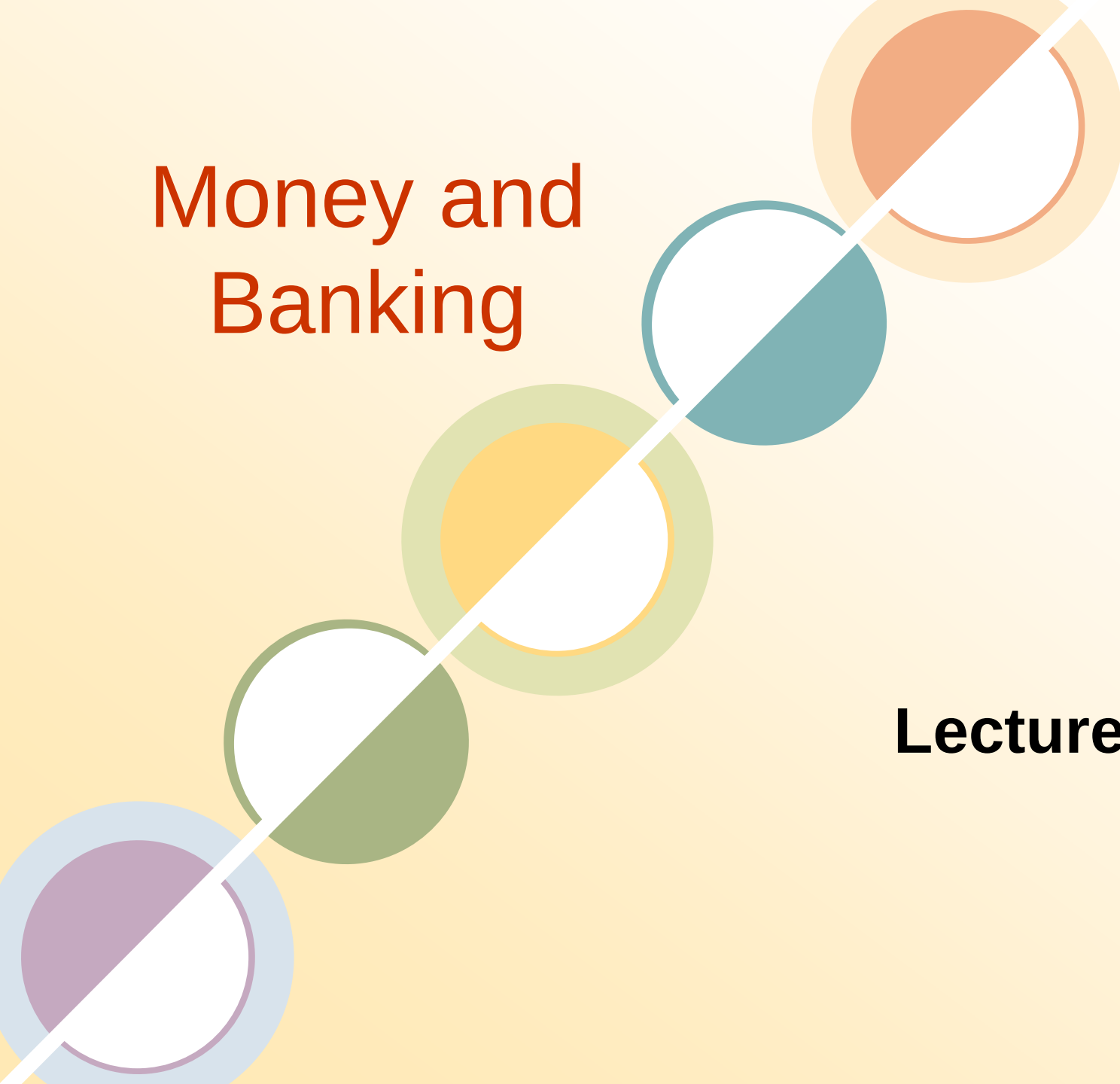


Money and Banking

Lecture 01



Text and Reference Material

- The Primary textbook for the course will be
 - “Money, Banking and Financial Markets” *by* Stephan G. Cecchetti
International Edition, McGraw Hill Publishers, ISBN 0-07-111565-X”

Text and Reference Material

- Reference books will be
 - “The Economics of Money, Banking and Financial Markets”, *by* Fredrick S. Mishkin
7th Edition Addison Wesley Longman Publishers
 - “Principles of Money, Banking and Financial Markets” *by* Lawrence S. Ritter, Willaim L. Silber and Gregory F. Udell,
Addison Wesley Longman Publishers

Course Contents

- Money and the Financial System
 - Money and the Payments System
 - Financial Instruments, Financial Markets, and Financial Institutions
- Interest rate, Financial instruments and financial markets
 - Future Value, Present Value and Interest Rates
 - Understanding Risk
 - Bonds, Pricing and Determination of Interest Rates
 - The Risk and Term Structure of Interest Rates
 - Stocks, Stock Markets and Market Efficiency

Course Contents

- Financial Institutions
 - Economics of Financial Intermediation
 - Depository Institutions: Banks and bank Management
 - Financial Industry Structure
 - Regulating the financial system
- Central Banks, Monetary Policy and Financial stability
 - Structure of central banks
 - Balance sheet and Money Supply process
 - Monetary policy
 - Exchange rate policy

Course Contents

- Modern Monetary Economics
 - Money growth and Money Demand
 - Aggregate demand
 - Business Cycle
 - Output and inflation in the short run
- Money and Banking in Islam
 - Monetary and financial policy and structure for an Interest-free economy
 - Islamic Banking in the contemporary world

Five Parts of the Financial System

1. Money
2. Financial Instruments
3. Financial Markets
4. Financial Institutions
5. Central Banks

Five Parts of the Financial System

1. Money

- To pay for purchases
- To store wealth
- Evolved from gold and silver coins to paper money to today's electronic funds transfers
- Traditional Paycheck system vs. ATM Withdrawals
- Mailed transactions vs. E-banking

Five Parts of the Financial System

2. Financial Instruments

- To transfer wealth from savers to borrowers
- To transfer risk to those best equipped to bear it.
- once investing was an activity reserved for the wealthy
 - Costly individual stock transactions through stockbrokers
 - Information collection was not so easy
- Now, small investors have the opportunity to purchase shares in “mutual funds.”

Five Parts of the Financial System

3. Financial Markets

- To buy and sell financial instruments quickly and cheaply
- evolved from coffeehouses to trading places (Stock exchanges) to electronic networks.
- Transactions are much more cheaper now
- Markets offer a broader array of financial instruments than were available even 50 years ago

Five Parts of the Financial System

4. Financial Institutions.

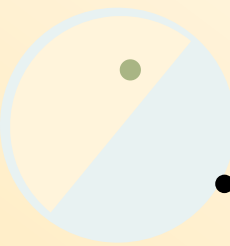
- Provide access to financial markets
- Banks evolved from Vaults and developed into deposits- and loans-agency
- Today's banks are more like financial supermarkets offering a huge assortment of financial products and services for sale.
 - Access to financial markets
 - Insurance
 - Home- and car-loans
 - Consumer credit
 - Investment advice

Five Parts of the Financial System

5. Central Banks

- Monitors financial Institutions
- stabilizes the Economy
- Initiated by Monarchs to finance the wars
- The govt. treasuries have evolved into the modern central bank
- control the availability of money and credit in such a way as to ensure
 1. low inflation,
 2. high growth, and
 3. the stability of the financial system.
- State Bank of Pakistan www.sbp.org.pk

Summary

- 
- Five Parts of the Financial System
 - Money
 - Financial Instruments
 - Financial Markets
 - Financial Institutions
 - Central Banks



Upcoming Topics

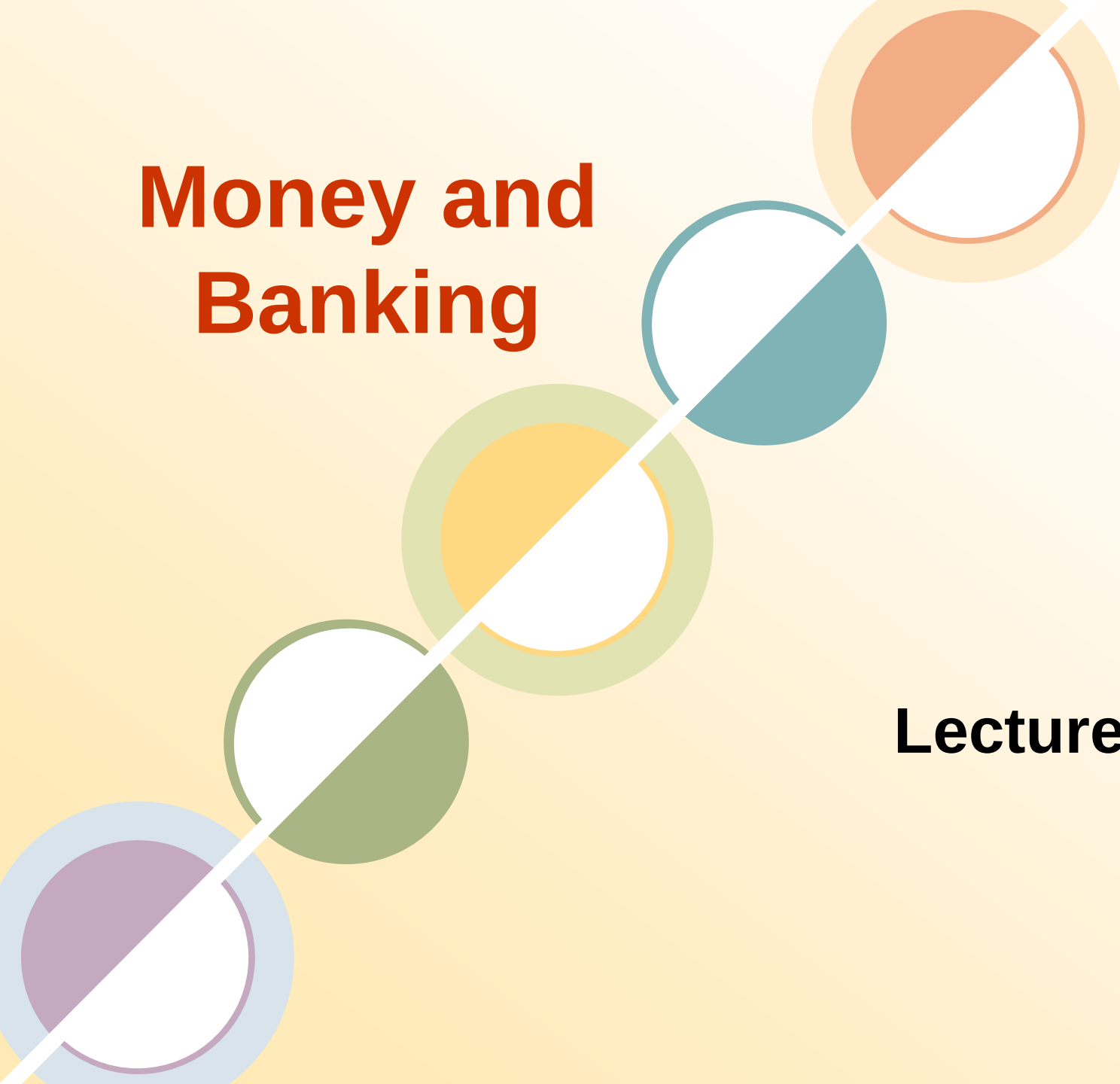
- Five Core Principles of Money and Banking
 - Time has Value
 - Risk Requires Compensation
 - Information is the basis for decisions
 - Markets set prices and allocate resources
 - Stability improves welfare
- Financial System Promotes Economic Efficiency
 - Facilitate Payments
 - Channel Funds From Savers to Borrowers
 - Enable Risk Sharing

Upcoming Topics

- Money
 - Characteristics of Money
 - Liquidity
- Payment system
 - Commodity vs. Fiat Money
 - Cheques
 - Other forms of payments
- Future of Money
- Measuring Money

Money and Banking

Lecture 02



Review of the Previous Lecture

- Five Parts of the Financial System
 - Money
 - Financial Instruments
 - Financial Markets
 - Financial Institutions
 - Central Banks

Topics under Discussion

- Five Core Principles of Money and Banking
 - Time has Value
 - Risk Requires Compensation
 - Information is the basis for decisions
 - Markets set prices and allocate resources
 - Stability improves welfare
- Financial System Promotes Economic Efficiency
 - Facilitate Payments
 - Channel Funds From Savers to Borrowers
 - Enable Risk Sharing

Topics under Discussion

- Money
 - Characteristics of Money
 - Liquidity

Five Core Principles of Money and Banking

1. Time has Value

- Time affects the value of financial instruments.
- Interest payments exist because of time properties of financial instruments

Five Core Principles of Money and Banking

1. Time has Value

• Example

- At 6% interest rate, 4 year loan of \$10,000 for a car
- Requires 48 monthly installments of \$235 each
- Total repayment = $\$235 \times 48 = \$11,280$
- | | | |
|-------------------|---|------------------|
| \$11,280 | > | \$10,000 |
| (total repayment) | | (amount of loan) |
- Reason: you are compensating the lender for the time during which you use the funds

Five Core Principles of Money and Banking

2. Risk Requires Compensation

- In a world of uncertainty, individuals will accept risk only if they are compensated in some form.
- The world is filled with uncertainty; some possibilities are welcome and some are not

Five Core Principles of Money and Banking

2. Risk Requires Compensation

- To deal effectively with risk we must consider the full range of possibilities:
 - eliminate some risks,
 - reduce others,
 - pay someone else to assume particularly onerous risks, and
 - just live with what's left
- Investors must be paid to assume risk, and the higher the risk the higher the required payment

Five Core Principles of Money and Banking

2. Risk Requires Compensation

- Car insurance is an example of paying for someone else to shoulder a risk you don't want to take. Both parties to the transaction benefit
 - Drivers are sure of compensation in the event of an accident
 - The insurance companies make profit by pooling the insurance premiums and investing them
- Now we can understand the valuation of a broad set of financial instruments
 - E.g., lenders charge higher rates if there is a chance the borrower will not repay.

Five Core Principles of Money and Banking

3. Information is the basis for decisions

- We collect information before making decisions
 - The more important the decision the more information we collect
- The collection and processing of information is the basis of foundation of the financial system.

Five Core Principles of Money and Banking

3. Information is the basis for decisions

- Some transactions are arranged so that information is NOT needed
 - stock exchanges are organized to eliminate the need for costly information gathering and thus facilitate the exchange of securities
- One way or another, information is the key to the financial system

Five Core Principles of Money and Banking

4. Markets set prices and allocate resources

- Markets are the core of the economic system; the place, physical or virtual,
 - Where buyers and sellers meet
 - where firms go to issue stocks and bonds,
 - where individuals go to purchase assets
- Financial markets are essential to the economy,
 - channeling its resources
 - minimizing the cost of gathering information
 - making transactions

Five Core Principles of Money and Banking

4. Markets set prices and allocate resources

- Well-developed financial markets are a necessary precondition for healthy economic growth
- The role of setting prices and allocation of resources makes the markets vital sources of information
- Markets provide the basis for the allocation of capital by attaching prices to different stocks or bonds

Five Core Principles of Money and Banking

4. Markets set prices and allocate resources

- Financial markets require rules to operate properly and authorities to police them
- The role of the govt. is to ensure investor protection
 - Investor will only participate if they perceive the markets are fair

Five Core Principles of Money and Banking

5. Stability improves welfare

- To reduce risk, the volatility must be reduced
- Govt. policymakers play pivotal role in reducing some risks
- A stable economy reduces risk and improves everyone's welfare.
 - By stabilizing the economy as a whole monetary policymakers eliminate risks that individuals can't and so improve everyone's welfare in the process.

Five Core Principles of Money and Banking

5. Stability improves welfare

- Stabilizing the economy is the primary function of central banks
- A stable economy grows faster than an unstable one

Financial System Promotes Economic Efficiency

- The Financial System Makes it Easier to Trade
 1. Facilitate Payments - bank checking accounts
 2. Channel Funds from Savers to Borrowers
 3. Enable Risk Sharing - Classic examples are insurance and forward markets

Financial System Promotes Economic Efficiency

- 1. Facilitate Payments
 - Cash transactions (Trade “value for value”). Could hold a lot of cash on hand to pay for things
 - Financial intermediaries provide checking accounts, credit cards, debit cards, ATMs
 - Make transactions easier.

Financial System Promotes Economic Efficiency

- 2. Channel Funds From Savers to Borrowers
 - Lending is a form of trade (Trade “value for a promise”)
 - Give up purchasing power today in exchange for purchasing power in the future.
 - Savers: have more funds than they currently need; would like to earn capital income
 - Borrowers: need more funds than they currently have; willing and able to repay with interest in the future.

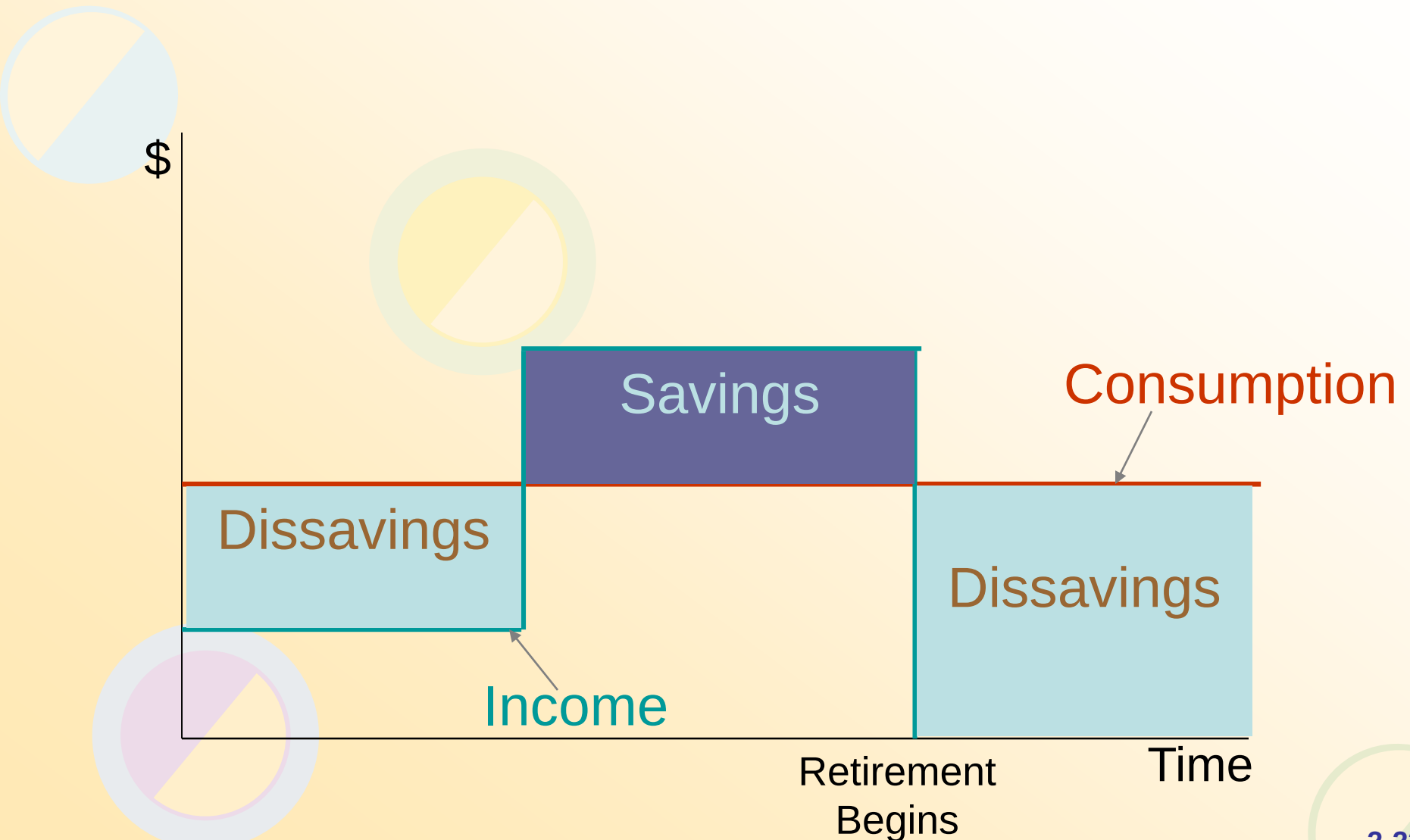
Financial System Promotes Economic Efficiency

- 2. Channel Funds From Savers to Borrowers
 - Why is this important?
 - A) Allows those without funds to exploit profitable investment opportunities.
 - Commercial loans to growing businesses;
 - Venture capital;
 - Student loans (investment in human capital);
 - Investment in physical capital and new products/processes to promote economic growth.

Financial System Promotes Economic Efficiency

- 2. Channel Funds From Savers to Borrowers
 - Why is this important?
 - B) Financial System allows the timing of income and expenditures to be decoupled.
 - Household earning potential starts low, grows rapidly until the mid 50s, then declines with age.
 - Financial system allows households to borrow when young to prop up consumption (house loans, car loans), repay and then accumulate wealth during middle age, then live off wealth during retirement.

Financial System Promotes Economic Efficiency



Financial System Promotes Economic Efficiency

• 3. Enable Risk Sharing

- The world is an uncertain place. The financial system allows **trade in risk**. (Trade “value for a promise”)
- Two principal forms of trade in risk are insurance and forward contracts.
 - Suppose everyone has a 1/1000 chance of dying by age 40 and one would need \$1 million to replace lost income to provide for their family.
 - What are your options to address this risk ?

Summary

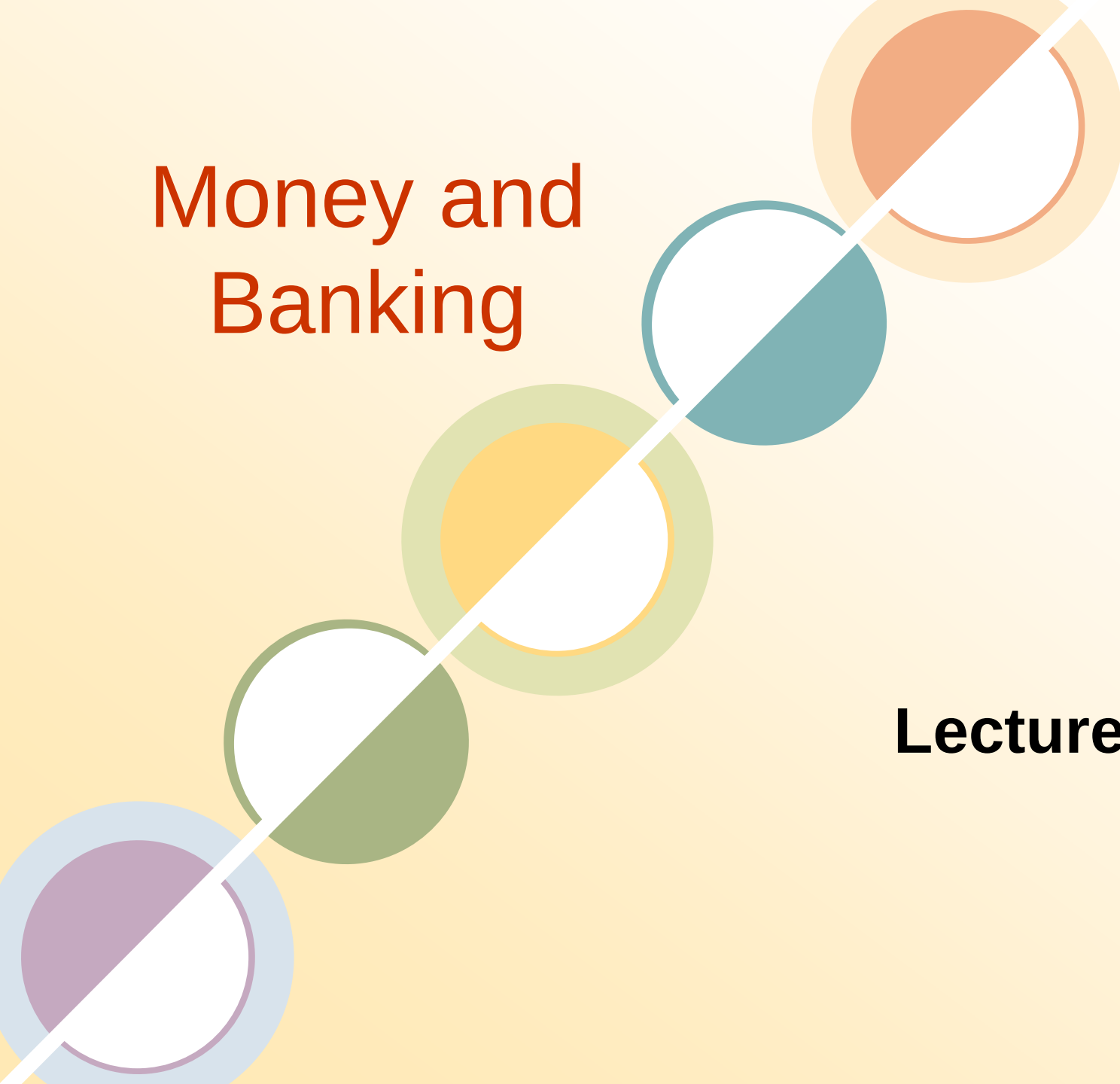
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Upcoming Topics

- Money
 - Characteristics of Money
 - Liquidity
- Payment system
 - Commodity vs. Fiat Money
 - Cheques
 - Other forms of payments
- Future of Money
- Measuring Money
- Financial Instruments
 - Uses
 - Characteristics
 - Valuation
- Financial Markets
 - Role & Structure

Money and Banking

Lecture 03



Review of the Previous Lecture

- Five Core Principles of Money and Banking
 - Time has Value
 - Risk Requires Compensation
 - Information is the basis for decisions
 - Markets set prices and allocate resources
 - Stability improves welfare
- Financial System Promotes Economic Efficiency
 - Channel Funds From Savers to Investors
 - Facilitate Payments
 - Enable Risk Sharing

Topics under Discussion

- Money
 - Characteristics of Money
 - Liquidity
- Payment system
 - Commodity vs. Fiat Money
 - Cheques
 - Other forms of payments
- Future of Money

Money

Money

- Money is an **asset** that is generally accepted as payment for goods and services or repayment of debt.
- Not the same as wealth or income

Money

- **Money** is a component of wealth that is held in a readily- spendable form
- Money is made up of
 - coin and currency
 - chequeing account balances
 - other assets that can be turned into cash or demand deposits nearly instantaneously, without risk or cost (liquid wealth)

Distinctions Among Money, Wealth, and Income

- While money, income and wealth are all measured in some currency unit, they differ significantly in their meaning.
 - People have *money* if they have large amounts of currency or big bank accounts at a point in time. (stock variable)
 - Someone earns *income* (not money) from work or investments over a period of time. (flow variable)
 - People have *wealth* if they have assets that can be converted into more currency than is necessary to pay their debts at a point in time. (stock variable)

Characteristics of Money

1. A means of payment.
2. A unit of Account
3. A Store of Value

Characteristics of Money

A means of payment.

- The primary use of money is as a means of payment.
- Money is accepted in economic exchanges.
- Barter is an alternative to using money but it doesn't work very well.
 - Barter requires a “double coincidence of wants,” meaning that in order for trade to take place both parties must want what the other has.

Characteristics of Money

A means of payment

- Money finalizes payments so that buyers and sellers have no further claim on each other.
- As economies have become more complex and physically dispersed the need for money has grown.
- Just as the division of labor and specialization allow for efficient *production*, money allows for efficient *exchange*.

Characteristics of Money

A unit of Account

- We measure value using rupees and paisas.
- Money is the unit of account that we use to quote prices and record debts.
- Money can be referred to as a standard of value.
- Using money makes comparisons of value easy

Characteristics of Money

- Under barter the general formula for n goods, we will have $n(n - 1) / 2$ prices
 - Two goods 1 price
 - 3 goods 3 prices
 - 100 goods 4,950 prices
 - 10,000 goods 50 million prices

Characteristics of Money

A Store of Value

- For money to function as a means of payment it has to be a store of value too because it must retain its worth from day to day.
 - The means of payment has to be durable and capable of transferring purchasing power from one day to the next.
 - Money is not the only store of value; wealth can be held in a number of other forms.
- Other stores of value can be preferable to money because they pay interest or deliver other services.

Characteristics of Money

A Store of Value

- However, we hold money because it is liquid, meaning that we can use it to make purchases.
- Liquidity is a measure of the ease with which an asset can be turned into a means of payment (namely money).
 - The more costly an asset is to turn into money, the less liquid it is.
 - Constantly transforming assets into money every time we wish to make a purchase would be extremely costly; hence we hold money

Liquidity

- Liquidity is a measure of the ease an asset can be turned into a means of payment, namely money
- An asset is *liquid* if it can be easily converted into money and illiquid if it is costly to convert.
 - Cash is perfectly liquid.
 - Stocks and bonds are somewhat less liquid.
 - Land is least liquid.

The Payments System

- The payment system is a web of arrangements that allows for the exchange of goods and services, as well as assets among different people
- Money is at the heart of payment system!

The Payments System

Types of Money

- **Commodity Money** – Things that have intrinsic value
- **Fiat Money** – Value comes from government decree (or fiat)

The Payments System

• Commodity Money

- The first means of payment were things with intrinsic value like silk or salt.
- Successful commodity monies had the following characteristics
 - they were usable in some form by most people;
 - they could be made into standardized quantities;
 - they were durable;
 - they had high value relative to their weight and size so that they were easily transportable; and
 - they were divisible into small units so that they were easy to trade
- For most of human history, gold has been the most common commodity money

The Payments System

Fiat Money

- Today we use paper money that is fiat money, meaning that its value comes from government decree (or fiat).
- A note costs about 0.04% its worth to produce.
- These notes are accepted as payment for goods or in settlement of debts for two reasons:
 1. We take them because we believe we can use them in the future.
 2. The law says we must accept them; that is what the words “legal tender” printed on the note means.
- As long as the government stands behind its paper money, and doesn't issue too much of it, we will use it. In the end, money is about trust.

The Payments System

- Fiat or Commodity Money
 - Does money need to be backed by a commodity at all?
 - The logical answer to this question is no.
 - If the monetary system is stable and functions effectively, “backing” is expensive, inconvenient, and unnecessary.
 - Today, money is only backed by confidence that government will responsibly limit the quantity of money to ensure that money in circulation will hold its value.

The Payments System

- Advantages of Fiat Money
 - Fewer resources are used to produce money.
 - The quantity of money in circulation can be determined by rational human judgment rather than by discovering further mineral deposits—like gold or diamonds
- Disadvantage
 - A corrupt or pressured government might issue excessive amounts of money, thereby unleashing severe inflation.

The Payments System

Cheques

- Cheques are another way of paying for things, but
 - they are not legal tender
 - they are not even money.
- Cheques are instructions to the bank to take funds from your account and transfer those funds to the person or firm whose name is written in the “Pay to the Order of” line.

The Payments System

- When you give someone a cheque in exchange for a good or service, it is not a final payment;
 - a series of transactions must still take place that lead to the final payment
 - Following are the steps in the process

The Payments System

1. You hand a paper cheque from **your bank** to a merchant in exchange for some good

A thick orange arrow pointing downwards from the first step to the second.

2. The merchant deposits the cheque into **merchant's bank** and merchant's account is credited

A thick orange arrow pointing downwards from the second step to the third.

3. The **merchant's bank** sends the cheque to the local central bank

A thick orange arrow pointing downwards from the third step, indicating the process continues.

The Payments System

4. The Central Bank

- a. Credits the merchant's bank's reserve account
- b. Debits your bank's reserve account

(This step involves money)



5. The Central Bank returns the cheque to your bank



6. Your bank debits your chequing account by the amount of the cheque

The Payments System

- The whole process is time consuming and expensive;
- Though cheque volumes have begun to fall, paper cheques are still with us because a cancelled cheque is legal proof of payment

Other Forms of Payments

- Debit Cards
- Credit Cards
- Electronic Funds transfers
- Stored Value Cards

Summary

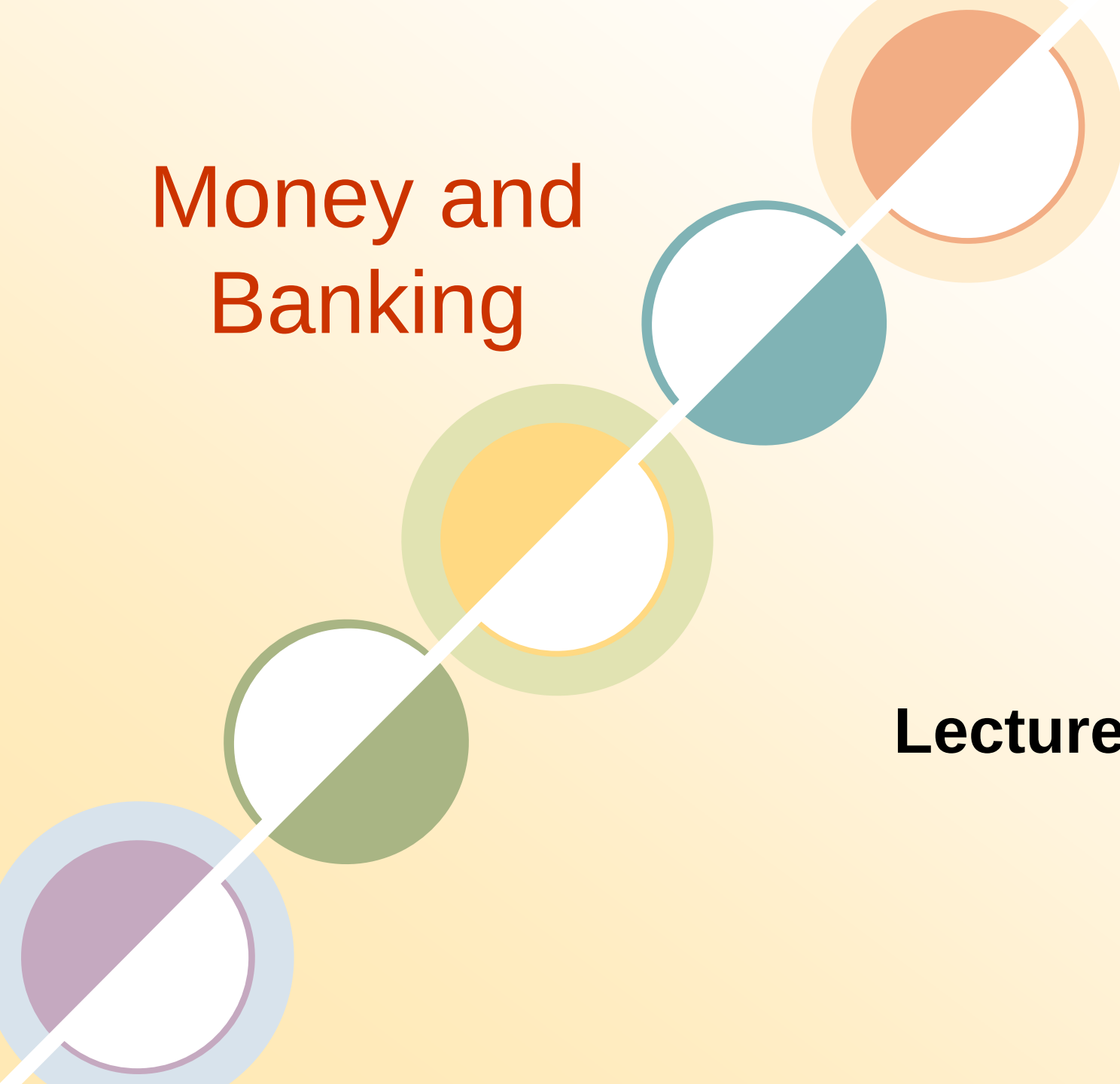
- Money
 - Characteristics of Money
 - Liquidity
- Payment system
 - Commodity vs. Fiat Money
 - Cheques
 - Other forms of payments

Upcoming Topics

- Payments System
 - Other forms of payments
- Future of Money
- Measuring Money
- Financial Instruments

Money and Banking

Lecture 04



Review of the Previous Lecture

- Money
 - Characteristics of Money
 - Liquidity
- Payment system
 - Commodity vs. Fiat Money
 - Cheques

Topics under Discussion

- Payment system
 - Other forms of payments
- Future of Money
- Measuring Money
 - Definitions
 - Monetary Aggregates
 - Measures of Inflation
- Financial Instruments

Other Forms of Payments

- Debit Cards
- Credit Cards
- Electronic Funds transfers
- Stored Value Cards

Other Forms of Payments

- Debit Card
 - The money in your account is used for payments
 - works like a cheque and there is usually a fee for the transaction

Other Forms of Payments

- Credit card
 - It is a promise by a bank to lend the cardholder money with which to make purchases.
 - When the card is used to buy merchandise the seller receives payment immediately
 - The money that is used for payment does not belong to the buyer
 - Rather, the bank makes the payment, creating a loan that the buyer must repay.
 - So, they do not represent money; rather, they represent access to someone else's money

Other Forms of Payments

Electronic Funds Transfer

- move funds directly from one account to another.
- Banks use these transfers to handle transactions among themselves
- Individuals may be familiar with such transfers through direct deposit of their paycheques and the payment of their utility bills, etc

Other Forms of Payments

E-money

- used for purchases on the Internet.
 - You open an account by transferring funds to the issuer of the e-money
 - When shopping online, instruct the issuer to send your e-money to the merchant
- It is really a form of private money.

Stored-value card

- Retail businesses are experimenting with new forms of electronic payment
 - Prepaid cellular cards, Internet scratch cards, calling cards etc..

The Future of Money

- The time is rapidly approaching when safe and secure systems for payment will use virtually no money at all
- We will also likely see
 - fewer “varieties” of currency, (*a sort of standardization of money*) and
 - a dramatic reduction in the number of units of account
- money as a store of value is clearly on the way out as many financial instruments have become highly liquid.

Measuring Money

Different Definitions of money based upon degree of liquidity.

Federal Reserve System defines monetary aggregates.

Measuring Money

- Changes in the amount of money in the economy are related to changes in interest rates, economic growth, and most important, inflation.
 - Inflation is a sustained rise in the general price level
 - With inflation you need more money to buy the same basket of goods because it costs more.
 - Inflation makes money less valuable
 - The primary cause of inflation is the issuance of too much money

Measuring Money

- Because money growth is related to inflation we need to be able to measure how much money is circulating
 - Money as a means of payments
 - We measure the quantity of money as the quantity of currency in circulation – an unrealistically limited measure, since there are other ways of payments

Measuring Money

- Alternatively, broadly categorize financial assets and sort them by the degree of liquidity
 - Sort them by the ease with which they can be converted into a means of payments
 - Arrange them from most liquid to least liquid
 - Draw a line and include everything on one side of the line in the measure of the money
 - Where to draw the line?

Measuring Money

- In reality, we draw line at different places and compute several measures of money called **the monetary aggregates**
 - M1, M2, and M3

Measuring Money

- M1 is the narrowest definition of money and includes only currency and various deposit accounts on which people can write cheques.
 - Currency in the hands of the public,
 - traveler's cheques,
 - demand deposits and
 - other chequeable deposits.

Measuring Money

- M2 includes everything that is in M1 plus assets that cannot be used directly as a means of payment and are difficult to turn into currency quickly,
 - small-denomination time deposits,
 - money market deposit accounts,
 - money market mutual fund shares.
- M2 is the most commonly quoted monetary aggregate since its movements are most closely related to interest rate and economic growth.

Measuring Money

- M3 adds to M2 other assets that are important to large institutions
 - large-denomination time deposits,
 - institutional money market mutual fund shares,
 - repurchase agreements and
 - Eurodollars.

Monetary Aggregates

<u>Symbol</u>	<u>Assets included</u>
C	Currency
M1	C + demand deposits, travelers' cheques, other chequeable deposits
M2	M1 + small time deposits, savings deposits, money market mutual funds, money market deposit accounts
M3	M2 + large time deposits, repurchase agreements, institutional money market mutual fund balances

Monetary Aggregates

Figures in millions as of March 2005

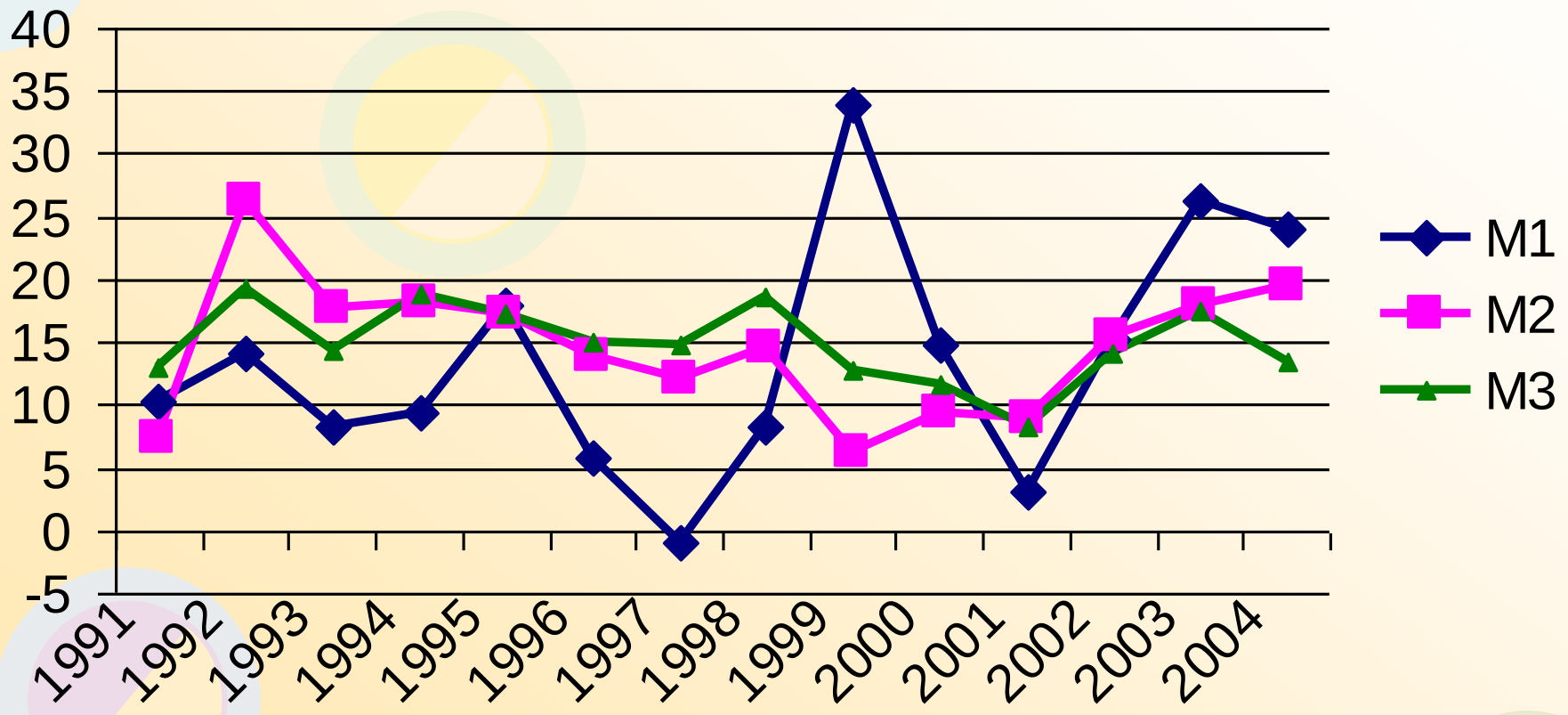
1. Currency issued	711,997	
2. Currency held by SBP	3,188	
3. Currency in tills of Scheduled Banks	43,914	
4. Currency in circulation (1 – 2 – 3)		664,895
5. Scheduled Banks demand deposits		93,272
6. Other Deposits with SBP		<u>4,826</u>
7. M1 (4+5+6)		1,602,423
8. Scheduled Banks Time Deposits		1,037,678
9. Resident Foreign Currency Deposits		<u>172,074</u>
10. Total Monetary Assets (M2)		2,812,175
11. M3		3,833,686

Monetary Aggregates

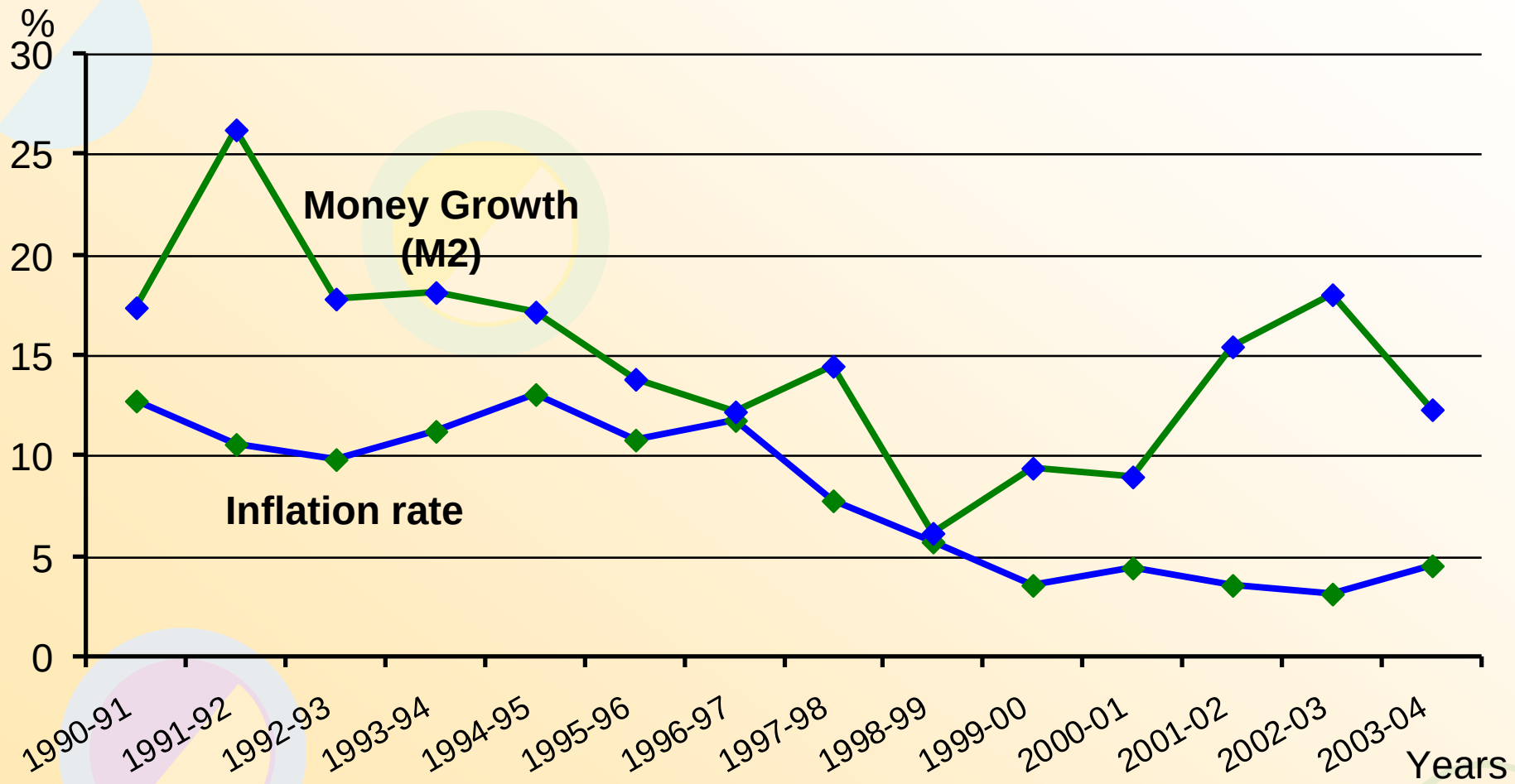
Table 2.1 The Monetary Aggregates

Monetary Aggregates	Value as of August 2004 (U.S.\$ billions)
M1 = Currency in the hands of the public	686.2
+ Travelers' checks	7.6
+ Demand deposits	315.3
+ Other checkable deposits	<u>328.5</u>
Total M1	1,337.6
M2 = M1	
+ Small-denomination time deposits	794.7
+ Savings deposits including money-market deposit accounts	3415.3
+ Retail money-market mutual fund shares	<u>735.5</u>
Total M2	6,283.1
M3 = M2	
+ Large-denomination time deposits	1,036.3
+ Institutional money-market mutual fund shares	1,104.7
+ Repurchase agreements	516.6
+ Eurodollars	<u>344.5</u>
Total M3	9,285.2

Growth Rates in Monetary Aggregates



Money Growth and Inflation



Measures of Inflation

- **Fixed-weight Index** - CPI
- **Deflator** – GDP or Personal Consumption Expenditure Deflator

Consumer Price Index (CPI)

- A measure of the overall level of prices
 - Used to
 - track changes in the typical household's cost of living
 - allow comparisons of dollar figures from different years

Consumer Price Index (CPI)

1. Survey consumers to determine composition of the typical consumer's "basket" of goods.
2. Every month, collect data on prices of all items in the basket; compute cost of basket
3. CPI in any month equals

$$100 \times \frac{\text{Cost of basket in that month}}{\text{Cost of basket in base period}}$$

Consumer Price Index (CPI)

The basket contains 20 pizzas and 10 compact discs.

prices:		
	pizza	CDs
2002	\$10	\$15
2003	\$11	\$15
2004	\$12	\$16
2005	\$13	\$15

- For each year, compute
- the cost of the basket
 - the CPI (use 2002 as the base year)
 - the inflation rate from the preceding year

Consumer Price Index (CPI)

	cost of basket	CPI	inflation rate
2002	\$350	100.0	n.a.
2003	370	105.7	5.7%
2004	400	114.3	8.1%
2005	410	117.1	2.5%

GDP Deflator

The GDP deflator, also called the implicit price deflator for GDP, measures the price of output relative to its price in the base year. It reflects what's happening to the overall level of prices in the economy

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}}$$

GDP Deflator

	Nom. GDP	Real GDP	GDP deflator	inflation rate
2001	Rs46,200	Rs46,200	100.0	<i>n.a.</i>
2002	51,400	50,000	102.8	2.8%
2003	58,300	52,000	112.1	9.1%

Financial Intermediaries

- The informal arrangements that were the mainstay of the financial system centuries ago have given way to the formal financial instruments of the modern world
- Today, the international financial system exists to facilitate the design, sale, and exchange of a broad set of contracts with a very specific set of characteristics.

Financial Intermediaries

- We obtain the financial resources we need from this system in two ways:
 - directly from lenders and
 - indirectly from financial institutions called financial intermediaries.

Summary

- Measuring Money
 - Definitions
 - Monetary Aggregates
 - Measures of Inflation

Upcoming Topics

- Financial Instruments
- Financial Markets
- Financial Institutions

Money and Banking

Lecture 5

Review of the Previous Lecture

- Five Parts of the Financial System
 - Money
 - Financial Instruments
 - Financial Markets
 - Financial Institutions
 - Central Banks
- Measuring Money
 - Definitions
 - Monetary Aggregates
 - Measures of Inflation

Topics under Discussion

- Financial Intermediaries
- Financial Instruments
 - Uses
 - Characteristics
 - Value
 - Examples

Financial Intermediaries

- The informal arrangements that were the mainstay of the financial system centuries ago have since given way to the formal financial instruments of the modern world
- Today, the international financial system exists to facilitate the design, sale, and exchange of a broad set of contracts with a very specific set of characteristics.

Financial Intermediaries

- We obtain the financial resources we need from this system in two ways:
 - directly from lenders and
 - indirectly from financial institutions called financial intermediaries.

Financial Intermediaries

Indirect Finance

- a financial institution (like a bank) borrows from the lender and then provides funds to the borrower.
- If someone borrows money to buy a car, the car becomes his or her asset and the loan a liability.

Financial Intermediaries

Direct Finance

- Borrowers sell securities directly to lenders in the financial markets.
- Governments and corporations finance their activities this way
- The securities become assets to the lenders who buy them and liabilities to the borrower who sells them

Financial and Economic Development

- Financial development is inextricably linked to economic growth
- There aren't any rich countries that have very low levels of financial development.

Financial Instruments

- A financial instrument is the written legal obligation of one party to transfer something of value – usually money – to another party at some future date, under certain conditions, such as stocks, loans, or insurance.

Financial Instruments

- Written legal obligation means that it is subject to government enforcement;
 - the enforceability of the obligation is an important feature of a financial instrument.
- The “party” referred to can be a person, company, or government
- The future date can be specified or can be when some event occurs

Financial Instruments

- Financial instruments generally specify a number of possible contingencies under which one party is required to make a payment to another
- Stocks, loans, and insurance are all examples of financial instruments

Characteristics of Financial Instruments

- Standardization
 - Standardized agreements are used in order to overcome the potential costs of complexity
 - Because of standardization, most of the financial instruments that we encounter on a day-to-day basis are very homogeneous
- Communicate Information
 - summarize certain essential information about the issuer
 - designed to handle the problem of “asymmetric information”,
 - borrowers have some information that they don't disclose to lenders

Classes of Financial Instruments

- Underlying Instruments (Primary or Primitive Securities)
 - e.g. Stocks and bonds
- Derivative Instruments
 - value and payoffs are “derived from” the behavior of the underlying instruments
 - Futures and options

Value of Financial Instruments

1. Size of the promised payment.
 - People will pay more for an instrument that obligates the issuer to pay the holder a greater sum.
 - The bigger the size of the promised payment, the more valuable the financial instrument
2. When the payment will be received.
 - The sooner the payment is made the more valuable is the promise to make it

Value of Financial Instruments

3. The likelihood the payment will be made (risk).
 - The more likely it is that the payment will be made, the more valuable the financial instrument
4. The conditions under which the payment will be made.
 - Payments that are made when we need them most are more valuable than other payments

Summary

- Financial Intermediaries
- Financial Instruments
 - Uses
 - Characteristics
 - Value

Upcoming Topics

- Financial Instruments
 - Examples
- Financial Markets
- Financial Institutions

Money and Banking

Lecture 6

Review of the Previous Lecture

- Financial Intermediaries
- Financial Instruments
 - Uses
 - Characteristics
 - Value

Topics under Discussion

- Financial Instruments
 - Examples
- Financial Markets
 - Roles
 - Structure
- Financial Institutions

Examples of Financial Instruments

Primarily Stores of Value

- Bank Loans
 - a borrower obtains resources from a lender immediately in exchange for a promised set of payments in the future
- Bonds
 - a form of a loan, whereby in exchange for obtaining funds today a government or corporation promises to make payments in the future

Examples of Financial Instruments

Primarily Stores of Value

- Home Mortgages
 - A loan that is used to purchase real estate.
 - The real estate is collateral for the loan,
 - it is a specific asset pledged by the borrower in order to protect the interests of the lender in the event of nonpayment.
 - If payment is not made the lender can foreclose on the property.
- Stocks
 - an owner of a share owns a piece of the firm and is entitled to part of its profits.

Examples of Financial Instruments

Primarily to transfer risk

- Insurance Contracts
 - the primary purpose is to assure that payments will be made under particular (and often rare) circumstances
- Futures Contracts
 - an agreement to exchange a fixed quantity of a commodity, such as wheat or corn, or an asset, such as a bond, at a fixed price on a set future date.
 - It is a derivative instrument since its value is based on the price of some other asset.
 - It is used to transfer the risk of price fluctuations from one party to another

Examples of Financial Instruments

Primarily to transfer risk

- Options
 - Derivative instruments whose prices are based on the value of some underlying asset;
 - They give the holder the right (but not the obligation) to purchase a fixed quantity of the underlying asset at a predetermined price at any time during a specified period.

Financial Markets

- Financial Markets are the places where financial instruments are bought and sold.
 - Enable both firms and individuals to find financing for their activities.
 - Promote economic efficiency by ensuring that resources are placed at the disposal of those who can put them to best use.
 - When they fail to function properly, resources are no longer channeled to their best possible use and the society suffers at large

Structure of Financial Markets

- Primary vs. Secondary Markets
 - In a primary market a borrower obtains funds from a lender by selling newly issued securities.
 - Most companies use an investment bank, which will determine a price and then purchase the company's securities in preparation for resale to clients; this is called **underwriting**.
 - In the secondary markets people can buy and sell existing securities

Structure of Financial Markets

- Centralized Exchanges vs. Over-the-counter Markets.
 - In the centralized exchange (e.g. Karachi Stock Exchange www.kse.com.pk), the trading is done “on the floor”
 - Over-the-counter (or OTC) market are electronic networks of dealers who trade with one another from wherever they are located

Structure of Financial Markets

- Debt and Equity vs. Derivative Markets
 - Equity markets are the markets for stocks, which are usually traded in the countries where the companies are based.
 - Debt instruments can be categorized as
 - money market (maturity of less than one year) or
 - bond markets (maturity of more than one year)

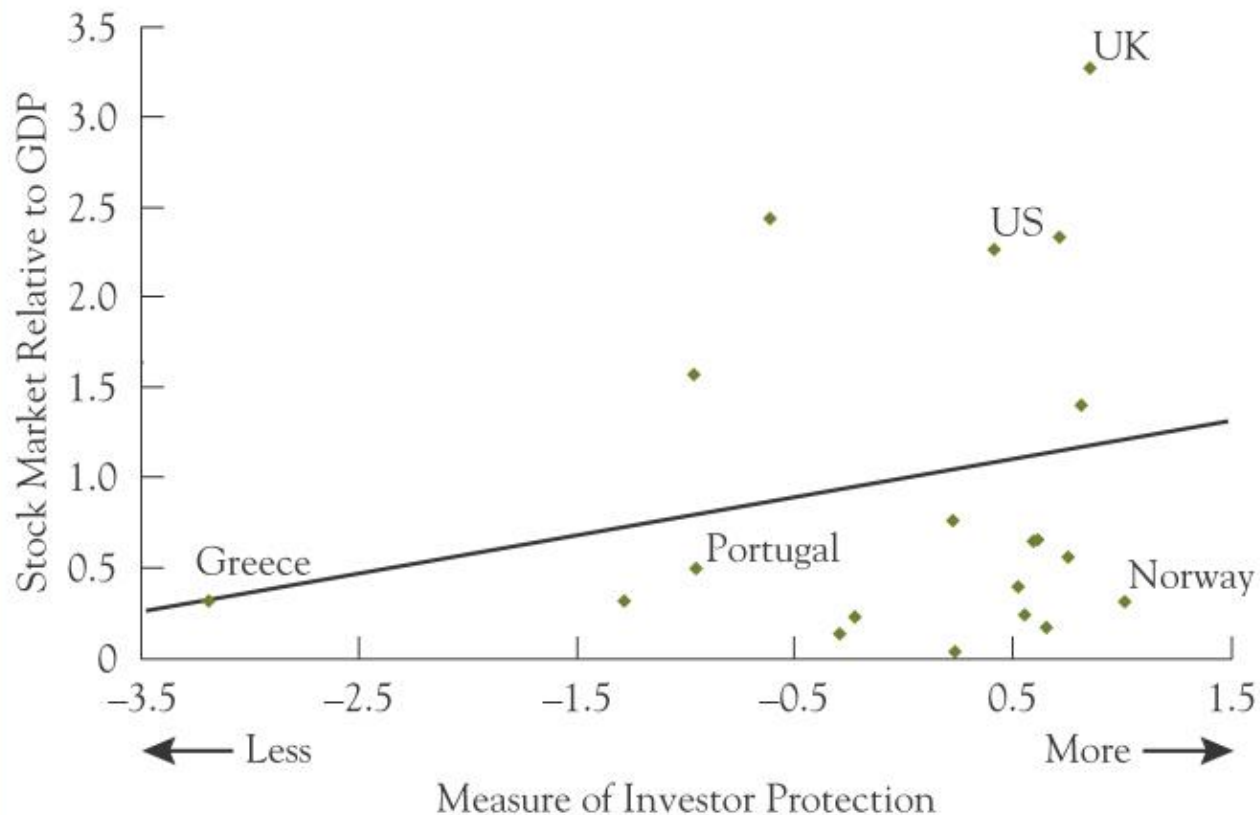
Financial Markets

Characteristics of a well-run financial market

1. Low transaction costs.
2. Information communicated must be accurate and widely available
 - If not, the prices will not be correct
 - prices are the link between the financial markets and the real economy
3. Investors must be protected.
 - A lack of proper safeguards dampens people's willingness to invest

Market Size and Investor Protection

Figure 3.2 Stock Market Size and Investor Protection



Summary

- Financial Instruments
 - Examples
- Financial Markets
 - Roles
 - Structure

Upcoming Topics

- Financial Institutions
- Time Value of Money
 - Future value,
 - Compound Interest
- Present Value
- Interest Rates

Money and Banking

Lecture 7

Review of the Previous Lecture

- Financial Instruments
 - Means of Payments
 - Store of Value
 - Transfer the Risk
- Financial Markets
 - Roles
 - Structure

Topics under Discussion

- Financial Institutions
 - Structure of Financial Industry
- Time Value of Money

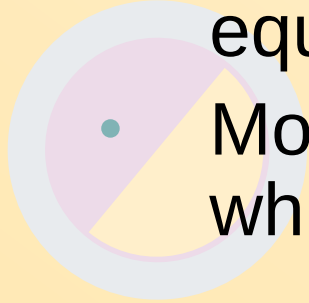
Financial Institutions

- Financial institutions are the firms that provide access to the financial markets;
- They sit between savers and borrowers and so are known as financial intermediaries.
- Banks, insurance companies, securities firms and pension funds.

Financial Institutions



A system without financial institutions would not work very well for three reasons

- Individual transactions between saver-lenders and borrower-spenders would be extremely expensive.
 - Lenders need to evaluate the creditworthiness of borrowers and then monitor them, and individuals are not equipped to do this.
 - Most borrowers want to borrow long term, while lenders favor short-term loans
- 
- 

Financial Institutions

Role of Financial Institutions

- Reduce transactions cost by specializing in the issuance of standardized securities
- Reduce information costs of screening and monitoring borrowers.
- Curb information asymmetries, helping to ensure that resources flow into their most productive uses

Financial Institutions

Role of Financial Institutions

- Make long-term loans but allow savers ready access to their funds.
- Provide savers with financial instruments (more liquid and less risky than the individual stocks and bonds) that savers would purchase directly in financial markets

The structure of the financial industry

The structure of the financial industry

- Financial institutions or intermediaries can be divided into two broad categories
 - Depository institutions - take deposits and make loans.
 - (commercial banks, savings banks, and credit unions)
 - Nondepository institutions
 - insurance companies, securities firms, mutual fund companies, finance companies, and pension funds

The structure of the financial industry

- Nondepository Institutions

- Insurance companies
 - accept premiums, which they invest in securities and real estate in return for promising compensation to policyholders should certain events occur (like death, property losses, etc.)
- Pension funds
 - invest individual and company contributions into stocks, bonds and real estate in order to provide payments to retired workers.

The structure of the financial industry

- Nondepository Institutions
 - Securities firms
 - They include brokers, investment banks, and mutual fund companies
 - Brokers and investment banks issue stocks and bonds to corporate customers, trade them, and advise clients.
 - Mutual fund companies pool the resources of individuals and companies and invest them in portfolios of bonds, stocks, and real estate.

The structure of the financial industry

- Nondepository Institutions

- Government Sponsored Enterprises:

- Federal credit agencies that provide loans directly for farmers and home mortgages, as well as guarantee programs that insure the loans made by private lenders.
 - HBFC
 - ZTBL
 - Khushhali bank
 - SME Bank
 - The government also provides retirement income and medical care to the elderly (and disabled) through Social Security and Medicare.

The structure of the financial industry

- Nondepository Institutions

- Finance Companies:

- Raise funds directly in the financial markets in order to make loans to individuals and firm

The monetary aggregates are made up of liabilities of commercial banks, so clearly the financial structure is tied to the availability of money and credit.

Summary

- Financial Institutions
 - Structure of Financial Industry

Upcoming Topics

- Future Value Concepts
- Present value
 - Application in financial environment

Money and Banking

Lecture 8

Review of the Previous Lecture

- Financial Institutions
 - Structure of Financial Industry

Topics under Discussion

- Time Value of Money
- Future Value Concepts
- Present value
 - Application in financial environment

Time Value of Money

- Credit is one of the critical mechanisms we have for allocating resources.
 - Even the simplest financial transaction, like saving some of your paycheck each month to buy a car, would be impossible.
 - Corporations, most of which survive from day to day by borrowing to finance their activities, would not be able to function.

Time Value of Money

- Yet even so, most people still take a dim view of the fact that lenders charge interest.
 - The main reason for the enduring unpopularity of interest comes from the failure to appreciate the fact that lending has an opportunity cost.
 - Think of it from the point of view of the lender.
 - Extending a loan means giving up the alternatives. While lenders can eventually recoup the sum they lend, neither the time that the loan was outstanding nor the opportunities missed during that time can be gotten back.
 - So interest isn't really "the breeding of money from money," as Aristotle put it; it's more like a rental fee that borrowers must pay lenders to compensate them for lost opportunities.

Time Value of Money

- It's no surprise that in today's world, interest rates are of enormous importance to virtually everyone
 - individuals, businesses, and governments.
- They link the present to the future, allowing us to compare payments made on different dates.
- Interest rates also tell us the future reward for lending today, as well as the cost of borrowing now and repaying later.
- To make sound financial decisions, we must learn how to calculate and compare different rates on various financial instruments

Future Value

- Future Value is the value on some future date of an investment made today.
- To calculate future value we multiply the present value by the interest rate and add that amount of interest to the present value.

Future Value

$$PV + \text{Interest} = FV$$

$$PV + PV*i = FV$$

$$\$100 + \$100(0.05) = \$105$$

PV = Present Value

FV = Future Value

i = interest rate (as a percentage)

- The higher the interest rate (or the amount invested) the higher the future value.

Future Value

Future Value in one year.

$$FV = PV \cdot (1+i)$$

Future Value

- Now we need to figure out what happens when the time to repayment varies
- When we consider investments with interest payments made for more than one year we need to consider compound interest, or the fact that interest will be paid on interest

Future Value

Future Value in two years

$$\begin{aligned} & \$100 + \$100(0.05) + \$100(0.05) + \$5(0.05) \\ & = \$110.25 \end{aligned}$$

Present Value of the Initial Investment
+ Interest on the initial investment in the 1st
Year + Interest on the initial investment in the
2nd Year
+ Interest on the Interest from the 1st Year in
the 2nd Year
= Future Value in Two Years

Future Value

General Formula for compound interest –
Future value of an investment of PV in n
years at interest rate i (measured as a
decimal, or 5% = .05)

$$FV_n = PV * (1+i)^n$$

Future Value

Computing the Future Value of \$100 at 5% Annual Interest

Years into Future	Computation	Future Value
1	$\$100(1.05)$	\$105.00
2	$\$100(1.05)^2$	\$110.25
3	$\$100(1.05)^3$	\$115.76
4	$\$100(1.05)^4$	\$121.55
5	$\$100(1.05)^5$	\$127.63
10	$\$100(1.05)^{10}$	\$162.89

Future Value

Note:

Both n and i must be measured in same time units—if i is annual, then n must be in years, So future value of \$100 in 18 months at 5% is

$$FV = 100 * (1 + .05)^{1.5}$$

Future Value

- How useful it is?
 - If you put \$1,000 per year into bank at 4% interest, how much would you have saved after 40 years?
 - Taking help of future value concept, the accumulated amount through the saving will be \$98,826 – more than twice the \$40,000 you invested
- How does it work?

Future Value

- The first \$1,000 is deposited for 40 years so its future value is
$$\$1,000 \times (1.04)^{40} = 4,801.02$$
- The 2nd \$1,000 is deposited for 39 years so its future value is
$$\$1,000 \times (1.04)^{39} = 4,616.37$$
- And so on.....upto the \$1,000 deposited in the 40th year
- Adding up all the future values gives you the amount of \$98,826

Present Value

Present Value (PV) is the value today (in the present) of a payment that is promised to be made in the future.

OR

Present Value is the amount that must be invested today in order to realize a specific amount on a given future date.

Present Value

- To calculate present value we invert the future value calculation;
 - we divide future value by one plus the interest rate (to find the present value of a payment to be made one year from now).
 - Solving the Future Value Equation

$$FV = PV \cdot (1+i)$$

- Present Value of an amount received in one year.

$$PV = \frac{FV}{(1+i)}$$

Present Value

Example:

\$100 received in one year, $i=5\%$

$$PV = \$100 / (1 + .05) = \$95.24$$

Note:

$$FV = PV * (1 + i) = \$95.24 * (1.05) = \$100$$

Present Value

- For payments to be made more than one year from now we divide future value by one plus the interest rate raised to the n^{th} power where n is the number of years
 - Present Value of \$100 received n years in the future:

$$PV = \frac{FV}{(1+i)^n}$$

Present Value

Example

Present Value of \$100 received in 2 ½ years and an interest rate of 8%.

$$PV = \$100 / (1.08)^{2.5} = \$82.50$$

Note:

$$FV = \$82.50 * (1.08)^{2.5} = \$100$$

Summary

- Future Value Concepts
- Present value
 - Application in financial environment

Upcoming Topics

- Internal Rate of Return
- Bond Pricing
- Real and Nominal Interest Rate

Money and Banking

Lecture 9

Review of the Previous Lecture

- Time Value of Money
- Future Value
- Present Value

Topics under Discussion

- Application of Present Value Concept
 - Compound Annual Rate
 - Interest Rates vs Discount Rate
 - Internal Rate of Return
 - Bond Pricing

Present Value

Important Properties of Present Value

Present Value is higher:

1. The higher the future value (FV) of the payment.
2. The shorter the time period until payment.(n)
3. The lower the interest rate.(i)

Present Value

1. The size of the payment (FV_n)

- Doubling the future value of the payment (without changing time of the payment or interest rate), doubles the present value
 - At 5% interest rate, \$100 payment has a PV of \$90.70
 - Doubling it to \$200, doubles the PV to \$181.40
- Increasing or decreasing FV_n by any percentage will change PV by the same percentage in the same direction

Present Value

2. The time until the payment is made (n)
 - Continuing with the previous example of \$100 at 5%, we allow the time to go from 0 to 30 years.
 - This process shows us that the PV payment is worth \$100 if it is made immediately, but gradually declines to \$23 for a payment made in 30 years

Present Value

- The rule of 72
 - For reasonable rates of return, the time it takes to double the money, is given approximately by
$$t = 72 / i\%$$
 - If we have an interest rate of 10%, the time it takes for investment to double is :
$$t = 72 / 10 = 7.2 \text{ years}$$
 - This rule is fairly applicable to discount rates in 5% to 20% range.

Present Value

3. The Interest rate (i)

- Higher interest rates are associated with lower present values, no matter what size or timing of the payment
- At any fixed interest rate, an increase in the time until a payment is made reduces its present value

Compound Annual Rates

- Comparing changes over days, months, years and decades can be very difficult.
- The way to deal with such problems is to turn the monthly growth rate into compound-annual rate.
 - An investment whose value grows 0.5% per month goes from 100 at the beginning of the month to 100.5 at the end of the month:
 - We can verify this as following

$$100 \frac{(100.5 - 100)}{100} = [(100.5/100) - 1] = 0.5\%$$

Compound Annual Rates

- What if the investment's value continued to grow at 0.5% per month for next 12 months?
 - We can't simply multiply 0.5 by 12
 - Instead we need to compute a 12 month compound rate
 - So the future value of 100 at 0.5%(0.005) per month compounded for 12 months will be:

$$Fv_n = PV(1+i)^n = 100(1.005)^{12} = 106.17$$

Compound Annual Rates

- An increase of 6.17% which is greater than 6%, had we multiplied 0.5% by 12
- The difference between the two answers grows as the interest rate grows
- At 1% monthly rate, 12 month compounded rate is 12.68%

Compound Annual Rates

- Another use for compounding is to compute the percentage change per year when we know how much an investment has grown over a number of years
- This rate is called average annual rate
 - If an investment has increased 20%, from 100 to 120 over 5 years
 - Is average annual rate is simply dividing 20% by 5?
 - This way we ignore compounding effect
 - Increase in 2nd year must be calculated as percentage of the investment worth at the end of 1st year

Compound Annual Rates

- To calculate the average annual rate, we revert to the same equation:

$$Fv_n = PV(1+i)^n$$

$$120 = 100(1+i)^5$$

Solving for i

$$i = [(120/100)^{1/5} - 1] = 0.0371$$

- 5 consecutive annual increases of 3.71% will result in an overall increase of 20%

Interest Rate and Discount Rate

- The interest rate used in the present value calculation is often referred to as the discount rate because the calculation involves discounting or reducing future payments to their equivalent value today.
- Another term that is used for the interest rate is yield
- Saving behavior can be considered in terms of a personal discount rate;
 - people with a low rate are more likely to save, while people with a high rate are more likely to borrow

Interest Rate and Discount Rate

- We all have a discount rate that describes the rate at which we need to be compensated for postponing consumption and saving our income
- If the market offers an interest rate higher than the individual's personal discount rate, we would expect that person to save (and vice versa)
- Higher interest rates mean higher saving

Applying Present Value

- To use present value in practice we need to look at a sequence or stream of payments whose present values must be summed. Present value is additive.
- To see how this is applied we will look at internal rate of return and the valuation of bonds

Internal Rate of Return

- The Internal Rate of Return is the interest rate that equates the present value of an investment with its cost.
- It is the interest rate at which the present value of the revenue stream equals the cost of the investment project.
 - In the calculation we solve for the interest rate

Internal Rate of Return

A machine with a price of \$1,000,000 that generates \$150,000/year for 10 years.

$$\$1,000,000 = \frac{\$150,000}{(1+i)^1} + \frac{\$150,000}{(1+i)^2} + \frac{\$150,000}{(1+i)^3} + \dots + \frac{\$150,000}{(1+i)^{10}}$$

Solving for i , $i = .0814$ or 8.14%

Internal Rate of Return

- The internal rate of return must be compared to a rate of interest that represents the cost of funds to make the investment.
 - These funds could be obtained from retained earnings or borrowing. In either case there is an interest cost
 - An investment will be profitable if its internal rate of return exceeds the cost of borrowing

Summary

- Application of Present Value Concept
 - Compound Annual Rate
 - Interest Rates vs Discount Rate
 - Internal Rate of Return

Upcoming Topics

- Application of Present Value Concept
 - Bond Pricing
- Real Vs Nominal Interest Rates
- Risk
 - Characteristics
 - Measurement

Money and Banking

Lecture 10

Review of the Previous Lecture

- Application of Present Value Concept
 - Compound Annual Rate
 - Interest Rates vs Discount Rate
 - Internal Rate of Return

Topics under Discussion

- Application of Present Value Concept
 - Bond Pricing
- Real Vs Nominal Interest Rates
- Risk
 - Characteristics
 - Measurement

Bond Pricing

- A bond is a promise to make a series of payments on specific future date.
 - It is a legal contract issued as part of an arrangement to borrow
- The most common type is a coupon bond, which makes annual payments called coupon payments
 - The percentage rate is called the coupon rate
 - The bond also specifies a maturity date (n) and has a final payment (F), which is the principal, face value, or par value of the bond

Bond Pricing

- The price of a bond is the present value of its payments
- To value a bond we need to value the repayment of principal and the payments of interest

Bond Pricing

- Valuing the Principal Payment:
 - a straightforward application of present value where n represents the maturity of the bond
- Valuing the Coupon Payments:
 - requires calculating the present value of the payments and then adding them; remember, present value is additive
- Valuing the Coupon Payments plus Principal:
 - means combining the above

Bond Pricing

Payment stops at the maturity date. (n)

A payment is for the face value (F) or principle of the bond

Coupon Bonds make annual payments called, Coupon Payments (C), based upon an interest rate, the coupon rate (i_c),

$$C = i_c * F$$

Bond Pricing

A bond that has a \$100 principle payment in n years. The present value (P_{BP}) of this is now:

$$P_{BP} = \frac{F}{(1+i)^n} = \frac{\$100}{(1+i)^n}$$

Bond Pricing

If the bond has n coupon payments (C), where $C = i_c * F$, the Present Value (P_{CP}) of the coupon payments is:

$$P_{CP} = \frac{C}{(1+i)^1} + \frac{C}{(1+i)^2} + \frac{C}{(1+i)^3} + \dots + \frac{C}{(1+i)^n}$$

Bond Pricing

Present Value of Coupon Bond (P_{CB}) =
Present value of Yearly Coupon Payments (P_{CP})
+
Present Value of the Principal Payment (P_{BP})

$$P_{CB} = P_{CP} + P_{BP} = \left[\frac{C}{(1+i)^1} + \frac{C}{(1+i)^2} + \frac{C}{(1+i)^3} + \dots + \frac{C}{(1+i)^n} \right] + \frac{F}{(1+i)^n}$$

Bond Pricing

Note:

- The value of the coupon bond rises when the yearly coupon payments rise and when the interest rate falls
- Lower interest rates mean higher bond prices and vice versa.
- The value of a bond varies inversely with the interest rate used to discount the promised payments

Real and Nominal Interest Rates

- So far we have been computing the present value using nominal interest rates (i), or interest rates expressed in current-dollar terms
- But inflation affects the purchasing power of a dollar, so we need to consider the real interest rate (r), which is the inflation-adjusted interest rate.
- The Fisher equation tells us that the nominal interest rate is equal to the real interest rate plus the expected rate of inflation

Real and Nominal Interest Rates

Fisher Equation:

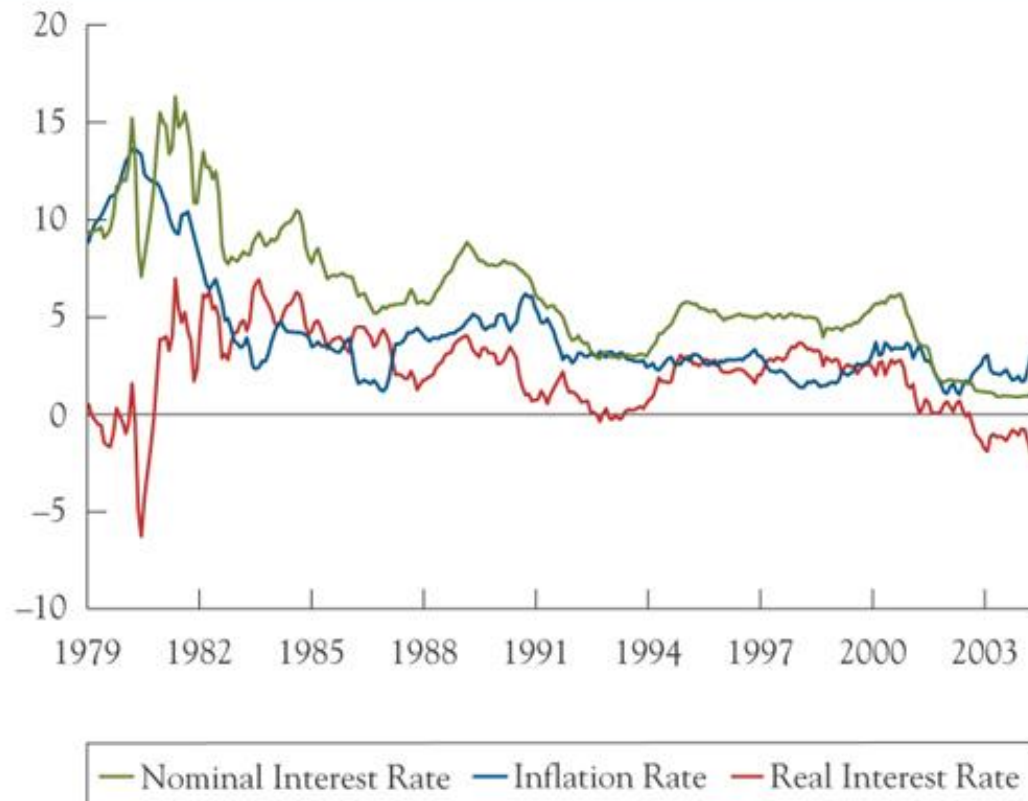
$$i = r + \pi^e$$

or

$$r = i - \pi^e$$

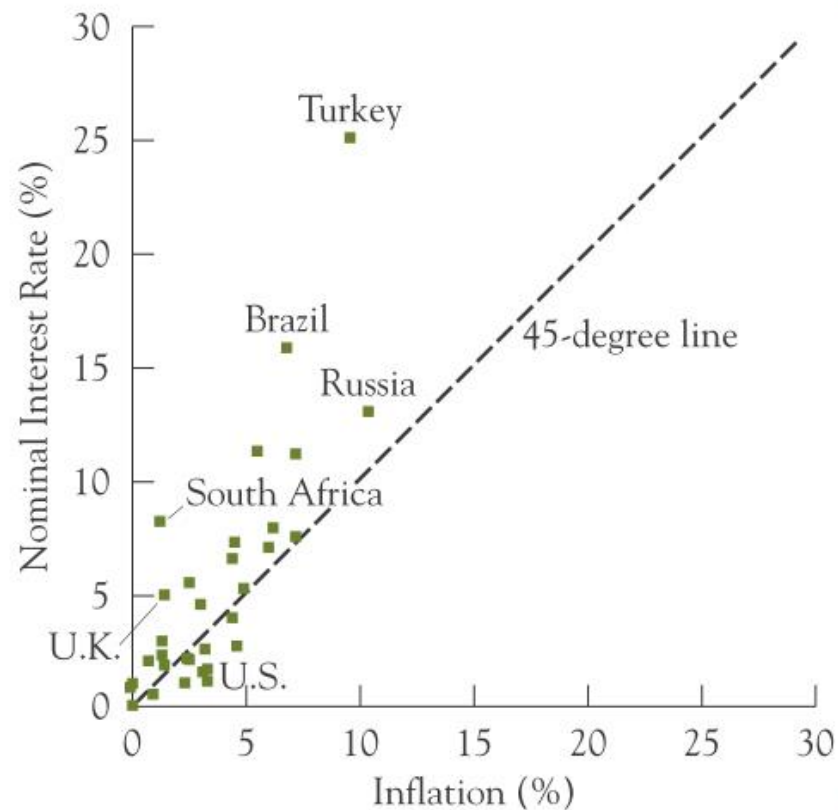
Real and Nominal Interest Rates

Figure 4.3 Nominal Interest Rates, Inflation, and Real Interest Rates



Real and Nominal Interest Rates

Figure 4.4 Inflation and Nominal Interest Rates, April 2004



Risk

- Every day we make decisions that involve financial and economic risk.
 - How much car insurance should we buy?
 - Should we refinance the home loan now or a year from now?
 - Should we save more for retirement, or spend the extra money on a new car?

Risk

- Interestingly enough, the tools we use today to measure and analyze risk were first developed to help players analyze games of chance.
- For thousands of years, people have played games based on a throw of the dice, but they had little understanding of how those games actually worked
- Since the invention of probability theory, we have come to realize that many everyday events, including those in economics, finance, and even weather forecasting, are best thought of as analogous to the flip of a coin or the throw of a die

Risk

- Still, while experts can make educated guesses about the future path of interest rates, inflation, or the stock market, their predictions are really only that—guesses.
- And while meteorologists are fairly good at forecasting the weather a day or two ahead, economists, financial advisors, and business gurus have dismal records.
- So understanding the possibility of various occurrences should allow everyone to make better choices. While risk cannot be eliminated, it can often be managed effectively.

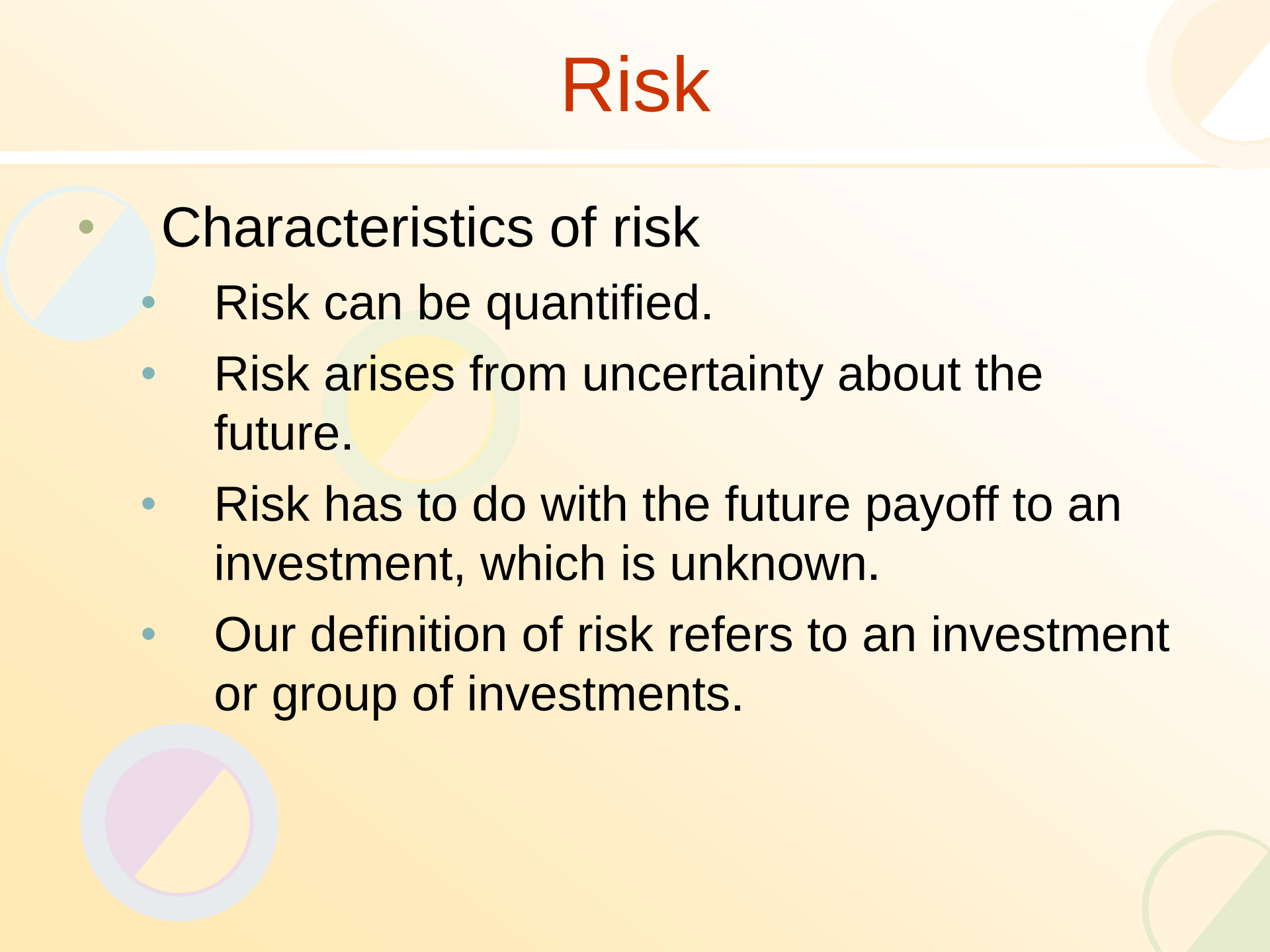
Risk

- Finally, while most people view risk as a curse to be avoided whenever possible, risk also creates opportunities.
 - The payoff from a winning bet on one hand of cards can often erase the losses on a losing hand.
 - Thus the importance of probability theory to the development of modern financial markets is hard to overemphasize.
 - People require compensation for taking risks. Without the capacity to measure risk, we could not calculate a fair price for transferring risk from one person to another, nor could we price stocks and bonds, much less sell insurance.
 - The market for options didn't exist until economists learned how to compute the price of an option using probability theory

Risk

- We need a definition of risk that focuses on the fact that the outcomes of financial and economic decisions are almost always unknown at the time the decisions are made.
- Risk is a measure of uncertainty about the future payoff of an investment, measured over some time horizon and relative to a benchmark.

Risk



Characteristics of risk

- Risk can be quantified.
- Risk arises from uncertainty about the future.
- Risk has to do with the future payoff to an investment, which is unknown.
- Our definition of risk refers to an investment or group of investments.

Risk

- Characteristics of risk
 - Risk must be measured over some time horizon.
 - Risk must be measured relative to some benchmark, not in isolation.
 - If you want to know the risk associated with a specific investment strategy, the most appropriate benchmark would be the risk associated with other investing strategies

Measuring Risk

Measuring Risk requires:

- List of all possible outcomes
- Chance of each one occurring.
- The tossing of a coin
 - What are possible outcomes?
 - What is the chance of each one occurring?
 - Is coin fair?

Measuring Risk

- Probability is a measure of likelihood that an event will occur
 - Its value is between zero and one
 - The closer probability is to zero, less likely it is that an event will occur.
 - No chance of occurring if probability is exactly zero
 - The closer probability is to one, more likely it is that an event will occur.
 - The event will definitely occur if probability is exactly one
- Probabilities can also be expressed as frequencies

Measuring Risk

Table 5.1 A Simple Example: All Possible Outcomes of a Single Coin Toss

Possibilities	Probability	Outcome
#1	$\frac{1}{2}$	Heads
#2	$\frac{1}{2}$	Tails

We must include **all possible outcomes** when constructing such a table

Measuring Risk

- The sum of the probabilities of all the possible outcomes must be 1, since one of the possible outcomes must occur (we just don't know which one)
- To calculate the expected value of an investment, multiply each possible payoff by its probability and then sum all the results. This is also known as the mean.

Measuring Risk

Case 1

An Investment can rise or fall in value.
Assume that an asset purchased for \$1000
is equally likely to fall to \$700 or rise to
\$1400

Measuring Risk

Table 5.2 Investing \$1,000: Case 1

Possibilities	Probability	Payoff	Payoff $\times$ Probability
#1	$\frac{1}{2}$	\$700	\$350
#2	$\frac{1}{2}$	\$1,400	\$700

Expected Value = Sum of (Probability times Payoff) = \$1,050

Expected Value = $\frac{1}{2} (\$700) + \frac{1}{2} (\$1400) = \$1050$

Measuring Risk

Case 2

The \$1,000 investment might pay off

- \$100 (prob=.1) or
- \$2000 (prob=.1) or
- \$700 (prob=.4) or
- \$1400 (prob=.4)

Measuring Risk

Table 5.3 Investing \$1,000: Case 2

Possibilities	Probability	Payoff	Payoff $\times$ Probability
#1	0.1	\$100	10
#2	0.4	\$700	280
#3	0.4	\$1,400	560
#4	0.1	\$2,000	200

Expected Value = Sum of (Probability times Payoff) = \$1,050

Measuring Risk

- Investment payoffs are usually discussed in percentage returns instead of in dollar amounts; this allows investors to compute the gain or loss on the investment regardless of its size
- Though both cases have the same expected return, \$50 on a \$1000 investment, or 5%, the two investments have different levels or risk.
- A wider payoff range indicates more risk.

Summary

- Application of Present Value Concept
 - Bond Pricing
- Real Vs Nominal Interest Rates
- Risk
 - Characteristics
 - Measurement

Money and Banking

Lecture 11

Review of the Previous Lecture

- Application of Present Value Concept
 - Internal Rate of Return
 - Bond Pricing
- Real Vs Nominal Interest Rates
- Risk
 - Characteristics

Topics under Discussion

- Measuring Risk
 - Variance and Standard Deviation
 - Value At Risk (VAR)
 - Risk Aversion & Risk Premium

Measuring Risk

- Probability is a measure of likelihood that an event will occur
 - Its value is between zero and one
 - The closer probability is to zero, less likely it is that an event will occur.
 - No chance of occurring if probability is exactly zero
 - The closer probability is to one, more likely it is that an event will occur.
 - The event will definitely occur if probability is exactly one
- Probabilities can also be expressed as frequencies

Measuring Risk

Table 5.1 A Simple Example: All Possible Outcomes of a Single Coin Toss

Possibilities	Probability	Outcome
#1	$\frac{1}{2}$	Heads
#2	$\frac{1}{2}$	Tails

We must include **all possible outcomes** when constructing such a table

Measuring Risk

- The sum of the probabilities of all the possible outcomes must be 1, since one of the possible outcomes must occur (we just don't know which one)
- To calculate the expected value of an investment, multiply each possible payoff by its probability and then sum all the results. This is also known as the mean.

Measuring Risk

Case 1

An Investment can rise or fall in value.
Assume that an asset purchased for \$1000
is equally likely to fall to \$700 or rise to
\$1400

Measuring Risk

Table 5.2 Investing \$1,000: Case 1

Possibilities	Probability	Payoff	Payoff $\times$ Probability
#1	$\frac{1}{2}$	\$700	\$350
#2	$\frac{1}{2}$	\$1,400	\$700

Expected Value = Sum of (Probability times Payoff) = \$1,050

Expected Value = $\frac{1}{2} (\$700) + \frac{1}{2} (\$1400) = \$1050$

Measuring Risk

Case 2

The \$1,000 investment might pay off

- \$100 (prob=.1) or
- \$2000 (prob=.1) or
- \$700 (prob=.4) or
- \$1400 (prob=.4)

Measuring Risk

Table 5.3 Investing \$1,000: Case 2

Possibilities	Probability	Payoff	Payoff $\times$ Probability
#1	0.1	\$100	10
#2	0.4	\$700	280
#3	0.4	\$1,400	560
#4	0.1	\$2,000	200

Expected Value = Sum of (Probability times Payoff) = \$1,050

Measuring Risk

- Investment payoffs are usually discussed in percentage returns instead of in dollar amounts; this allows investors to compute the gain or loss on the investment regardless of its size
- Though both cases have the same expected return, \$50 on a \$1000 investment, or 5%, the two investments have different levels or risk.
- A wider payoff range indicates more risk.

Measuring Risk

- Most of us have an intuitive sense for risk and its measurement;
 - the wider the range of outcomes the greater the risk.
- A financial instrument with no risk at all is a risk-free investment or a risk-free asset;
 - its future value is known with certainty and
 - its return is the risk-free rate of return

Measuring Risk

- If the risk-free return is 5 percent, a \$1000 risk-free investment will pay \$1050, its expected value, with certainty.
- If there is a chance that the payoff will be either more or less than \$1050, the investment is risky.

Measuring Risk

- We can measure risk by measuring the spread among an investment's possible outcomes. There are two measures that can be used:
 - Variance and Standard Deviation
 - measure of spread
 - Value At Risk (VAR)
 - Measure of riskiness of worst case

Variance

- The *variance* is defined as the probability weighted average of the squared deviations of the possible outcomes from their expected value
- To calculate the variance of an investment,
 1. Compute expected value
 2. Subtract expected value from each possible payoff
 3. Square each result
 4. multiply by its probability
 5. Add up the results

Variance

1. Compute the expected value:
 $(\$1400 \times \frac{1}{2}) + (\$700 \times \frac{1}{2}) = \$1050.$
2. Subtract this from each of the possible payoffs:
 $\$1400 - \$1050 = \$350$
 $\$700 - \$1050 = -\$350$
3. Square each of the results:
 $\$350^2 = 122,500(\text{dollars})^2$ and
 $(-\$350)^2 = 122,500(\text{dollars})^2$

Variance

4. Multiply each result times its probability and add up the results:

$$\frac{1}{2} [122,500(\text{dollars})^2] +$$

$$\frac{1}{2} [122,500(\text{dollars})^2] = 122,500(\text{dollars})^2$$

Variance

More compactly;

$$\begin{aligned}\text{Variance} &= \frac{1}{2}(\$1400 - \$1050)^2 + \\ &\quad \frac{1}{2}(\$700 - \$1050)^2 \\ &= 122,500(\text{dollars})^2\end{aligned}$$

Standard Deviation

The standard deviation is the square root of the variance, or:

Standard Deviation (case 1) = \$350

Standard Deviation (case 2) = \$528

The greater the standard deviation, the higher the risk.

It more useful because it is measured in the same units as the payoffs (that is, dollars and not squared dollars)

Standard Deviation

- The standard deviation can then also be converted into a percentage of the initial investment, providing a baseline against which we can measure the risk of alternative investments
- Given a choice between two investments with the same expected payoff, most people would choose the one with the lower standard deviation because it would have less risk

Value At Risk

- Sometimes we are less concerned with the spread of possible outcomes than we are with the value of the worst outcome.
- To assess this sort of risk we use a concept called “value at risk.”
- Value at risk measures risk at the maximum potential loss

Risk Aversion

- Most people don't like risk and will pay to avoid it; most of us are risk averse
- A risk-averse investor will always prefer an investment with a certain return to one with the same *expected* return, but any amount of uncertainty.
- Buying insurance is paying someone to take our risks, so if someone wants us to take on risk we must be paid to do so

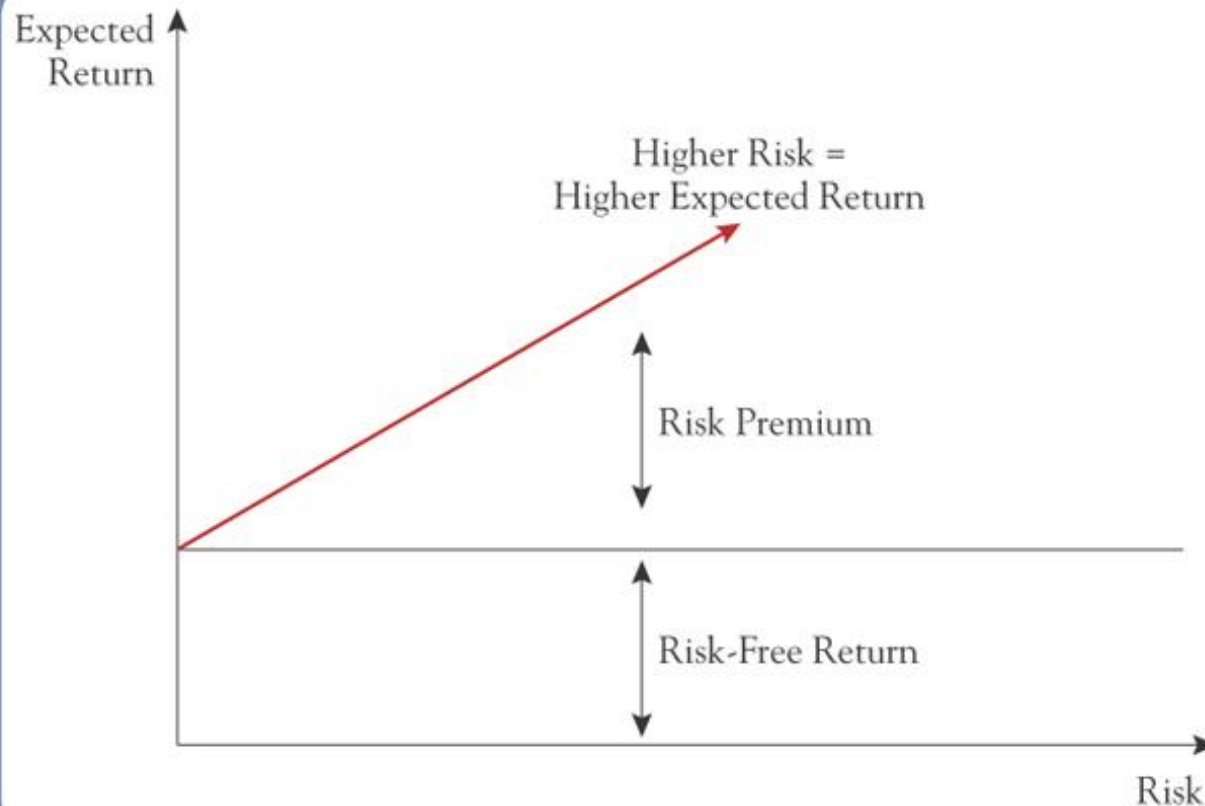
Risk Premium

- The riskier an investment – the higher the compensation that investors require for holding it – the higher the risk premium.
- Riskier investments must have higher expected returns
- There is a trade-off between risk and expected return;
 - you can't get a high return without taking considerable risk.

Risk and Expected Return

Figure 5.2 The Trade-off between Risk and Expected Return

The higher the risk, the higher the expected return. The risk premium equals the expected return on the risky investment minus the risk-free return.



Summary

- Measuring Risk
 - Variance and Standard Deviation
 - Value At Risk (VAR)
 - Risk Aversion and Risk Premium

Money and Banking

Lecture 12

Review of the Previous Lecture

- Measuring Risk
 - Variance and Standard Deviation
 - Value At Risk (VAR)
 - Risk Aversion & Risk Premium

Topics under Discussion

- Sources of Risk
 - Idiosyncratic
 - Systematic
- Reducing Risk Through Diversification
 - Hedging Risk
 - Spreading Risk
- Bond and Bond Pricing

How to Evaluate Risk

- Lets go back to our previous example where \$1,000 yields either \$1,400 and \$700 with equal probability
- If we think about this investment in terms of gains and losses, this investment offers an equal chance of gaining \$400 or loosing \$300
- Should you take the risk?

How to Evaluate Risk

Evaluating the Risk of a \$1,000 investment

A. The Gain

Payoff	Probability
+ \$400	$\frac{1}{2}$
\$0	$\frac{1}{2}$

B. The Loss

Payoff	Probabilities
\$0	$\frac{1}{2}$
- \$300	$\frac{1}{2}$

How to Evaluate Risk

- Deciding if a risk is worth taking
 1. List all the possible outcomes or payoffs
 2. Assign a probability to each possible payoff
 3. Divide the payoffs into gains and losses
 4. Ask how much you would be willing to pay to receive the gain
 5. Ask how much you would be willing to pay to avoid the loss
 6. If you are willing to pay more to receive the gain than to avoid the loss, you should take the risk

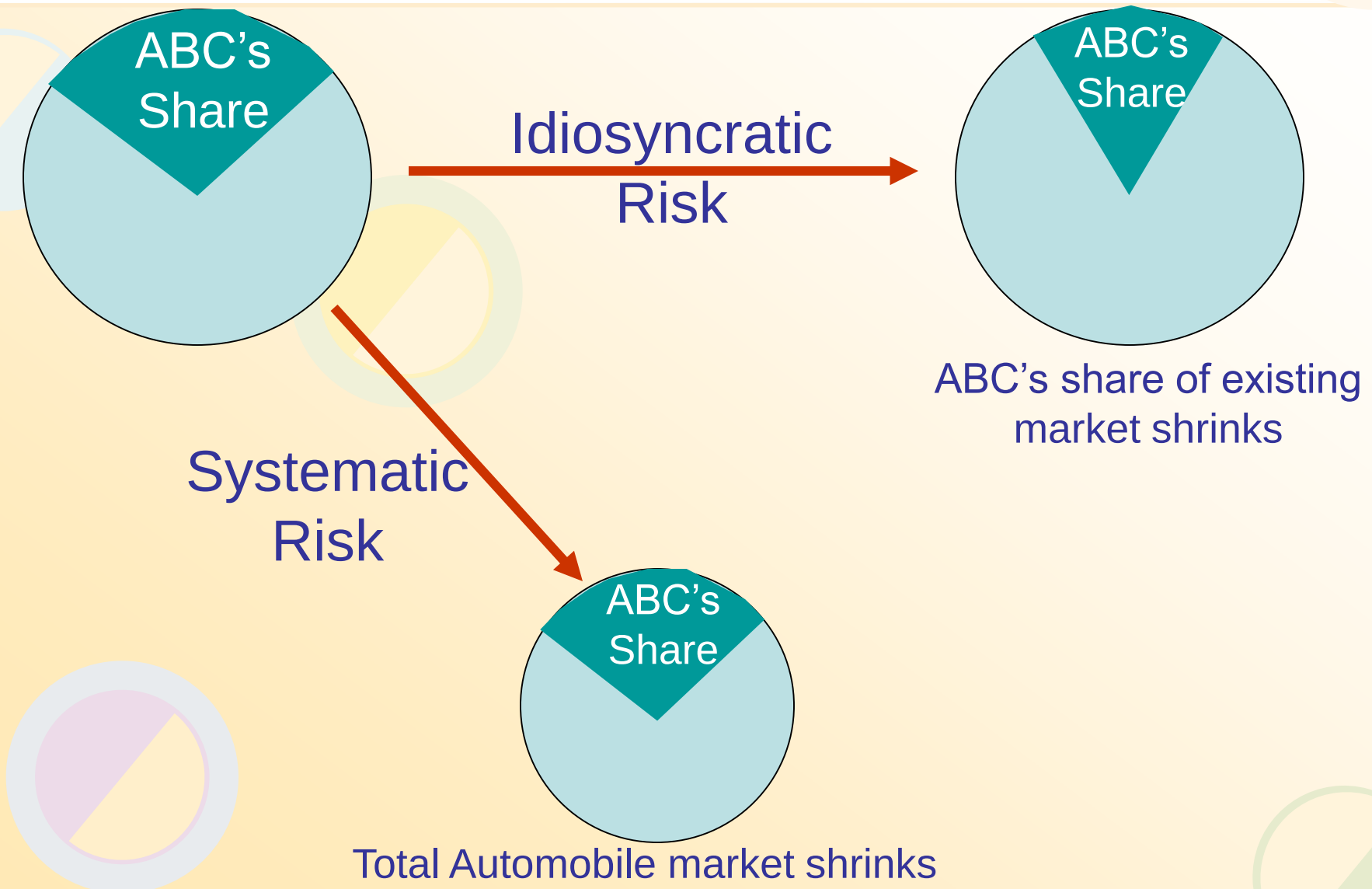
Sources of Risk

- Risk is everywhere. It comes in many forms and from almost every imaginable place
- Regardless of the source, risks can be classified as either idiosyncratic or systematic
 - Idiosyncratic, or unique, risks affect only a small number of people.
 - Systematic risks affect everyone.

Sources of Risk

- In the context of the entire economy,
 - higher oil prices would be an idiosyncratic risk and
 - changes in general economic conditions would be systematic risk.

Sources of Risk



Reducing Risk through Diversification

- Risk can be reduced through diversification, the principle of holding more than one risk at a time.
 - Holding several different investments reduces the overall risk that an investor bears
 - A combination of risky investments is often less risky than any one individual investment
- There are two ways to diversify your investments:
 - you can hedge risks or
 - you can spread them among the many investments

Reducing Risk through Diversification

Hedging Risk

- Hedging is the strategy of reducing overall risk by making two investments with opposing risks.
 - When one does poorly, the other does well, and vice versa.
 - So while the payoff from each investment is volatile, together their payoffs are stable.

Reducing Risk through Diversification

Table 5.6 Payoffs on Two Separate Investments of \$100

	Payoff from Owning Only		
Possibility	ABC Electric	XYZ Oil	Probability
Oil prices rise	\$100	\$120	$\frac{1}{2}$
Oil prices fall	\$120	\$100	$\frac{1}{2}$

Reducing Risk through Diversification

Let's compare three strategies for investing \$100, given the relationships shown in the table:

1. Invest \$100 in ABC Electric
2. Invest \$100 in XYZ Oil
3. Invest half in each company – \$50 in ABC and \$50 in XYZ

Reducing Risk through Diversification

**Table 5.7 Results of Possible Investment Strategies:
Hedging Risk**

Initial Investment = \$100

Investment Strategy	Expected Payoff	Standard Deviation
ABC only	\$110	\$10
XYZ only	\$110	\$10
$\frac{1}{2}$ and $\frac{1}{2}$	\$110	\$0

Reducing Risk through Diversification

Spreading Risk

- Investments don't always move predictably in opposite directions, so you can't always reduce risk through hedging
- You can lower risk by simply spreading it around and finding investments whose payoffs are completely unrelated
 - The more independent sources of risk you hold the lower your overall risk

Reducing Risk through Diversification

- Adding more and more independent sources of risk reduces the standard deviation until it becomes negligible.
- Consider three investment strategies:
 - (1) ABC Electric only,
 - (2) EFG Soft only, and
 - (3) half in ABC and half in EFG.

Reducing Risk through Diversification

- The expected payoff on each of these strategies is the same: \$110.
 - For the first two strategies, \$100 in either company, the standard deviation is still 10, just as it was before.
 - But for the third strategy, \$50 in ABC and \$50 in EFG, the analysis is more complicated.
 - There are four possible outcomes, two for each stock

Reducing Risk through Diversification

Table 5.8 Payoffs from Investing \$50 in Each of Two Stocks
Initial Investment = \$100

Possibilities	ABC	EFG Soft	Total Payoff	Probability
#1	\$60	\$60	\$120	$\frac{1}{4}$
#2	\$60	\$50	\$110	$\frac{1}{4}$
#3	\$50	\$60	\$110	$\frac{1}{4}$
#4	\$50	\$50	\$100	$\frac{1}{4}$

Reducing Risk through Diversification

Table 5.9 Results of Possible Investment Strategies: Spreading Risk
Initial Investment = \$100

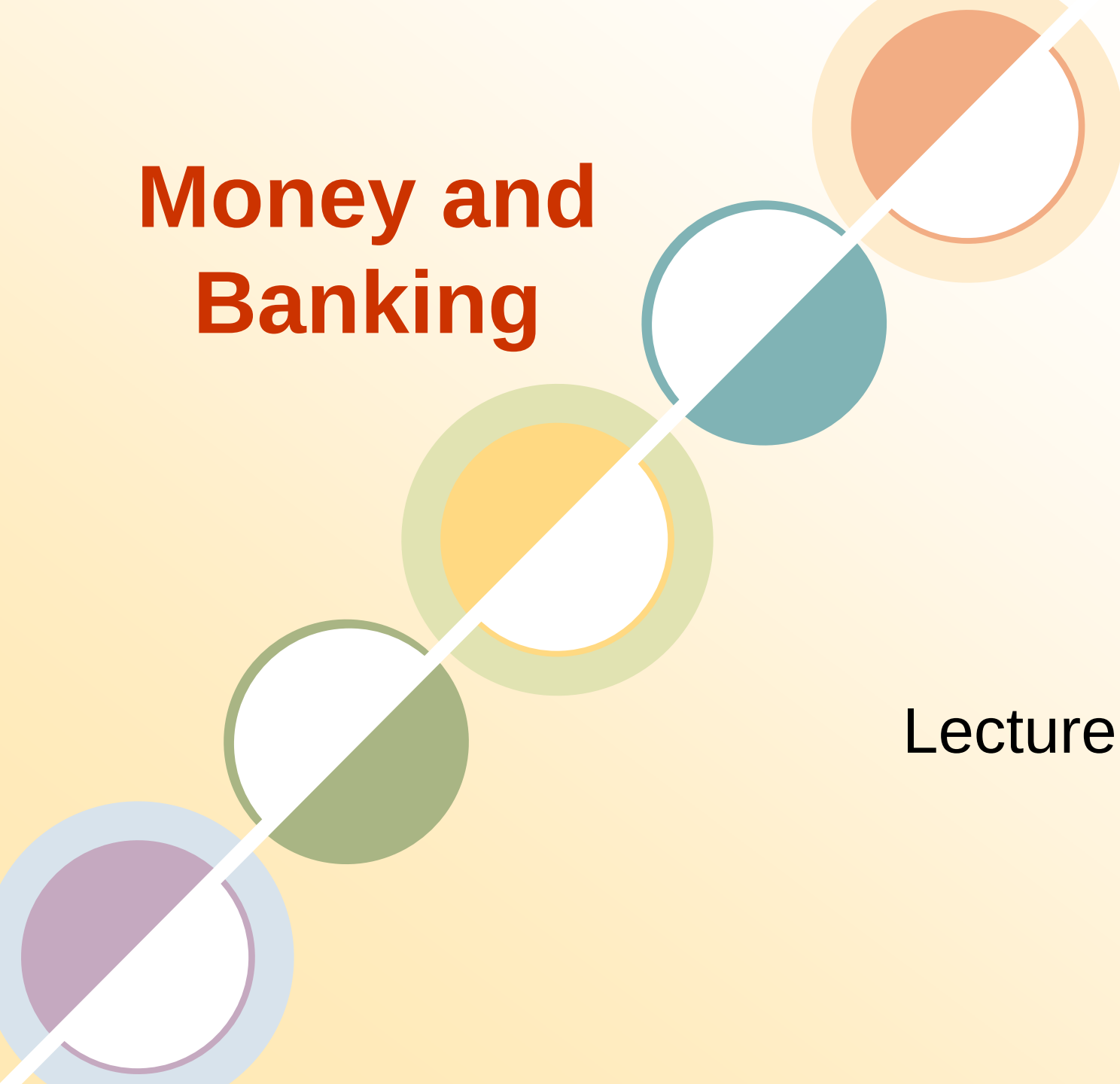
Investment Strategy	Expected Payoff	Standard Deviation
ABC	\$110	\$10
EFG Soft	\$110	\$10
$\frac{1}{2}$ and $\frac{1}{2}$	\$110	\$7.1

Summary

- Sources of Risk
 - Idiosyncratic
 - Systematic
- Reducing Risk Through Diversification
 - Hedging Risk
 - Spreading Risk
- Bond and Bond Pricing

Money and Banking

Lecture 13



Review of the Previous Lecture

- Risk
 - Characteristics
 - Measurement
 - Sources
- Reducing Risk
 - Hedging
 - Spreading

Topics Under Discussion

- Bond & Bond pricing
 - Zero Coupon Bond
 - Fixed Payment Loan
 - Coupon Bonds
 - Consols
- Bond Yield
 - Yield to Maturity
 - Current Yield

Bonds

- Virtually any financial arrangement involving the current transfer of resources from a lender to a borrower, with a transfer back at some time in the future, is a form of bond.
 - Car loans, home mortgages, even credit card balances all create a loan from a financial intermediary to an individual making a purchase
 - Governments and large corporations sell bonds when they need to borrow

Bonds

- The ease with which individuals, corporations, and governments borrow is essential to the functioning of our economic system.
- Without this free flow of resources through the bond markets, the economy would grind to a halt.
- Historically, we can trace the concept of using bonds to borrow to monarchs' almost insatiable appetite for resources.

Bonds

- The Dutch invented modern bonds to finance their lengthy war of independence
- The British refined the use of bonds to finance government activities.
- The practice was soon popular among other countries

Bonds

- A standard bond specifies the fixed amount to be paid and the exact dates of the payments
- How much should you be paying for a bond?
 - The answer depends on bond's characteristics

Bond Prices

- Zero-coupon bonds,
 - promise a single future payment, such as a Treasury Bill.
- Fixed payment loans,
 - conventional mortgages.
 - Car loans
- Coupon Bonds,
 - make periodic interest payments and repay the principal at maturity.
 - Treasury Bonds and most corporate bonds are coupon bonds.
- Consols,
 - make periodic interest payments forever, never repaying the principal that was borrowed.

Zero-Coupon Bonds

- These are pure discount bonds since they sell at a price below their face value
- The difference between the selling price and the face value represents the interest on the bond
- The price of such a bond, like a Treasury bill (called “T-bill”), is the present value of the future payment

Zero-Coupon Bonds

Price of a \$100 face value zero-coupon bond

$$= \frac{\$100}{(1 + i)^n}$$

Where

i is the interest rate in decimal form and
 n is time until the payment is made in the
same time units as the interest rate

Zero-Coupon Bonds

- Given n , the price of a bond and the interest rate move in opposite directions
- The most common maturity of a T-bill is 6 months; the Treasury does not issue them with a maturity greater than 1 year
- The shorter the time until the payment is made the higher the price of the bond, so 6 month T-bills have a higher price than a one-year T-bill

Zero-Coupon Bonds

Examples. Assume $i=4\%$

Price of a One-Year Treasury Bill.

$$= \frac{100}{(1 + 0.04)} = \$96.15$$

Price of a Six-Month Treasury Bill

$$= \frac{100}{(1 + 0.04)^{1/2}} = \$98.06$$

Zero-Coupon Bonds

- The interest rate and the price for the T-bill move inversely.
- If we know the face value and the price then we can solve for the interest rate

Fixed Payment Loans

- they promise a fixed number of equal payments at regular intervals
 - Home mortgages and car loans are examples of fixed payment loans;
- These loans are amortized, meaning that the borrower pays off the principal along with the interest over the life of the loan.
- Each payment includes both interest and some portion of the principal
- The price of the loan is the present value of all the payments

Fixed Payment Loans

Value of a Fixed Payment Loan =

$$\frac{\textit{FixedPayment}}{(1+i)} + \frac{\textit{FixedPayment}}{(1+i)^2} + \dots + \frac{\textit{FixedPayment}}{(1+i)^n}$$

Coupon Bond

- The value of a coupon bond is the present value of the periodic interest payments plus the present value of the principal repayment at maturity

$$P_{CB} = \left[\frac{\text{CouponPayment}}{(1+i)^1} + \frac{\text{CouponPayment}}{(1+i)^2} + \dots + \frac{\text{CouponPayment}}{(1+i)^n} \right] + \frac{\text{FaceValue}}{(1+i)^n}$$

- The latter part, the repayment of the principal, is just like a zero-coupon bond

Consols

- A consol offers only periodic interest payments; the borrower never repays the principal
- There are no privately issued consols because only governments can credibly promise to make payments forever
- The price of a consol is the present value of all the future interest payments, which is a bit complicated because there are an infinite number of payments

$$P_{\text{Consol}} = \frac{\text{Yearly Coupon Payment}}{i}$$

Bond Yields

- Now that we know how to price a bond while interest rate is known; we now move to other direction and calculate the interest rate or return to an investor
- So combining information about the promised payments with the price to obtain what is called the yield – a measure of cost of borrowing or reward for lending.
- Interest rate and yield are used interchangeably

Yield To Maturity

- The most useful measure of the return on holding a bond is called the yield to maturity (YTM).
- This is the yield bondholders receive if they hold the bond to its maturity when the final principal payment is made
- It can be calculated from the present value formula

Yield To Maturity

Price of One-Year 5 percent Coupon Bond =

$$\frac{\$5}{(1+i)} + \frac{\$100}{(1+i)}$$

- The value of i that solves this equation is the yield to maturity

Yield To Maturity

- If the price of the bond is \$100, then the yield to maturity equals the coupon rate.
- Since the price rises as the yield falls, when the price is above \$100, the yield to maturity must be below the coupon rate.
- Since the price falls as the yield rises, when the price is below \$100, the yield to maturity must be above the coupon rate.

Yield To Maturity

- Considering 5% coupon bond

- If YTM is 5% then price is

$$\frac{\$5}{(1+.05)} + \frac{\$100}{(1+.05)} = \$100$$

- If YTM is 4% then price is

$$\frac{\$5}{(1+.04)} + \frac{\$100}{(1+.04)} = \$100.96$$

- If YTM is 6% then price is

$$\frac{\$5}{(1+.06)} + \frac{\$100}{(1+.06)} = \$99.06$$

Yield To Maturity

- Generally
 - If the yield to maturity equals the coupon rate, the price of the bond is the same as its face value.
 - If the yield is greater than the coupon rate, the price is lower;
 - if the yield is below the coupon rate, the price is greater

Summary

- Bond & Bond pricing
 - Zero Coupon Bond
 - Fixed Payment Loan
 - Coupon Bonds
 - Consols
- Bond Yield
 - Yield to Maturity
 - Current Yield

Money and Banking

Lecture 14

Review of the Previous Lecture

- Bond & Bond pricing
 - Zero Coupon Bond
 - Fixed Payment Loan
 - Coupon Bonds
 - Consols
- Bond Yield
 - Yield to Maturity
 - Current Yield

Topics under Discussion

- Yield to Maturity
- Current Yield
- Holding Period Returns
- Bond Supply & Demand
- Factors affecting Bond Supply
- Factors affecting Bond Demand

Yield To Maturity

- General Relationships
 - If the yield to maturity equals the coupon rate, the price of the bond is the same as its face value.
 - If the yield is greater than the coupon rate, the price is lower;
 - if the yield is below the coupon rate, the price is greater

Yield To Maturity

- If you buy a bond at a price less than its face value you will receive its interest and a capital gain, which is the difference between the price and the face value.
- As a result you have a higher return than the coupon rate
- When the price is above the face value, the bondholder incurs a capital loss and the bond's yield to maturity falls below its coupon rate

Relationship Between Price and Yield to Maturity

Table 1 Yields to Maturity on a 10%-Coupon-Rate Bond Maturing in Ten Years (Face Value = \$1,000)

Price of Bond (\$)	Yield to Maturity (%)
1,200	7.13
1,100	8.48
1,000	10.00
900	11.75
800	13.81

Three Interesting Facts in Table 1

1. When bond is at par, yield equals coupon rate
2. Price and yield are negatively related
3. Yield greater than coupon rate when bond price is below par value

Current Yield

- Current yield is a commonly used, easy-to-compute measure of the proceeds the bondholder receives for making a loan
- It is the yearly coupon payment divided by the price

$$\text{Current Yield} = \frac{\text{Yearly Coupon Payment}}{\text{Price Paid}}$$

Current Yield

- The current yield measures that part of the return from buying the bond that arises solely from the coupon payments;
 - it ignores the capital gain or loss that arises when the bond's price differs from its face value

Current Yield

- Let's return to 1-year 5% coupon bond assuming that it is selling for \$99.
 - Current yield is $5/99 = 0.0505$ or 5.05%
 - YTM for this bond is calculated to be 6.06% through the following calculations

$$\frac{\$5}{(1+i)} + \frac{\$100}{(1+i)} = \$99$$

- If you buy the bond for \$99, one year later you get not only the \$5 coupon payment but also a guaranteed \$1 capital gain, totaling to \$6
- Repeating this process for the bond selling for \$101, current yield is 4.95% and YTM is 3.96%

Current Yield

- The current yield moves inversely to the price;
 - If the price is above the face value, the current yield falls below the coupon rate.
 - When the price falls below the face value, the current yield rises above the coupon rate.
 - If the price and the face value are equal the current yield and the coupon rate are equal.

Current Yield

- Since the yield to maturity takes account of capital gains (and losses),
 - when the bond price is less than its face value the yield to maturity is higher than the current yield,
 - if the price is greater than face value, the yield to maturity is lower than the current yield, which is lower than the coupon rate

Relationship Between a Bond's Price and Its Coupon Rate, Current Yield and Yield to Maturity

- **Bond Price < Face Value:**
Coupon Rate < Current Yield < Yield to Maturity
- **Bond Price = Face Value:**
Coupon Rate = Current Yield = Yield to Maturity
- **Bond Price > Face Value:**
Coupon Rate > Current Yield > Yield to Maturity

Holding Period Returns

- The investor's return from holding a bond need not be the coupon rate
- Most holders of long-term bonds plan to sell them well before they mature, and because the price of the bond may change in the time since its purchase, the return can differ from the yield to maturity
- The holding period return – the return to holding a bond and selling it before maturity.

Holding Period Returns

- The holding period return can differ from the yield to maturity
- The longer the term of the bond, the greater the price movements and associated risk can be

Holding Period Returns

Examples:

- You pay for \$100 for a 10-year 6% coupon bond with a face value of \$100, You intend to hold the bond for one year, i.e. buy a 10 year bond and sell a 9 year bond an year later
 - If interest rate does not change your return will be $\$6/100 = 0.06 = 6\%$

Holding Period Returns

- If interest rate falls to 5% over the year then through using bond pricing formula we can see that
 - You bought a 10-year bond for \$100 and sold a 9-year bond for \$107.11
 - Now the one year holding return has two parts
 - \$6 coupon payment and
 - \$7.11 capital gain

Holding Period Returns

So now, one year holding Period return =

$$\frac{\$6}{\$100} + \frac{\$107.11 - \$100}{\$100} = \frac{\$13.11}{\$100} = .1311$$

or 13.11%

Holding Period Returns

If the interest rate in one year is 7%

One year holding Period return =

$$\frac{\$6}{\$100} + \frac{\$93.48 - \$100}{\$100} = \frac{-\$0.52}{\$100} = -.0052$$

or -.52%

Holding Period Returns

- Generalizing, 1-year holding return is

$$= \frac{\text{Yearly Coupon Payment}}{\text{Price Paid}} + \frac{\text{Change in Price of the Bond}}{\text{Price of the Bond}}$$

$$= \text{Current Yield} + \text{Capital Gain (as a \%)}$$

Bond Market and Interest Rates

- To find out how bond prices are determined and why they change we need to look at the supply and demand in the bond market.
- Let's consider the market for existing bonds at a particular time (the stock of bonds) and consider prices and not interest rates.
- One Year Zero-coupon (discount) Bond.


$$P = \frac{\$100}{1+i} \quad \text{or} \quad i = \frac{\$100 - P}{P}$$

Bond Supply, Demand and Equilibrium

Bond Supply

- The Bond supply curve is the relationship between the price and the quantity of bonds people are willing to sell, all other things being equal.

Bond Supply, Demand and Equilibrium

Bond Supply

- From the point of view of investors, the higher the price, the more tempting it is to sell a bond they currently hold.
- From the point of view of companies seeking finance for new projects, the higher the price at which they can sell bonds, the more advantageous it is to do so.

Bond Supply, Demand and Equilibrium

Bond Supply

For a \$100 one-year zero-coupon bond, the supply will be higher at \$95 than it will be at \$90, all other things being equal.

Bond Supply, Demand and Equilibrium

Bond Demand

- The bond demand curve is the relationship between the price and quantity of bonds that investors demand, all other things being equal.
- As the price falls, the reward for holding the bond rises, so the demand goes up

Bond Supply, Demand and Equilibrium

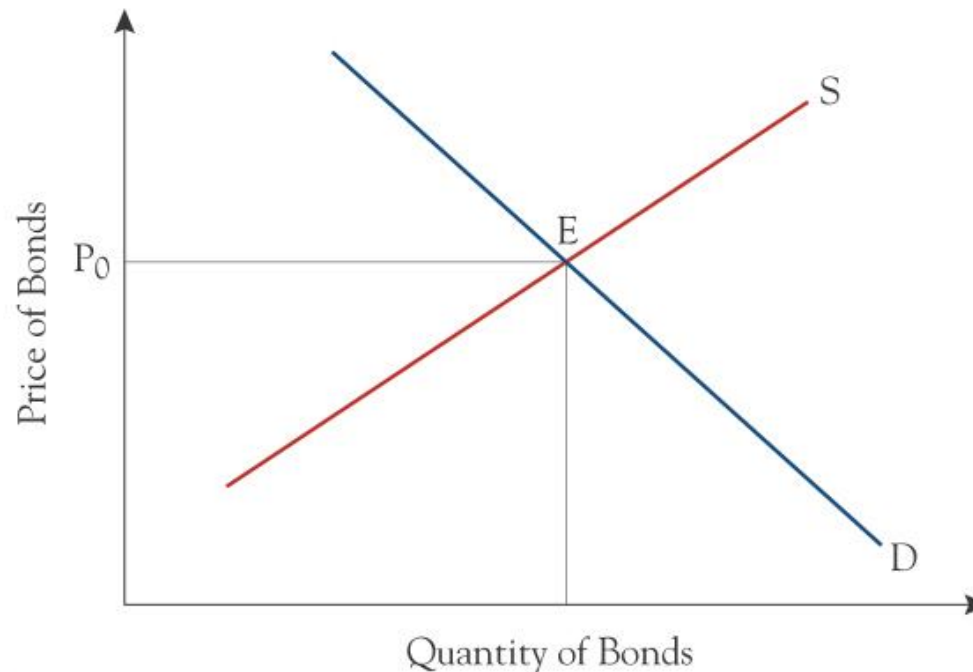
Bond Demand

- The lower the price potential bondholders must pay for a fixed-dollar payment on a future date, the more likely they are to buy a bond
- The zero-coupon bond promising to pay \$100 in one year will be more attractive at \$90 than it will at \$95, all other things being equal.

Bond Supply, Demand and Equilibrium

Figure 6.2 Supply, Demand, and Equilibrium in the Bond Market

The supply of bonds from borrowers slopes up and the demand for bonds from lenders slopes down. Equilibrium in the bond market is determined by the intersection of supply and demand.



Equilibrium in the bond market is the point at which supply equals demand

Bond Supply, Demand and Equilibrium

- If the price is too high (above equilibrium) the excess supply of bonds will push the price back down.
- If the price is too low (below equilibrium) the excess demand for bonds will push it up
- Over time the supply and demand curves can shift, leading to changes in the equilibrium price

Factors that shift Bond Supply

- Changes in government borrowing
 - Any increase in the government's borrowing needs increases the quantity of bonds outstanding, shifting the bond supply curve to the right.
 - This reduces price and increases the interest rate on the bond.

Factors that shift Bond Supply

- Changes in business conditions
 - business-cycle expansions mean more investment opportunities, prompting firms to increase their borrowing and increasing the supply of bonds
 - As business conditions improve, the bond supply curve shifts to the right.
 - This reduces price and increases the interest rate on the bond.
 - By the same logic, weak economic growth can lead to rising bond prices and lower interest rates

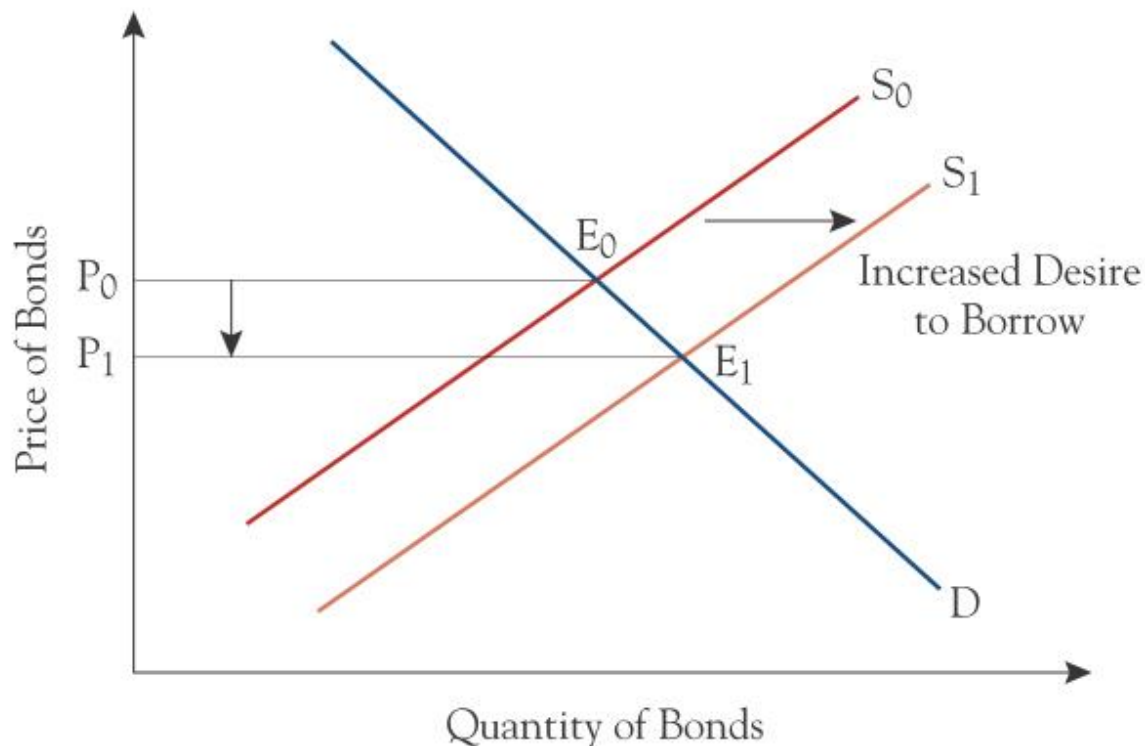
Factors that shift Bond Supply

- Changes in expected inflation
 - Bond issuers care about the real cost of borrowing,
 - So if inflation is expected to increase then the real cost falls and the desire to borrow rises, resulting in the bond supply curve shifting to the right
 - This reduces price and increases the interest rate on the bond.

Factors that shift Bond Supply

Figure 6.3 A Shift in the Supply of Bonds

When borrowers' desire for funds increases, the bond supply curve shifts to the right, lowering bond prices and raising interest rates.



Factors that shift Bond Supply

Table 6.3 Factors That Increase Bond Supply, Lower Bond Prices, and Raise Interest Rates

Change	Effect on Bond Supply, Bond Prices, and Interest Rates
An increase in the government's desired expenditure relative to its revenue	Bond supply shifts to the right, Bond prices ↓, and interest rates ↑
An improvement in general business conditions	Bond supply shifts to the right, Bond prices ↓, and interest rates ↑
An increase in expected inflation, reducing the real cost of repayment	Bond supply shifts to the right, Bond prices ↓, and interest rates ↑

Factors that shift Bond Demand

- wealth
 - An increase in wealth shifts the demand for bonds to the right as wealthier people invest more.
 - This will happen as the economy grows during an expansion.
 - This will increase Bond Prices and lower yields.

Factors that shift Bond Demand

- Expected inflation
 - A fall in expected inflation shifts the bond demand curve to the right, increasing demand at each price and lowering the yield and increasing the Bond's price.

Factors that shift Bond Demand

- Expected return on stocks and other assets
 - If the return on bonds rises relative to the return on alternative investments, the demand for bonds will rise.
 - This will increase bond prices and lower yields.

Factors that shift Bond Demand

- Risk relative to alternatives
 - If a bond becomes less risky relative to alternative investments, the demand for the bond shifts to the right.

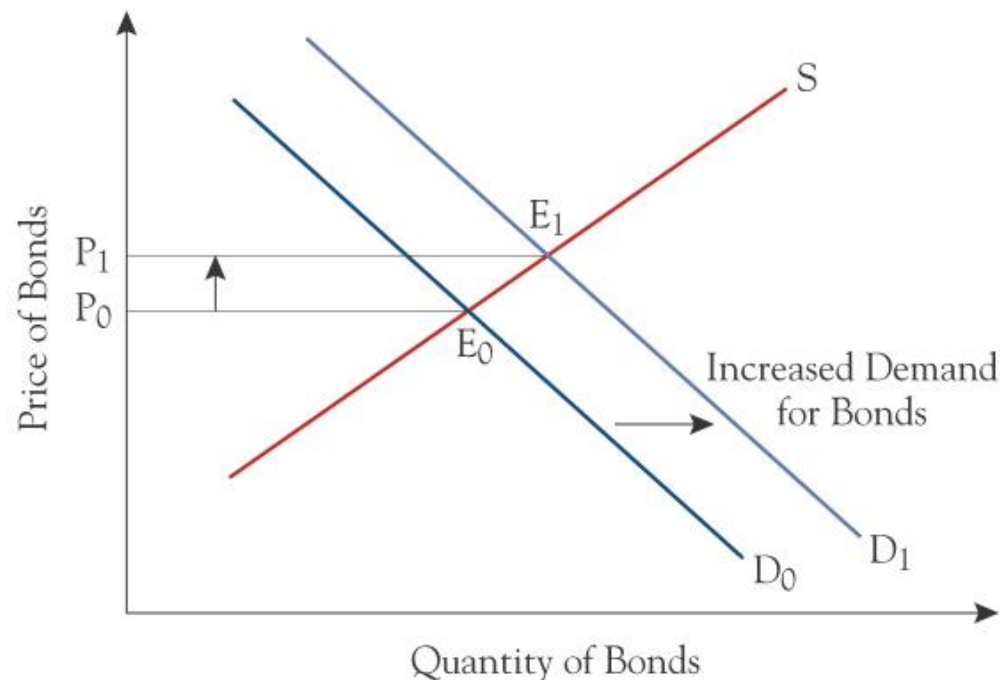
Factors that shift Bond Demand

- Liquidity of bonds relative to alternatives
 - When a bond becomes more liquid relative to alternatives, the demand curve shifts to the right.

Factors that shift Bond Demand

Figure 6.4 Shift in Bond Demand

When there is an increase in investors' willingness to hold bonds, the bond demand curve shifts to the right, increasing bond prices and reducing interest rates.



Factors that shift Bond Demand

Table 6.4 Factors That Increase Bond Demand, Raise Bond Prices, and Lower Interest Rates

Change	Effect on Bond Demand
An increase in wealth increases demand for all assets, including bonds.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓
A reduction in expected inflation makes bonds with fixed nominal payments more desirable.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓
An increase in the expected return on the bond relative to the expected return on alternatives makes bonds more attractive.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓
A decrease in the expected future interest rate makes bonds more attractive.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓
A fall in the riskiness of the bond relative to the riskiness of alternatives makes bonds more attractive.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓
An increase in the liquidity of the bond relative to the liquidity of alternatives makes bonds more attractive.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓

Bonds and Risk

Sources of Bond Risk

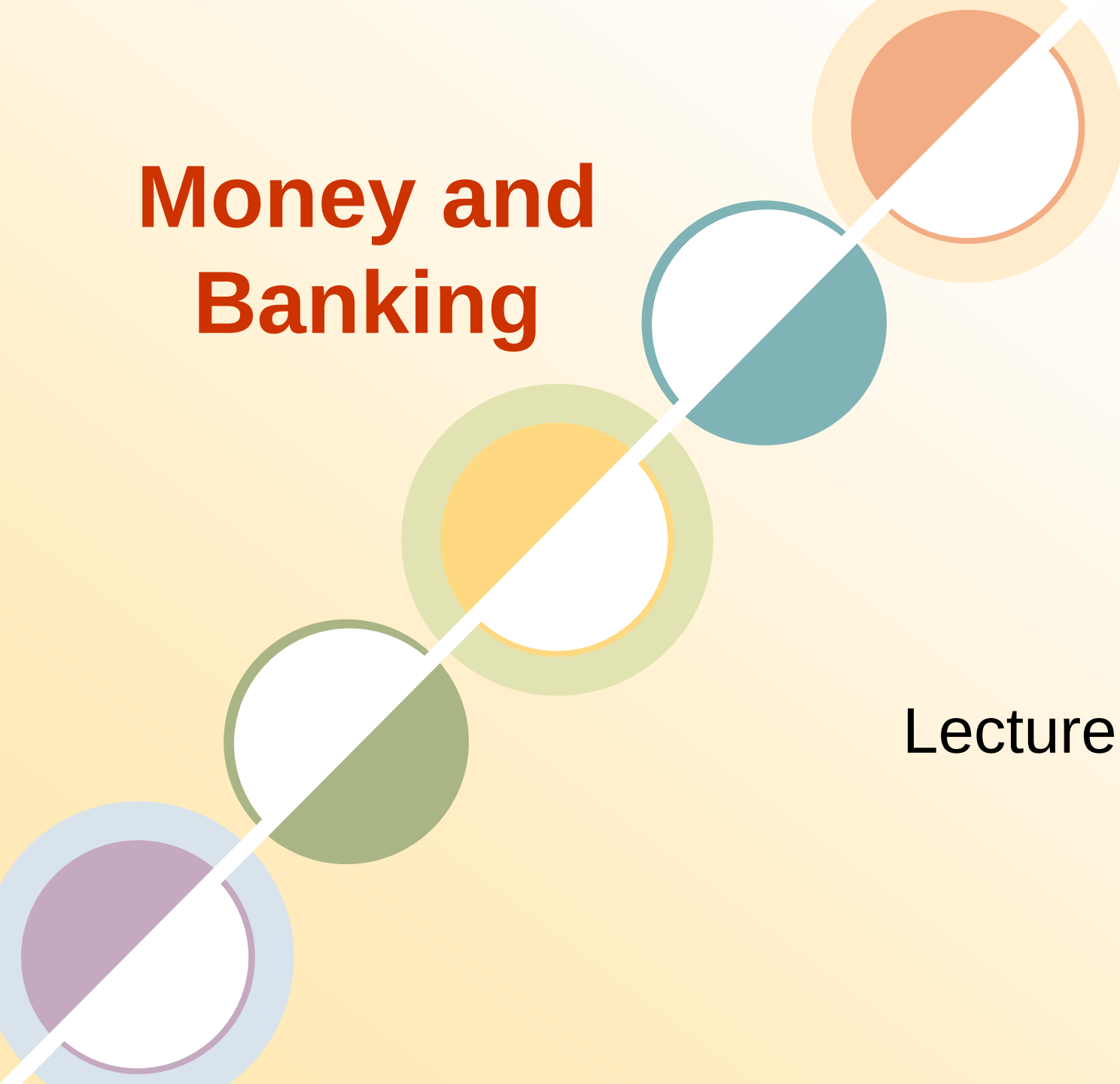
- Default Risk
- Inflation Risk
- Interest-Rate Risk

Summary

- Yield to Maturity
- Current Yield
- Holding Period Returns
- Bond Supply
- Bond Demand

Money and Banking

Lecture 15



Review of the Previous Lecture

- Bonds
 - Yield to Maturity
 - Current Yield
 - Holding Period Returns
 - Bond Supply, Demand and Equilibrium

Topics under Discussion

- Factors Influencing Bond Supply
- Factors Influencing Bond Demand
- Bond and Risk
 - Default Risk
 - Inflation Risk
 - Interest Rate Risk

Factors that shift Bond Supply

- Changes in government borrowing
 - Any increase in the government's borrowing needs increases the quantity of bonds outstanding, shifting the bond supply curve to the right.
 - This reduces price and increases the interest rate on the bond.

Factors that shift Bond Supply

- Changes in business conditions
 - business-cycle expansions mean more investment opportunities, prompting firms to increase their borrowing and increasing the supply of bonds
 - As business conditions improve, the bond supply curve shifts to the right.
 - This reduces price and increases the interest rate on the bond.
 - By the same logic, weak economic growth can lead to rising bond prices and lower interest rates

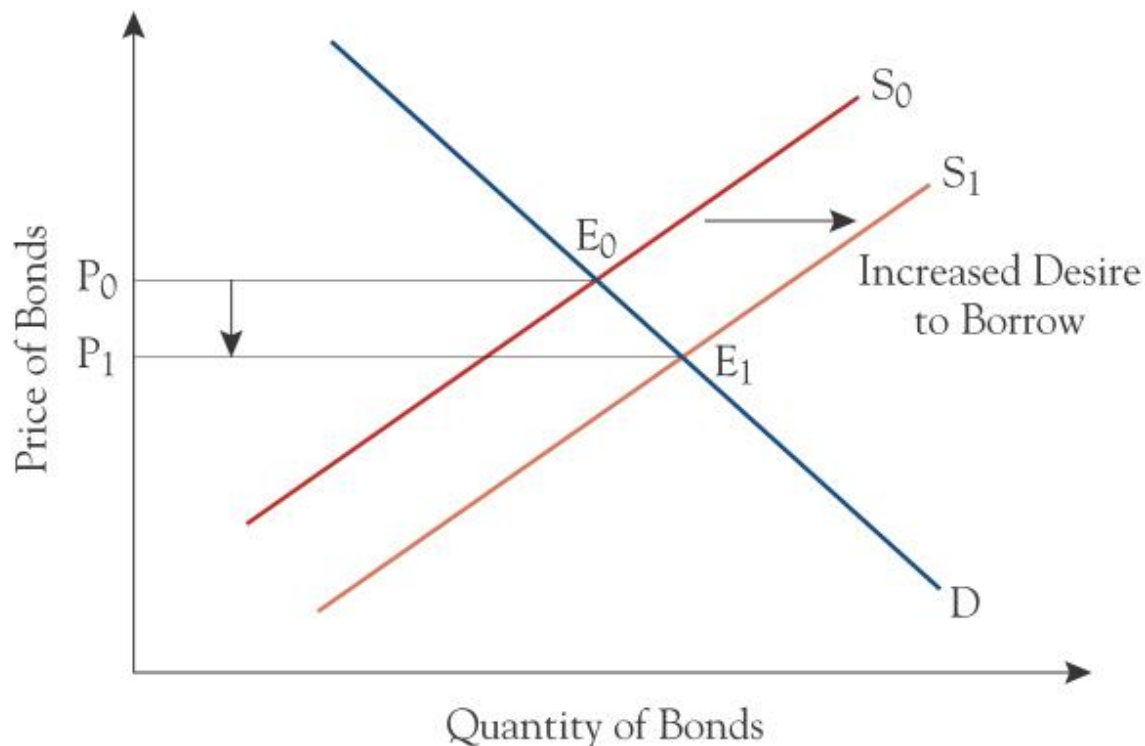
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Factors that shift Bond Supply

Figure 6.3 A Shift in the Supply of Bonds

When borrowers' desire for funds increases, the bond supply curve shifts to the right, lowering bond prices and raising interest rates.



Factors that shift Bond Supply

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Factors that shift Bond Demand

- wealth
 - An increases in wealth shift the demand for bonds to the right as wealthier people invest more.
 - This will happen as the economy grows during an expansion.
 - This will increase Bond Prices and lower yields.

Factors that shift Bond Demand

- Expected inflation
 - A fall in expected inflation shifts the bond demand curve to the right, increasing demand at each price and lowering the yield and increasing the Bond's price.

Factors that shift Bond Demand

- Expected return on stocks and other assets
 - If the return on bonds rises relative to the return on alternative investments, the demand for bonds will rise.
 - This will increase bond prices and lower yields.

Factors that shift Bond Demand

- Risk relative to alternatives
 - If a bond becomes less risky relative to alternative investments, the demand for the bond shifts to the right.

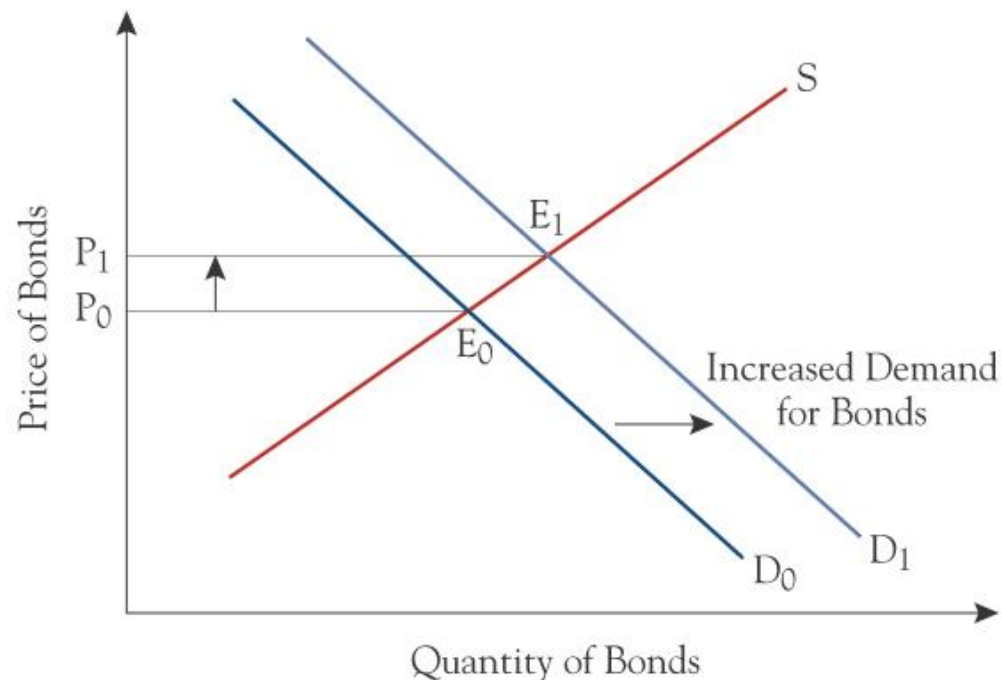
Factors that shift Bond Demand

- Liquidity of bonds relative to alternatives
 - When a bond becomes more liquid relative to alternatives, the demand curve shifts to the right.

Factors that shift Bond Demand

Figure 6.4 Shift in Bond Demand

When there is an increase in investors' willingness to hold bonds, the bond demand curve shifts to the right, increasing bond prices and reducing interest rates.



Factors that shift Bond Demand

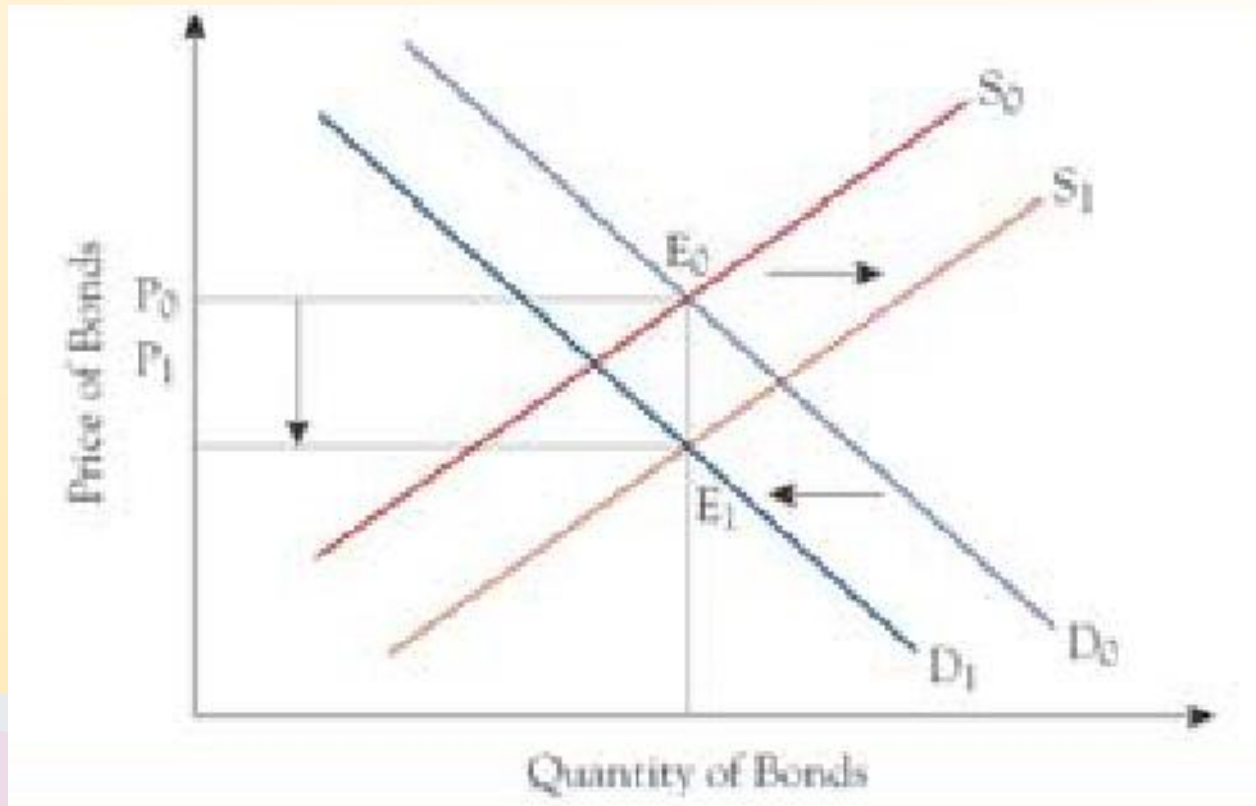
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An increase in the expected return on the bond relative to the expected return on alternatives makes bonds more attractive.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓
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A fall in the riskiness of the bond relative to the riskiness of alternatives makes bonds more attractive.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓
An increase in the liquidity of the bond relative to the liquidity of alternatives makes bonds more attractive.	Bond demand shifts to the right, bond prices ↑, and interest rates ↓

Equilibrium

- An increase in expected inflation shifts bond supply to the right and bond demand to the left.
- The two effects reinforce each other, resulting in a lower bond price and a higher interest rate

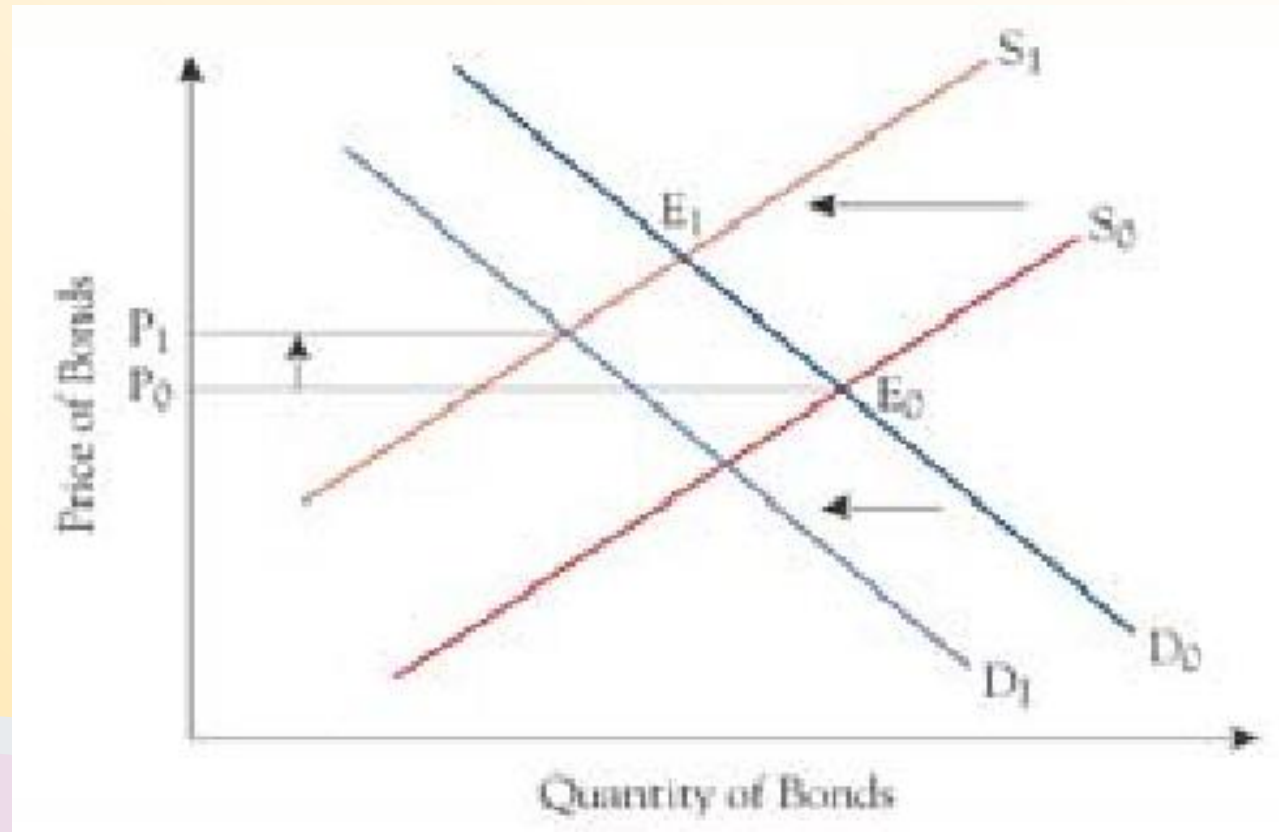
Equilibrium



Equilibrium

- A business-cycle downturn shifts the bond supply to the left and the bond demand to the left.
- In this case the bond price can rise or fall, depending on which shift is greater.
- But interest rates tend to fall in recessions, so bond prices are likely to increase

Equilibrium



Bonds and Risk

Sources of Bond Risk

- Default Risk
- Inflation Risk
- Interest-Rate Risk

Summary

- Factors Influencing Bond Supply
- Factors Influencing Bond Demand
- Equilibrium Conditions

Money and Banking

Lecture 16

Review of the Previous Lecture

- Factors Influencing Bond Supply
- Factors Influencing Bond Demand
- Equilibrium Conditions

Topics under Discussion

- Bonds and Risk
 - Default Risk
 - Inflation Risk
 - Interest Rate Risk
- Bond Ratings
- Bond Ratings and Risk
- Tax Effect

Bonds and Risk

Sources of Bond Risk

- Default Risk
- Inflation Risk
- Interest-Rate Risk

Bonds and Risk

- Default Risk
 - There is no guarantee that a bond issuer will make the promised payments
 - Investors who are risk averse require some compensation for bearing risk; the more risk, the more compensation they demand
 - The higher the default risk the higher the probability that bondholders will not receive the promised payments and thus, the higher the yield

Bonds and Risk

- Suppose risk-free rate is 5%
- ZEDEX Corp. issues one-year bond at 5%
- Price without risk = $(\$100 + \$5)/1.05 = \$100$
- Suppose there is 10% probability that ZEDEX Corp. goes bankrupt, get nothing
- Two possible payoffs: \$105 and \$0

Bonds and Risk

Table 6.5 Expected Value of ZEDEX Bond Payment

Possibilities	Payoff	Probability	Payoff × Probabilities
Full payment	\$105	0.90	\$94.50
Default	\$0	0.10	\$0
Expected Value = Sum of Payoffs times Probabilities = \$94.50			

- Expected PV of ZEDEX bond payment = $\$94.5 / 1.05 = \90
- If the promised payment is \$105, YTM will be $\$105 / 90 - 1 = 0.1667$ or 16.67%
- Default risk premium = 16.67% - 5% = 11.67%

Bonds and Risk

- Inflation Risk
 - Bonds promise to make fixed-dollar payments, and bondholders are concerned about the purchasing power of those payments
 - The nominal interest rate will be equal to the real interest rate plus the expected inflation rate plus the compensation for inflation risk
 - The greater the inflation risk, the larger will be the compensation for it

Bonds and Risk

- Assuming real interest rate is 3% with the following information

Table 6.6 Inflation Risk

Inflation	Probabilities		
	Case I	Case II	Case III
1 percent	0.50	0.25	0.10
2 percent	–	0.50	0.80
3 percent	0.50	0.25	0.10
Expected inflation	2 percent	2 percent	2 percent
Standard deviation	1.0 percent	0.71 percent	0.45 percent

- Nominal rate = 3% real rate + 2% expected inflation + compensation for inflation risk

Bonds and Risk

- Interest-Rate Risk
 - Interest-rate risk arises from the fact that investors don't know the holding period yield of a long-term bond.
 - If you have a short investment horizon and buy a long-term bond you will have to sell it before it matures, and so you must worry about what happens if interest rates change
 - Because the price of long-term bonds can change dramatically, this can be an important source of risk

Bond Ratings

- The risk of default (i.e., that a bond issuer will fail to make a bond's promised payments) is one of the most important risks a bondholder faces, and it varies among issuers.
- Credit rating agencies have come into existence to assess the default risk of different issuers

Bond Ratings

- The bond ratings are an assessment of the creditworthiness of the corporate issuer.
- The definitions of creditworthiness used by the rating agencies are based on how likely the issuer firm is to default and the protection creditors have in the event of a default.

Bond Ratings

- These ratings are concerned only with the possibility of the default. Since they do not address the issue of interest rate risk, the price of a highly rated bond may be quite volatile.

Bond Ratings

Long Term Ratings by PACRA

Investment Grades:

- **AAA: Highest credit quality.** 'AAA' ratings denote the lowest expectation of credit risk.
- **AA: Very high credit quality.** 'AA' ratings denote a very low expectation of credit risk.
- **A: High credit quality.** 'A' ratings denote a low expectation of credit risk.
- **BBB: Good credit quality.** 'BBB' ratings indicate that there is currently a low expectation of credit risk.

Bond Ratings

- Long Term Ratings by PACRA

Speculative Grades:

BB: Speculative. 'BB' ratings indicate that there is a possibility of credit risk developing,

B: Highly speculative. 'B' ratings indicate that significant credit risk is present, but a limited margin of safety remains.

CCC, CC, C: High default risk. Default is a real possibility.

Bond Ratings

- Short Term Ratings by PACRA
- **A1+:** highest capacity for timely repayment.
- A1:** strong capacity for timely repayment.
- A2:** satisfactory capacity for timely repayment, may be susceptible to adverse economic conditions.
- A3:** an adequate capacity for timely repayment. more susceptible to adverse economic conditions.

Bond Ratings

- Short Term Ratings by PACRA
- **B:** timely repayment is susceptible to adverse changes in business, economic, or financial conditions.
- **C:** an inadequate capacity to ensure timely repayment.
- **D:** high risk of default or which are currently in default.

Bond Ratings and Risk

Bond Ratings -

- Moody's and Standard and Poor's

Ratings Groups

- Investment Grade
- Non-Investment – Speculative Grade
- Highly Speculative

Bond Ratings and Risk

Commercial Paper Ratings

- Moody's and Standard and Poor's

Rating Groups

- Investment
- Speculative
- Default

Bond Ratings and Risk

Bond (credit) ratings

S&P	Moody's	What it means
AAA	Aaa	Highest quality and creditworthiness
AA	Aa	Slightly less likely to pay principal + interest
A	A	Strong capacity to make payments, upper medium grade
BBB	Baa	Medium grade, adequate capacity to make payments
BB	Ba	Moderate ability to pay, speculative element, vulnerable
B	B	Not desirable investment, long term payment doubtful
CCC	Caa	Poor standing, known vulnerabilities, doubtful payment
CC	Ca	Highly speculative, high default likelihood, known reasons
C	C	Lowest rated class, most unlikely to reach investment grade
D		Already defaulted on payments
NR		No public rating has been requested
+ or -	& 1,2,3	Within-class refinement of AA to CCC ratings

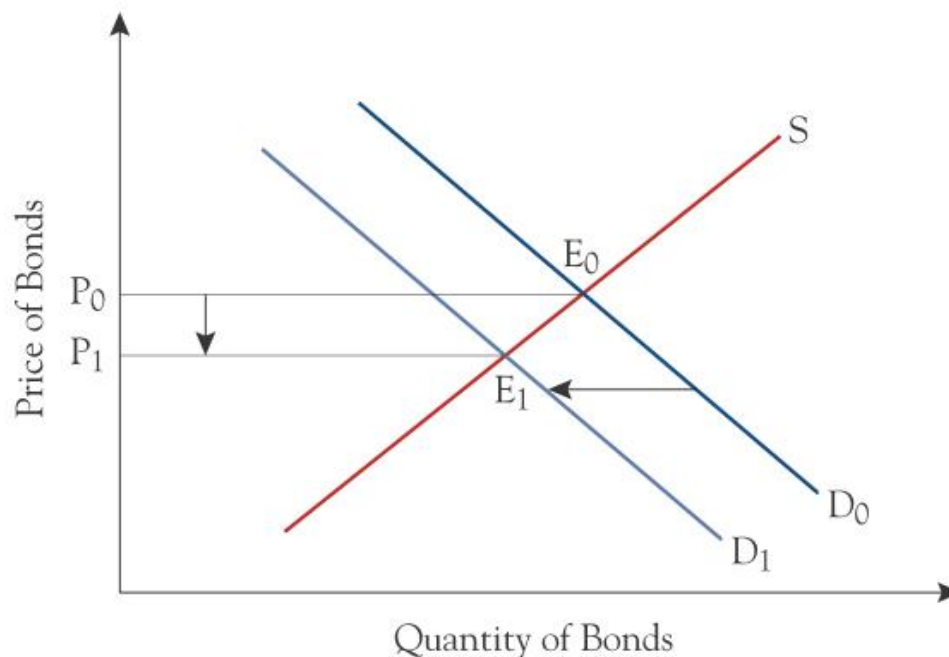
Bond Ratings and Risk

- The lower a bond's rating the lower its price and the higher its yield

Bond Ratings and Risk

Figure 7.1 The Effect of an Increase in Risk on Equilibrium in the Bond Market

Increased risk reduces the demand for the bond at every price, shifting the demand curve to the left from D_0 to D_1 . The result is a decline in the equilibrium price and quantity in the market. Importantly, the price falls from P_0 to P_1 , so the yield on the bond must rise.



Bond Ratings and Risk

- Increased Risk reduces Bond Demand.
- The resulting shift to the left causes a decline in equilibrium price and an increase in the bond yield.
- A bond yield can be thought of as the sum of two parts:
 - the yield on the Treasury bond (called “benchmark bonds” because they are close to being risk-free) and
 - a risk spread or default risk premium

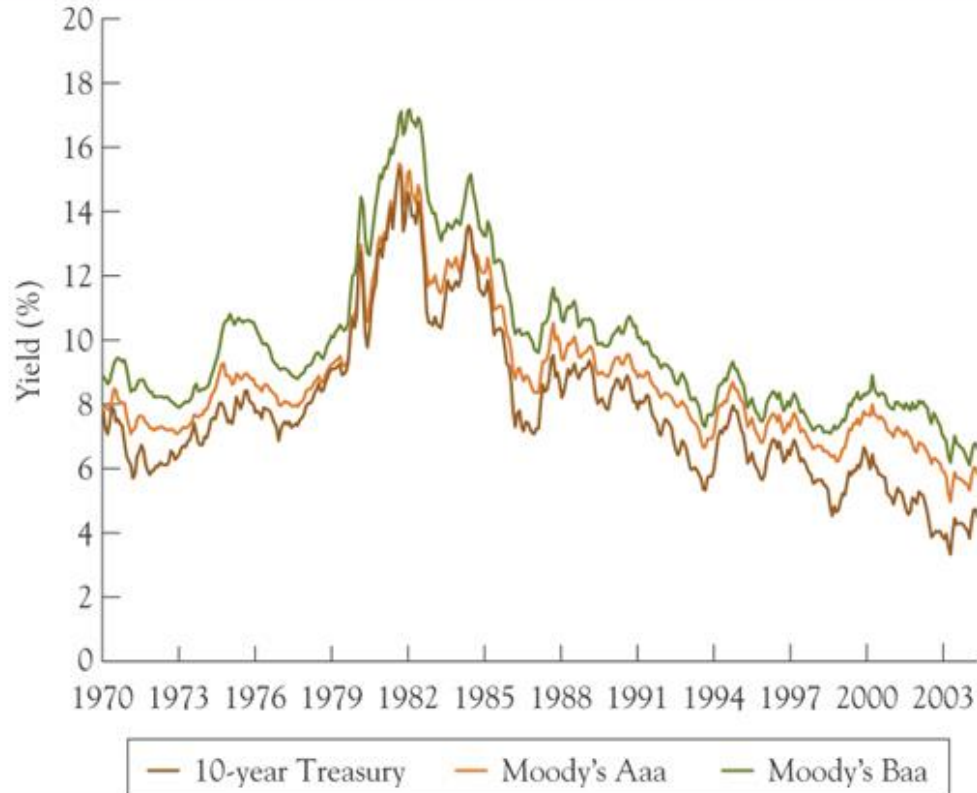
Bond Ratings and Risk

- If the bond ratings properly reflect the probability of default, then lower the rating of the issuer, the higher the default risk premium
- So we may conclude that when treasury bond yields change, all other yields will change in the same direction

Bond Ratings and Risk

Long-Term Bond Interest Rates and Ratings

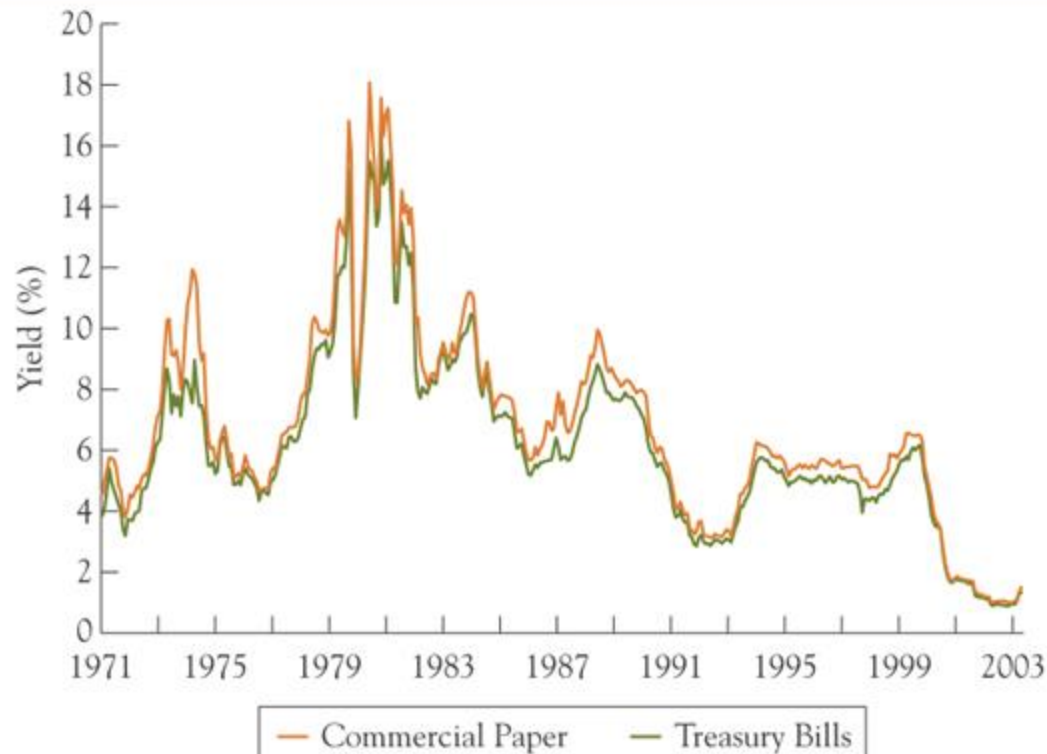
Figure 7.2 Comparing Long-Term Interest Rates



Bond Ratings and Risk

Short-Term Interest Rates and Risk

Figure 7.3 Comparing Short-Term Interest Rates

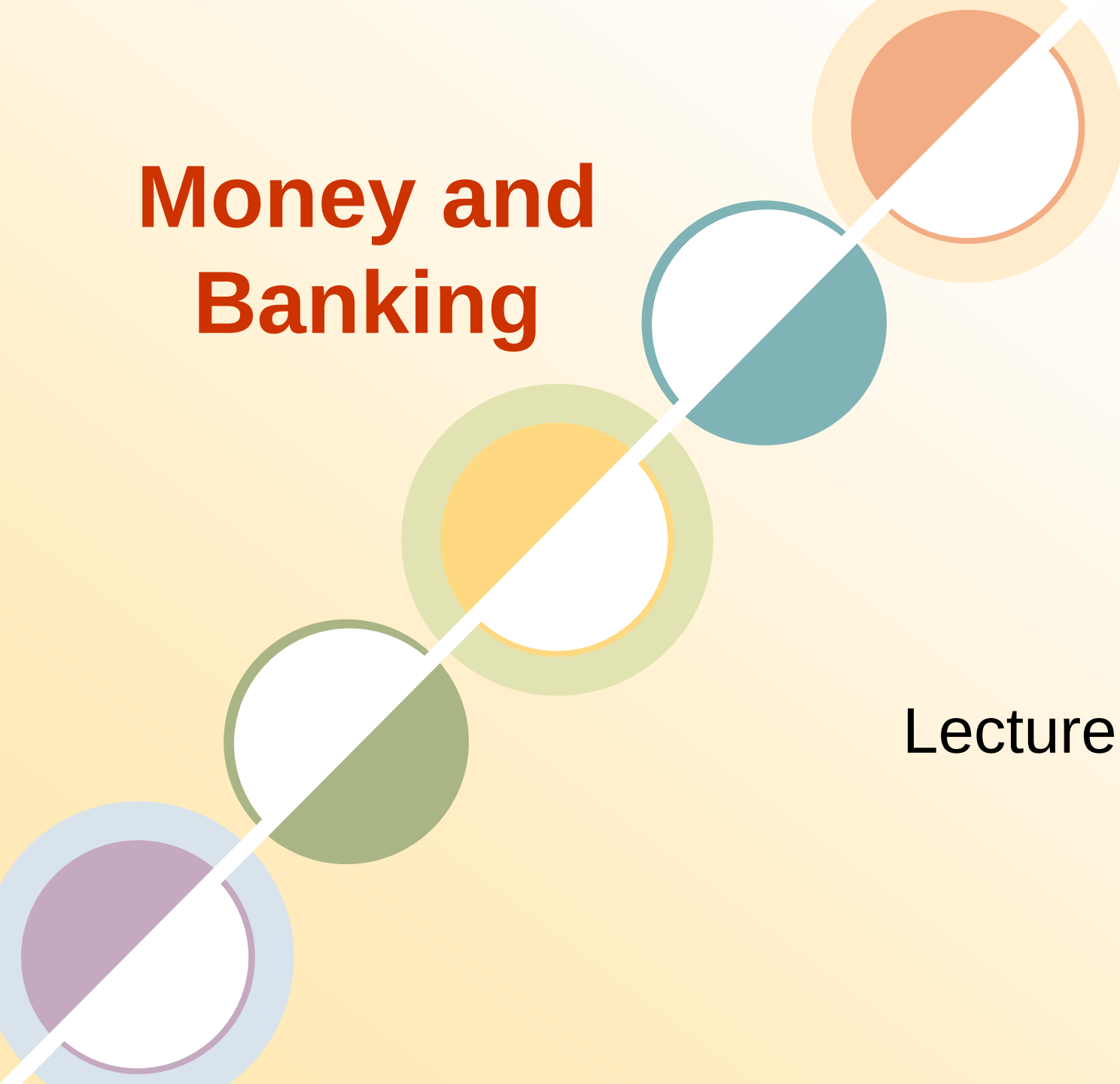


Summary

- Bonds and Risk
 - Default Risk
 - Inflation Risk
 - Interest Rate Risk
- Bond Ratings
- Bond Ratings and Risk

Money and Banking

Lecture 17



Review of the Previous Lecture

- Bonds and Risk
 - Default Risk
 - Inflation Risk
 - Interest Rate Risk
- Bond Ratings
- Bond Ratings and Risk

Topics under Discussion

- Tax Effect
- Term Structure of Interest Rate
 - Expectations Hypothesis
 - Liquidity Premium

Tax Effect

- The second important factor that affects the return on a bond is taxes
- Bondholders must pay income tax on the interest income they receive from privately issued bonds (taxable bonds), but government bonds are treated differently
- Interest payments on bonds issued by state and local governments, called “municipal” or “tax-exempt” bonds are specifically exempt from taxation

Tax Effect

- A tax exemption affects a bond's yield because it affects how much of the return the bondholder gets to keep
- Tax-Exempt Bond Yield
= (Taxable Bond Yield) x (1- Tax Rate).

Term Structure of Interest Rates

The relationship among bonds with the same risk characteristics but different maturities is called the term structure of interest rates.

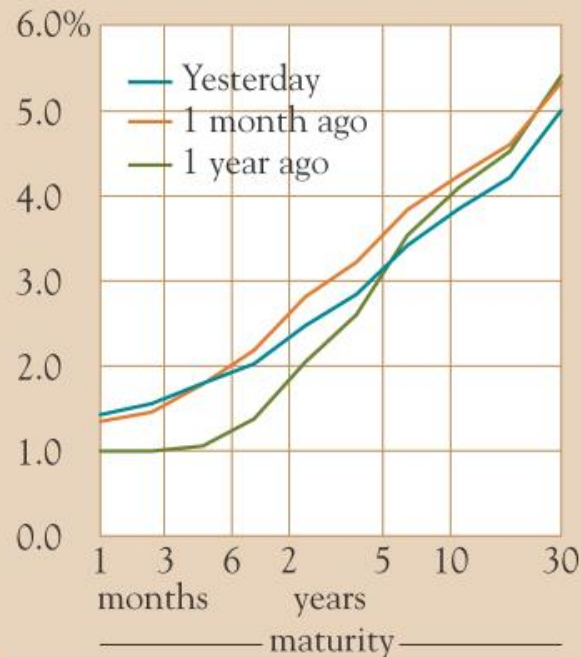
A plot of the term structure, with the yield to maturity on the vertical axis and the time to maturity on the horizontal axis, is called the yield curve.

Term Structure of Interest Rates

Figure 7.4 The U.S. Treasury Yield Curve

The figure plots the yields on Treasury bills and bonds for August 27, 2004.

Yield to maturity of current bills, notes, and bonds.



Term Structure of Interest Rates

Term Structure of Treasury Interest Rates

Figure 7.5 The Term Structure of Treasury Interest Rates



Term Structure of Interest Rates

Term Structure “Facts”

- Interest Rates of different maturities tend to move together
- Yields on short-term bond are more volatile than yields on long-term bonds
- Long-term yields tend to be higher than short-term yields.

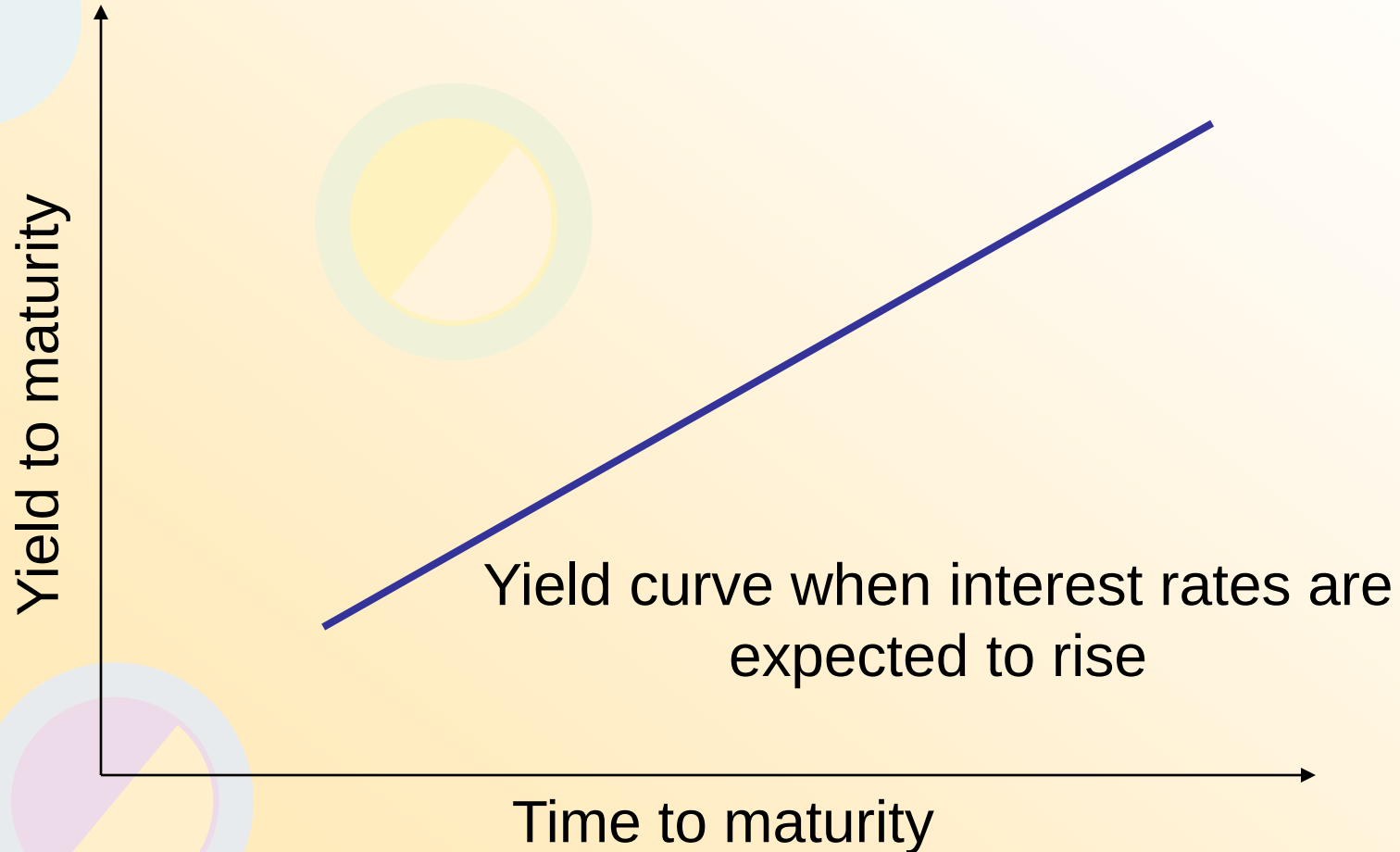
Expectations Hypothesis

- The risk-free interest rate can be computed, assuming that there is no uncertainty about the future
- Since certainty means that bonds of different maturities are perfect substitutes for each other, an investor would be indifferent between holding
 - a two-year bond or
 - a series of two one-year bonds
- Certainty means that bonds of different maturities are perfect substitutes for each other

Expectations Hypothesis

- Assuming that current 1-year interest rate is 5%. The expectations hypothesis implies that the current 2-year interest rate should equal the average of 5% and 1-year interest rate one year in future.
 - If future interest rate is 7%, then current 2-year interest rate will be $(5+7) / 2 = 6\%$
- Therefore, when interest rates are expected to rise long-term rates will be higher than short-term rates and the yield curve will slope up (and vice versa)

Expectations Hypothesis



Expectations Hypothesis

- From this we can construct investment strategies that must have the same yield.
- Assuming the investor has a two-year horizon, the investor can:
 - invest in a two-year bond and hold it to maturity
 - Interest rate will be i_{2y}
 - Investment will yield $(1 + i_{2y}) (1 + i_{2y})$ two years later
 - invest in a one-year bond today and a second one a year from now when the first one matures
 - Interest rate will be i_{y+1}^e
 - Investment will yield $(1 + i_{1y}) (1 + i_{y+1}^e)$ in two years

Expectations Hypothesis

- The hypothesis tells us that investors will be indifferent between the two strategies, so the strategies must have the same return

Total return from 2 year bonds over 2 years

$$(1+i_{2y})(1+i_{2y})$$

Return from one year bond and then another one year bond

$$(1+i_{1y})(1+i_{1y}^e)$$

Expectations Hypothesis

If one and two year bonds are perfect substitutes, then:

$$(1+i_{2y})(1+i_{2y}) = (1+i_{1y})(1+i_{1y}^e)$$

Or

$$i_{2y} = \frac{i_{1y} + i_{1y}^2}{2}$$

Or in general terms

$$i_{nt} = \frac{i_{1t} + i_{1t+1}^e + i_{1t+2}^e + \dots + i_{1t+n-1}^e}{n}$$

Expectations Hypothesis

- Therefore the rate on the two-year bond must be the average of the current one-year rate and the expected future one-year rate
- Implications would be the same old
 - a. Interest rates of different maturities tend to move together.
 - b. Yields on short-term bonds are more volatile than those on long-term bonds.
 - c. Long-term yields tend to be higher than short-term yields

Expectations Hypothesis

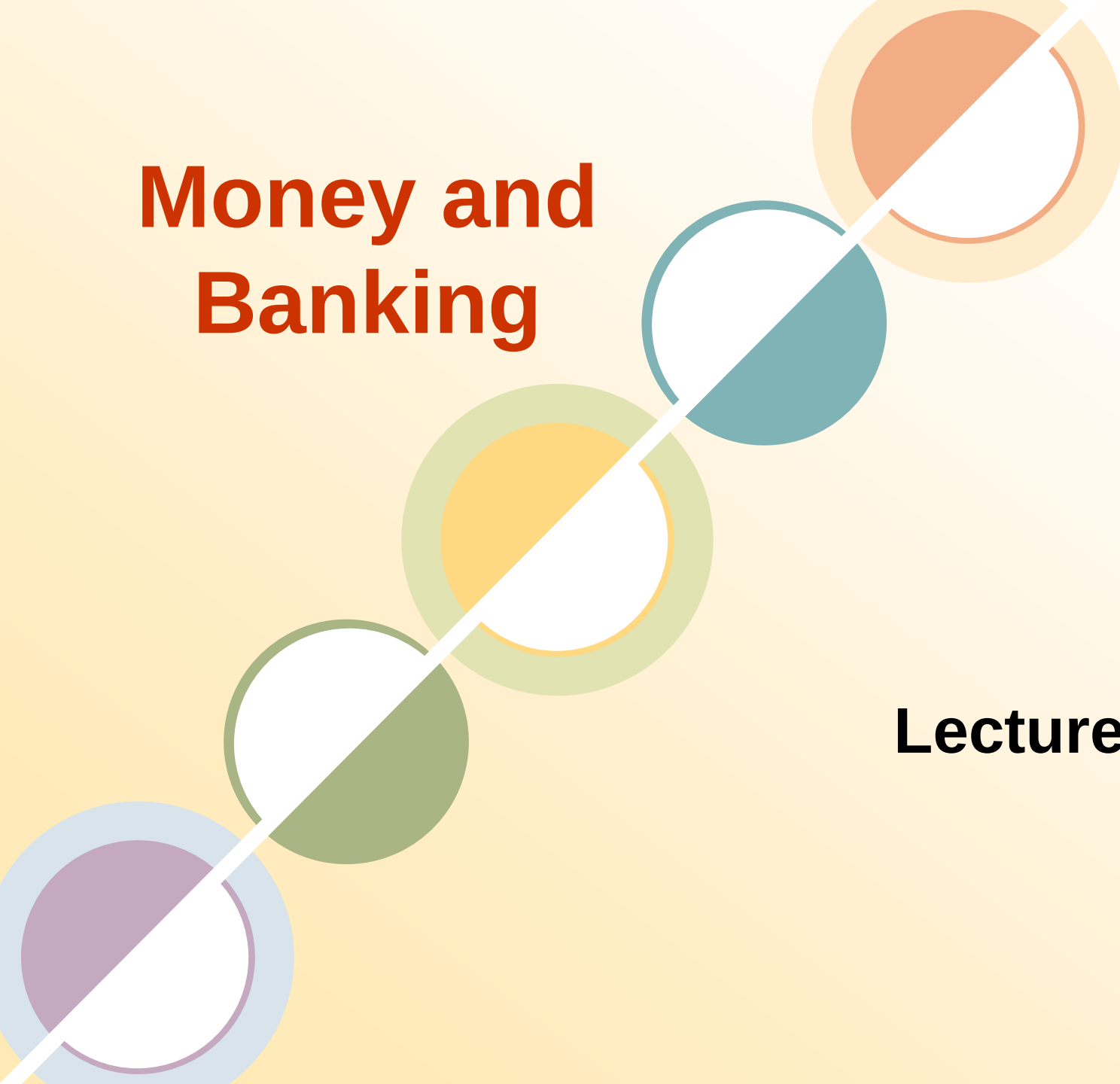
- However, expectations theory can not explain why long-term rates are usually above short term rates
- In order to explain why the yield curve normally slopes upward, we need to extend the hypothesis to include risk

Summary

- Bonds
 - Tax Effect
 - Term Structure of Interest Rate
 - Expectations Hypothesis
 - Liquidity Premium

Money and Banking

Lecture 18



Review of the Previous Lecture

- Bond
 - Tax Effect
 - Term Structure of Interest Rate
 - Expectations Hypothesis

Topics under Discussion

- Bonds
 - Liquidity Premium Theory
- Stocks
 - Essential Characteristics
 - Process
- Measuring Level of a Stock Market
- Valuing Stocks

The Liquidity Premium Theory

- Risk is the key to understanding the slope of the yield curve
- The yield curve's upward slope is due to long-term bonds being riskier than short-term bonds
- Bondholders face both inflation and interest-rate. The longer the term the greater the inflation and interest-rate risk

The Liquidity Premium Theory

- a. Inflation risk increases over time because investors, who care about the real return, must forecast inflation over longer periods.
- b. Interest-rate risk arises when an investor's horizon and the bond's maturity do not match. If holders of long-term bonds need to sell them before maturity and interest rates have increased, the bonds will lose value

The Liquidity Premium Theory

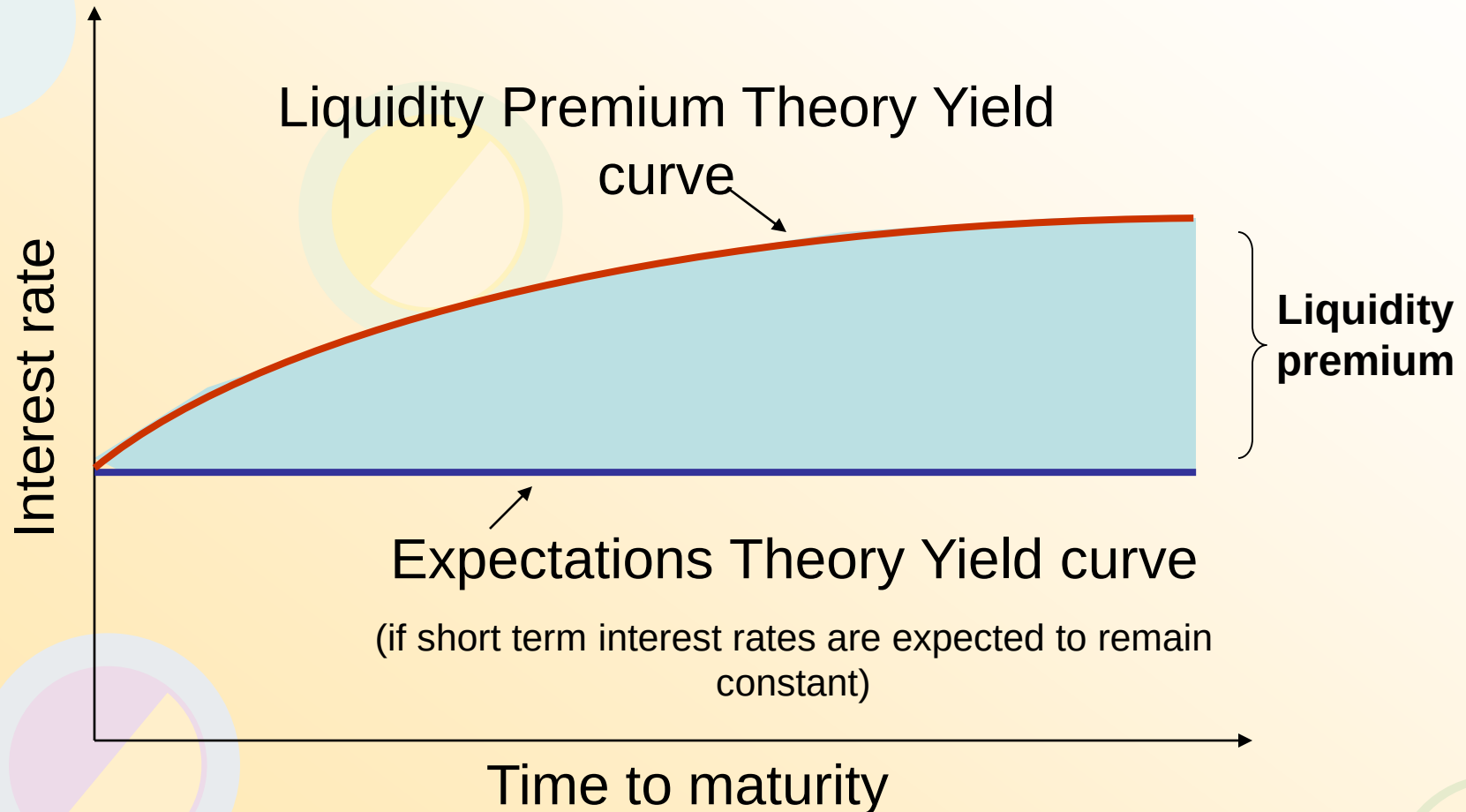
- Including risk in the model means that we can think of yield as having two parts:
 - risk-free and
 - risk premium

$$i_{nt} = rp_n + \frac{i_{1t} + i_{1t+1}^e + i_{1t+2}^e + \dots + i_{1t+n-1}^e}{n}$$

Risk premium

Pure expectations theory

Relationship Between the Liquidity Premium and Expectations Theories



The Liquidity Premium Theory

- Again, we arrive at the same three conclusions about the term structure of interest rates
 - a. Interest rates of different maturities tend to move together.
 - b. Yields on short-term bonds are more volatile than those on long-term bonds.
 - c. Long-term yields tend to be higher than short-term yields

Stocks: An Introduction

- Stocks provide a key instrument for holding personal wealth as well as a way to diversify, spreading and reducing the risks that we face
- For companies, they are one of several ways to obtain financing.

Stocks: An Introduction

- Additionally,
 - stocks and **stock markets** are one of the central links between the financial world and the real economy.
 - Stock prices are fundamental to the functioning of a market-based economy
 - They indicate the value of the companies that issued the stocks and,
 - They allocate scarce investment resources

Stocks: An Introduction

- The firms deemed most valuable in the marketplace for stocks are the ones that will be able to obtain financing for growth. When resources flow to their most valued uses, the economy operates more efficiently

Stocks: An Introduction

- Most people see stock market as a place where fortunes are easily made or lost, and they recoil at its unfathomable booms and busts.
 - Great American Depression (1929)
 - Post-September 11, 2001 scenario
 - Pakistan stock market on roller-coaster-ride (March 2005)

Stocks: An Introduction

- What happens in reality?
 - stock prices tend to rise steadily and slowly, and
 - collapse rarely when normal market mechanisms are out of alignment
- For most people the experience of losing or gaining wealth *suddenly* is more memorable than the experience of making it *gradually*.
- By being preoccupied with the potential *short-term* losses associated with crashes, we lose sight of the gains we could realize if we took a *longer-term view*

Essential Characteristics of Common Stock

- Stocks, also known as common stock or equity, are shares in a firm's ownership
- From their early days, stocks had two important characteristics that today are taken for granted:
 - the shares are issued in small denominations and
 - the shares are transferable
- Until recently, stockowners received a certificate from the issuing company, but now it is a computerized process where the shares are registered in the names of brokerage firms that hold them on the owner's behalf

Essential Characteristics of Common Stock

- The ownership of common stock conveys a number of rights
 - A stockholder is entitled to participate in the shares of the enterprise, but this is a residual claim i.e. meaning the leftovers after all other creditors have been paid.
 - Stockholders also have limited liability,
 - Even if a company fails, the maximum amount that the stockholder can lose is the initial investment

Essential Characteristics of Common Stock

- Stockholders are entitled to vote at the firm's annual meeting including voting to elect (or remove) the firm's board of directors

Essential Characteristics of Common Stock

- Following are some salient features of stock trading
 - a. an individual share represents only a small fraction of the value of the company that issued it
 - b. a large number of shares are outstanding
 - c. prices of individual shares are low, allowing individuals to make relatively small investments

Essential Characteristics of Common Stock

- d. as residual claimants, stockholders receive the proceeds of a firm's activities only after all other creditors have been paid
- e. because of limited liability, investor's losses cannot exceed the price they paid for the stock; and
- f. shareholders can replace managers who are doing a bad job

Measuring the Level of the Stock Market

- Stocks are one way in which we choose to hold our wealth, so when stock values rise we get richer and when they fall we get poorer
- These changes affect our consumption and saving patterns, causing general economic activity to fluctuate
- We need to understand the dynamics of the stock market, in order to
 - manage our personal finances and
 - see the connections between stock values and economic conditions

Measuring the Level of the Stock Market

- Stock market indexes
 - Designed to give us a sense of the extent to which stock prices are going up or down
 - Tell us both how much the value of an average stock has changed, and how much total wealth has gone up or down
 - Provide benchmarks for performance of money managers, comparing how they have done to the market as a whole

Measuring the Level of the Stock Market

- Every major country in the world has a stock market, and each of these markets has an index
- For the most part, these are value-weighted indices
- To analyze the performance of these different markets it is useful to look at percentage changes, but percentage change isn't everything

Measuring the Level of the Stock Market

- The Dow Jones Industrial Average
- The Standard & Poor's 500 Index
- Nasdaq Composite index
- Financial Times Stock Exchange 100 Index
- Hang Seng 100
- Nikkei 225
- KSE 100 Index

Measuring the Level of the Stock Market

- The KSE100

- It contains a representative sample of common stock that trade on the Karachi Stock Exchange.
- The KSE stocks that comprise the index have a total market value of around Rs. 1,197 Billion compared to total market value of Rs. 1,365 Billion for over 679 stocks listed on the Karachi Stock Exchange.
- This means that the KSE100 Index represents 88 percent of the total market capitalization of the Karachi Stock Exchange, as of February, 2004

Table 8.2 World Stock Markets

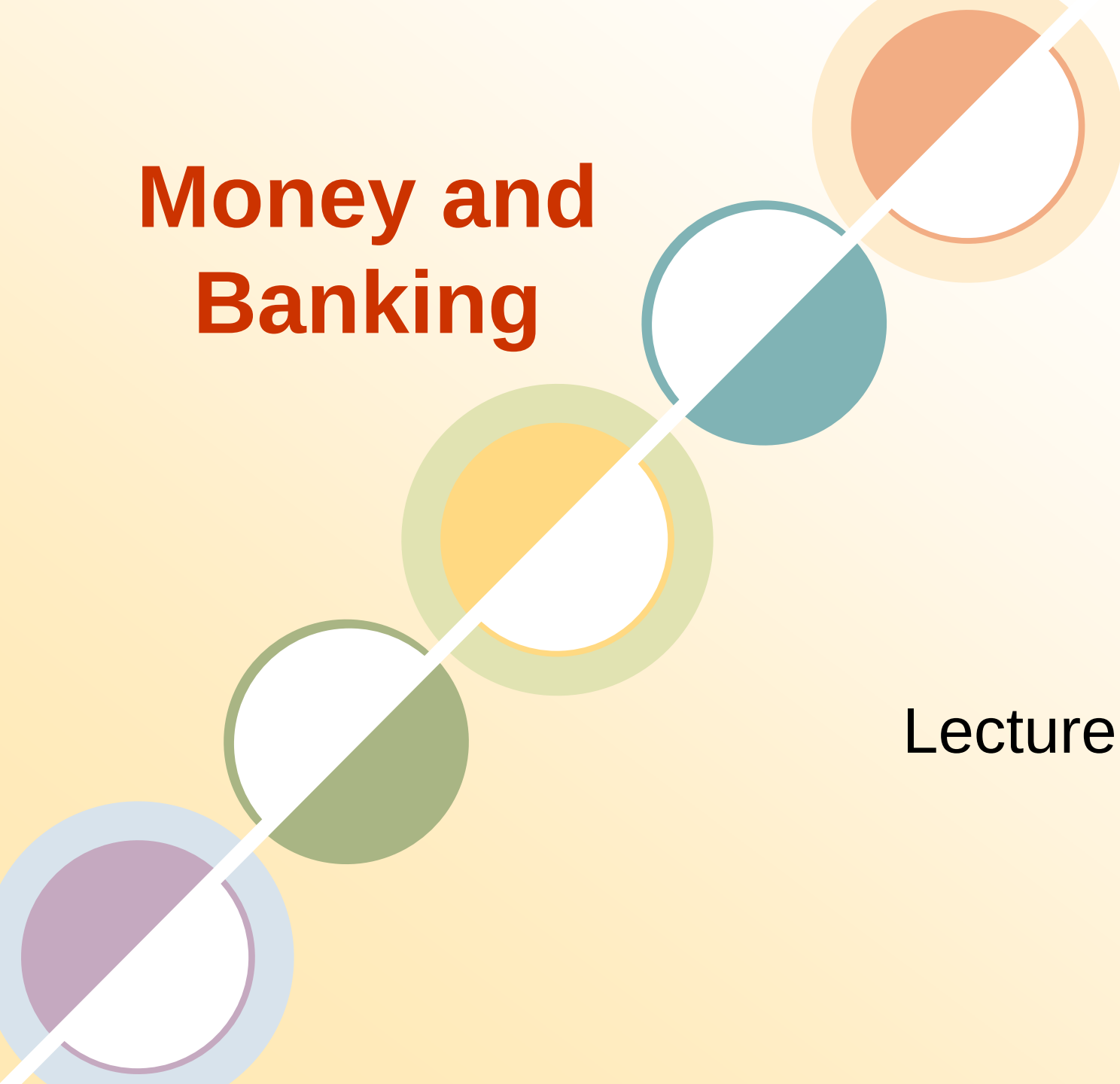
Country	Index Name	Index Level 9/2001	Change from 2001 Peak	Change over the year
Argentina	General	11,213.7	-7,250.4	-39%
Australia	All Ordinaries	2,941.6	-483.6	-14%
Brazil	Bovespa	10,404.7	-7,484.4	-42%
Britain	FTSE 100	4,763.7	-1,570.8	-25%
Canada	Toronto 300 Comp.	6,669.4	-2,679.0	-29%
France	CAC 40	4,013.2	-1,985.3	-33%
Germany	DAX	4,184.5	-2,610.5	-38%
Hong Kong	Hang Seng	9,600.8	-6,563.2	-41%
Italy	MibTel General	19,877.0	-11,271.0	-36%
Japan	Nikkei 225	9,696.5	-4,833.0	-33%
Mexico	IPC All-Shares	5,306.1	-1,662.8	-24%
Singapore	Straits Times	1,310.9	-480.4	-27%
United States	S&P 500	1,018.6	-355.1	-26%

Summary

- Bonds
 - Liquidity Premium Theory
- Stocks
 - Essential Characteristics
 - Process
- Measuring Level of a Stock Market

Money and Banking

Lecture 19



Review of the Previous Lecture

- Bonds and Their Characteristics
- Stocks
 - Essential Characteristics
 - Process
- Measuring Level of a Stock Market

Topics under Discussion

- Valuing Stocks
 - Fundamental Value and Dividend Discount Model
 - Risk and Value of Stocks

Valuing Stocks

- People differ in their opinions of how stocks should be valued
 - Chartists believe that they can predict changes in a stock's price by looking at patterns in its past price movements
 - Behavioralists estimate the value of stocks based on their perceptions of investor psychology and behavior

Valuing Stocks

- Others estimate stock values based on a detailed study of the fundamentals, which can be analyzed by examining the firm's financial statements.
 - In this view the value of a firm's stock depends both on its current assets and estimates of its future profitability

Valuing Stocks

- The fundamental value of stocks can be found by using the present value formula to assess how much the promised payments are worth, and then adjusting to allow for risk
- Chartists and Behavioralists focus instead on estimates of the deviation of stock prices from those fundamental values

Fundamental Value and the Dividend-Discount Model

- As with all financial instruments, a stock represents a promise to make monetary payments on future dates, under certain circumstances
- With stocks the payments are in the form of dividends, or distributions of the firm's profits
- The price of a stock today is equal to the present value of the payments the investor will receive from holding the stock

Fundamental Value and the Dividend-Discount Model

- This is equal to
 - the selling price of the stock in one year's time plus
 - the dividend payment received in the interim
- Thus the current price is the present value of next year's price plus the dividend

Fundamental Value and the Dividend-Discount Model

- If P_{today} is the purchase price of stock, $P_{\text{next year}}$ is the sales price one year later and $D_{\text{next year}}$ is the size of the dividend payment, we can say:

$$P_{\text{today}} = \frac{D_{\text{next year}}}{(1 + i)} + \frac{P_{\text{Next year}}}{(1 + i)}$$

Fundamental Value and the Dividend-Discount Model

- What if investor plans to hold stock for two years?

$$P_{today} = \frac{D_{next\ year}}{(1 + i)} + \frac{D_{In\ two\ years}}{(1 + i)^2} + \frac{P_{In\ two\ years}}{(1 + i)^2}$$

Fundamental Value and the Dividend-Discount Model

- Generalizing for n years:

$$P_{today} = \frac{D_{next_year}}{(1+i)} + \frac{D_{in_two_years}}{(1+i)^2} + \dots + \frac{D_{n_years_from_now}}{(1+i)^n} + \frac{P_{n_years_from_now}}{(1+i)^n}$$

Fundamental Value and the Dividend-Discount Model

- If a stock does not pay dividends the calculation can still be performed; a value of zero is used for the dividend payments

Fundamental Value and the Dividend-Discount Model

- Future dividend payments can be estimated assuming that current dividends will grow at a constant rate of g per year.

$$D_{next_year} = D_{today} (1 + g)$$

Fundamental Value and the Dividend-Discount Model

- For multiple periods:

$$D_{n \text{ years from now}} = D_{\text{today}} (1 + g)^n$$

Fundamental Value and the Dividend-Discount Model

- Price equation can now be re-written as:

$$P_{today} = \frac{D_{today} (1+g)}{(1+i)} + \frac{D_{today} (1+g)^2}{(1+i)^2} + \dots + \frac{D_{today} (1+g)^n}{(1+i)^n} + \frac{P_{n_years_from_now}}{(1+i)^n}$$

Fundamental Value and the Dividend-Discount Model

- Assuming that the firm pays dividends forever solves the problem of knowing the selling price of the stock; the assumption allows us to treat the stock as we did a consol

$$P_{today} = \frac{D_{today}}{i - g}$$

- This relationship is the dividend discount model

Fundamental Value and the Dividend-Discount Model

- The model tells us that stock price should be high when
 - dividends are high
 - Dividend growth is rapid, or
 - Interest rate is low

Why stocks are risky?

- Stockholders receive profits only after the firm has paid everyone else, including bondholders
- It is as if the stockholders bought the firm by putting up some of their own wealth and borrowing the rest
- This borrowing creates leverage, and leverage creates risk

Why stocks are risky?

- Imagine a software business that needs only one computer costing \$1,000 and purchase can be financed by any combination of stocks (equity) and bonds (debt). Interest rate on bonds is 10%. Company earns \$160 in good years and \$80 in bad years with equal probability

Why stocks are risky?

Returns distributed to debt and equity holders under different financing assumptions

Percent Equity (%)	Percent Debt (%)	Required payments on 10% bonds	Payment to equity holders	Equity Return (%)	Expected Equity Return (%)	St. Dev. of Equity Return
100%	0	0	\$80-160	8-16%	12%	4%
50%	50%	\$50	\$30-110	6-22%	14%	8%
30%	70%	\$70	\$10-90	3.3-30%	16.67%	13.3%
20%	80%	\$80	\$0-80	0-40%	20%	20%

Why stocks are risky?

- If the firm were only 10% equity financed, shareholders' liability could come into play.
 - Issuing \$900 worth of bonds means \$90 for interest payments.
 - If the business turned out to be bad, the \$80 revenue would not be enough to pay the interest
 - Without their limited liability, stockholders will be liable for \$10 shortfall. But actually, they will lose only \$100 investment and not more and the firm goes bankrupt.

Why stocks are risky?

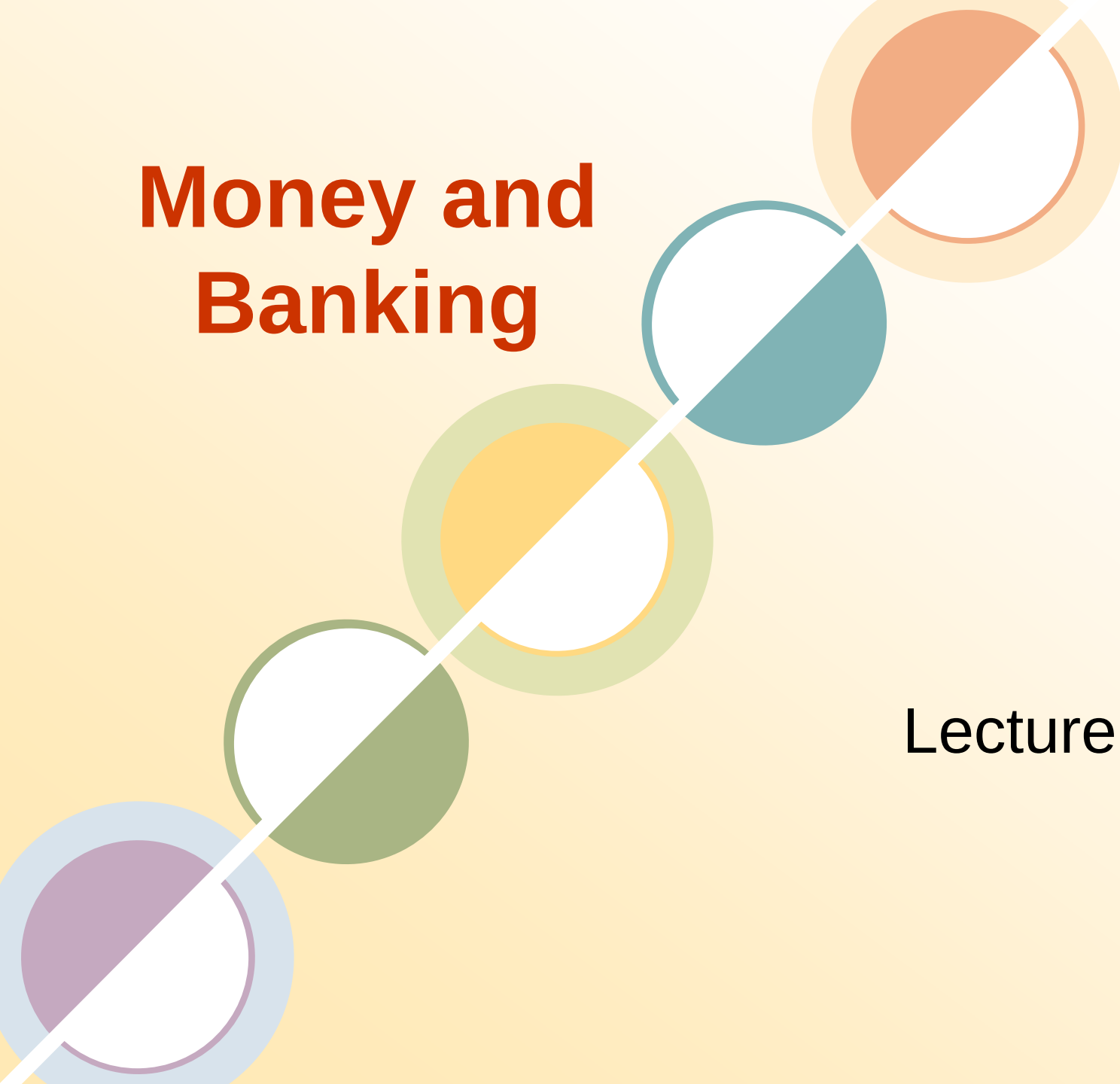
- Stocks are risky because the shareholders are residual claimants. Since they are paid last, they never know for sure how much their return will be.
- Any variation in the firm's revenue flows through to stockholders dollar for dollar, making their returns highly volatile

Summary

- Valuing Stocks
 - Fundamental Value and Dividend Discount Model
 - Why Stocks are risky?

Money and Banking

Lecture 20



Review of the Previous Lecture

- Stocks
- Valuing Stocks
 - Dividend Discount Model
 - Why Stocks are Risky?

Topics under Discussion

- Stocks
 - Risk and the Value of Stocks
 - Theory of Efficient Markets
 - Investing in Stocks for Long Run
 - Stock Markets' Role in the Economy
- Financial Intermediation
 - Role of Financial Intermediaries

Risk and value of stocks

- The dividend-discount model must be adjusted to include compensation for a stock's risk

Return to Holding Stock for One Year =

$$\frac{D_{next_year}}{P_{today}} + \frac{P_{next_year} - P_{today}}{P_{today}}$$

Risk and value of stocks

- Since the ultimate future sale price is unknown the stock is risky,
- The investor will require compensation in the form of a risk premium
- Required Stock Return (i) =
Risk-free Return (r_f) + Risk Premium (r_p)
- The risk-free rate can be thought of as the interest rate on a treasury security with a maturity of several months

Risk and value of stocks

- Our dividend discount model becomes:

$$P_{today} = \frac{D_{today}}{rf + rp - g}$$

Risk and value of stocks

- Stock Prices are High when
 - Current dividends are high (D_{today} is high)
 - Dividends are expected to grow quickly (g is high)
 - The risk-free rate is low (r_f is low)
 - The risk premium on equity is low (r_p is low)

Risk and value of stocks

- The S&P 500 index finished the year 2003 at just over 1,100. was this level warranted by fundamentals?
 - Risk free real interest rate is about 2% or $r_f = 0.02$
 - Risk premium is assumed to be 4% or $r_p = 0.04$
 - Dividend growth rate is around 2% or $g = 0.02$
 - The owner of a \$1,000 portfolio would have received \$30 in dividends during 2003

Risk and value of stocks

- Substituting the information in our adjusted dividend discount model:

$$P_{today} = \frac{\$30}{0.02+0.04-0.02} = \$750$$

- But the actual stock prices were substantially higher than this calculated figure
- This may be due to wrong assumption on risk premium. The investors may have been demanding lower risk premium in 2003.
- To compute it, we use the same equation

Risk and value of stocks

$$1,100 = \frac{\$30}{(0.02 + r_p) - 0.02}$$

- The answer is approximately 2.75%

The Theory of Efficient Markets

- The basis for the *theory of efficient markets* is the notion that the prices of all financial instruments, including stocks, reflect all available information
- As a result, markets adjust immediately and continuously to changes in fundamental values
- When markets are efficient, the prices at which stocks currently trade reflect all available information, so that future price movements are unpredictable.

The Theory of Efficient Markets

- If the theory is correct then no one can consistently beat the market average; active portfolio management will not yield a return that is higher than that of a broad stock-market index
- If managers claim to exceed the market average year after year, it may be because
 - They must be taking on risk,
 - They are lucky,
 - They have private information (which is illegal), or
 - Markets are not efficient

Investing in Stocks For the Long Run

- Stocks appear to be risky, and yet many people hold substantial proportions of their wealth in the form of stock
- This is due to the difference between the short term and the long term;
 - Investing in stocks is risky only if you hold them for a short time
 - In fact, when held for the long term, stocks are less risky than bonds.

Investing in Stocks For the Long Run

Figure 8.3 S&P 1-Year Stock Returns, 1871 to 2003

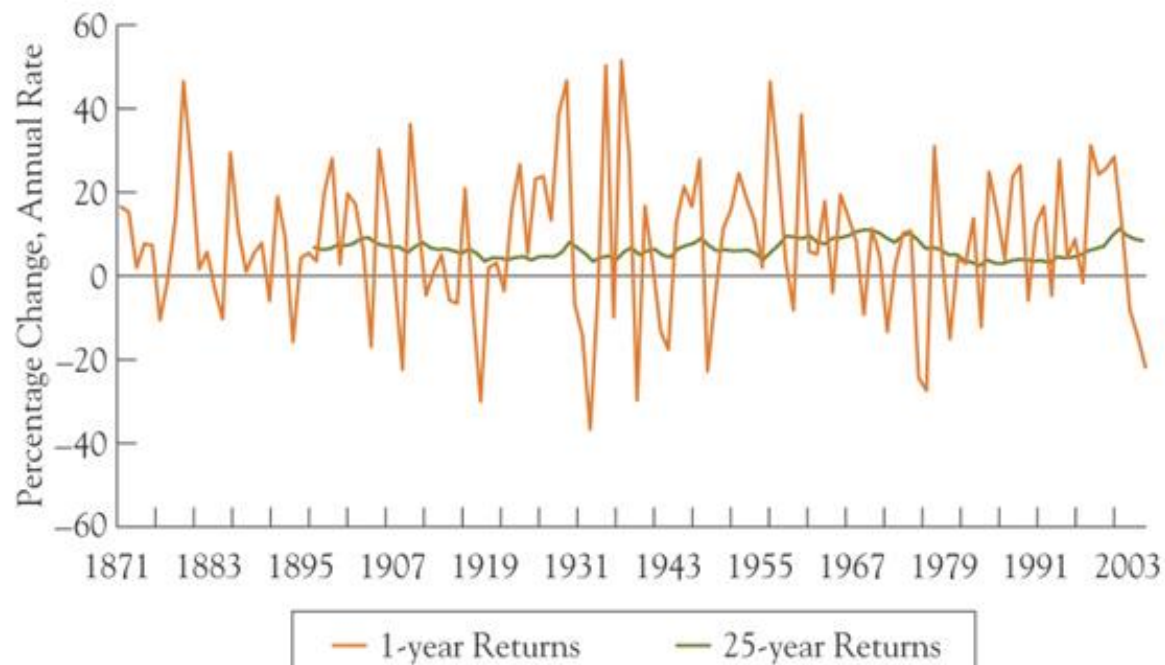
(Returns Are Real, Adjusted for Inflation Using the CPI)



Investing in Stocks For the Long Run

Figure 8.4 S&P Long-Run Stock Returns, 1871 to 2003

(Returns Are Real, Adjusted for Inflation Using the CPI)



The Stock Market's Role in the Economy

- The stock market plays a crucial role in every modern capitalist economy.
- The prices determined there tell us the market value of companies, which determines the allocation of resources.
 - Firms with a high stock market value are the ones investors prize, so they have an easier time garnering the resources they need to grow.
 - In contrast, firms whose stock value is low have difficulty financing their operations

The Stock Market's Role in the Economy

- So long as stock prices accurately reflect fundamental values, this resource allocation mechanism works well.
- At times, however, stock prices deviate significantly from the fundamentals and prices move in ways that are difficult to attribute to changes in the real interest rate, the risk premium, or the growth rate of future dividends.

The Stock Market's Role in the Economy

- Shifts in investor psychology may distort prices; both euphoria and depression are contagious
- When investors become unjustifiably exuberant about the market's future prospects, prices rise regardless of the fundamentals, and such mass enthusiasm creates bubbles

The Stock Market's Role in the Economy

- Bubbles

- Bubbles are persistent and expanding gaps between actual stock prices and those warranted by the fundamentals.
- These bubbles inevitably burst, creating crashes.
- They affect all of us because they distort the economic decisions companies and consumers make

The Stock Market's Role in the Economy

- If bubbles result in real investment that is both excessive and inefficiently distributed, crashes do the opposite; the shift to excessive pessimism causes a collapse in investment and economic growth
- When bubbles grow large enough and result in crashes the stock market can destabilize the real economy

Financial Intermediation

- Economic well-being is essentially tied to the health of the financial intermediaries that make up the financial system.
- We know that financial intermediaries are the businesses whose assets and liabilities are primarily financial instruments.
 - Various sorts of banks, brokerage firms, investment companies, insurance companies, and pension funds all fall into this category.
 - These are the institutions that pool funds from people and firms who save and lend them to people and firms who need to borrow

How can we get resources from people that have them to people who need them?

A.

Where Do Funds
Come From?

Students & fast food
workers

Factory & office
workers

Doctors & lawyers

Sports & movie stars

Extremely wealthy
individuals

?

How Are Funds Used?

Buy a refrigerator

Pay for a vacation

Purchase a new car

Purchase a home

Buy an oil tanker

Build a new factory

We use financial institutions to channel resources from people that have them to people who need them and can use them most efficiently.

B.

Where Do Funds
Come From?

Students & fast food
workers

Factory & office
workers

Doctors & lawyers

Sports & movie stars

Extremely wealthy
individuals

Bank or Other
Financial
Intermediary

How Are Funds Used?

Buy a refrigerator

Pay for a vacation

Purchase a new car

Purchase a home

Buy an oil tanker

Build a new factory

Financial Intermediation

- Financial intermediaries funnel savers' surplus resources into home mortgages, business loans, and investments.
- They are involved in both
 - direct finance—in which borrowers sell securities directly to lenders in the financial markets
 - indirect finance—in which a third party stands between those who provide funds and those who use them.

Financial Intermediation

- Intermediaries investigate the financial condition of the individuals and firms who want financing to figure out which have the best investment opportunities.
 - As providers of indirect finance, banks want to make loans only to the highest-quality borrowers.
 - When they do their job correctly, financial intermediaries increase investment and economic growth at the same time that they reduce investment risk and economic volatility

Role of Financial Intermediaries

- As a general rule, indirect finance through financial intermediaries is much more important than direct finance through the stock and bond markets
- In virtually every country for which we have comprehensive data, credit extended by financial intermediaries is larger as a percentage of GDP than stocks and bonds combined

Role of Financial Intermediaries

Table 11.1 The Relative Importance of Direct and Indirect Finance

(Averages for 1980-1995)

Country	Direct Finance		Indirect Finance	
	Stock Market Capitalization as Percent of GDP (A)	Outstanding Domestic Debt Securities as Percent of GDP (B)	Credit Extended by Banks & Other Financial Institutions as Percent of GDP (C)	Ratio of Indirect to Direct Finance C/(A+B) (D)
Industrialized Countries				
France	19.8%	41.2%	90.9%	1.5
Germany	18.6	37.4	92.3	1.6
Greece	8.1	4.3	40.2	3.2
Italy	11.9	28.1	50.5	1.3
Japan	73.0	30.0	169.3	1.6
United Kingdom	76.3	14.4	74.4	0.8
United States	58.2	52.6	130.7	1.2
Emerging Markets Countries				
Argentina	4.8	5.9	15.0	1.4
Brazil	11.9	4.0	24.7	1.6
India	13.2	5.7	26.8	1.4

Role of Financial Intermediaries

- Around the world, firms and individuals draw their financing primarily from banks and other financial intermediaries
- The reason for this is information;

Role of Financial Intermediaries

- Just think of an online store
 - You can buy virtually EVERYTHING – from \$5 dinner plates to \$300,000 sports car
 - But you will notice an absence of financial products, like student loans, car loans, credit cards or home mortgages
 - You can not bonds on which issuer is still making payments, nor can you have the services of checking account.

Role of Financial Intermediaries

- Why such online store does not deal in mortgages?

- Suppose a company needs a mortgage of \$100,000 and the store can (if at all) establish a system in which 100 people sign up to lend \$1,000 each to the company
- But the store has to do more
 - Collecting the payments
 - Figuring out how to repay the lenders
 - Writing legal contracts
 - Evaluating the creditworthiness of the company and feasibility of the mortgaged project
- Can it do it all?

Role of Financial Intermediaries

- financial intermediaries exist so that individual lenders don't have to worry about getting answers to all of the important questions concerning a loan and a borrower
- Lending and borrowing involve transactions costs and information costs, and financial intermediaries exist to reduce these costs

Role of Financial Intermediaries

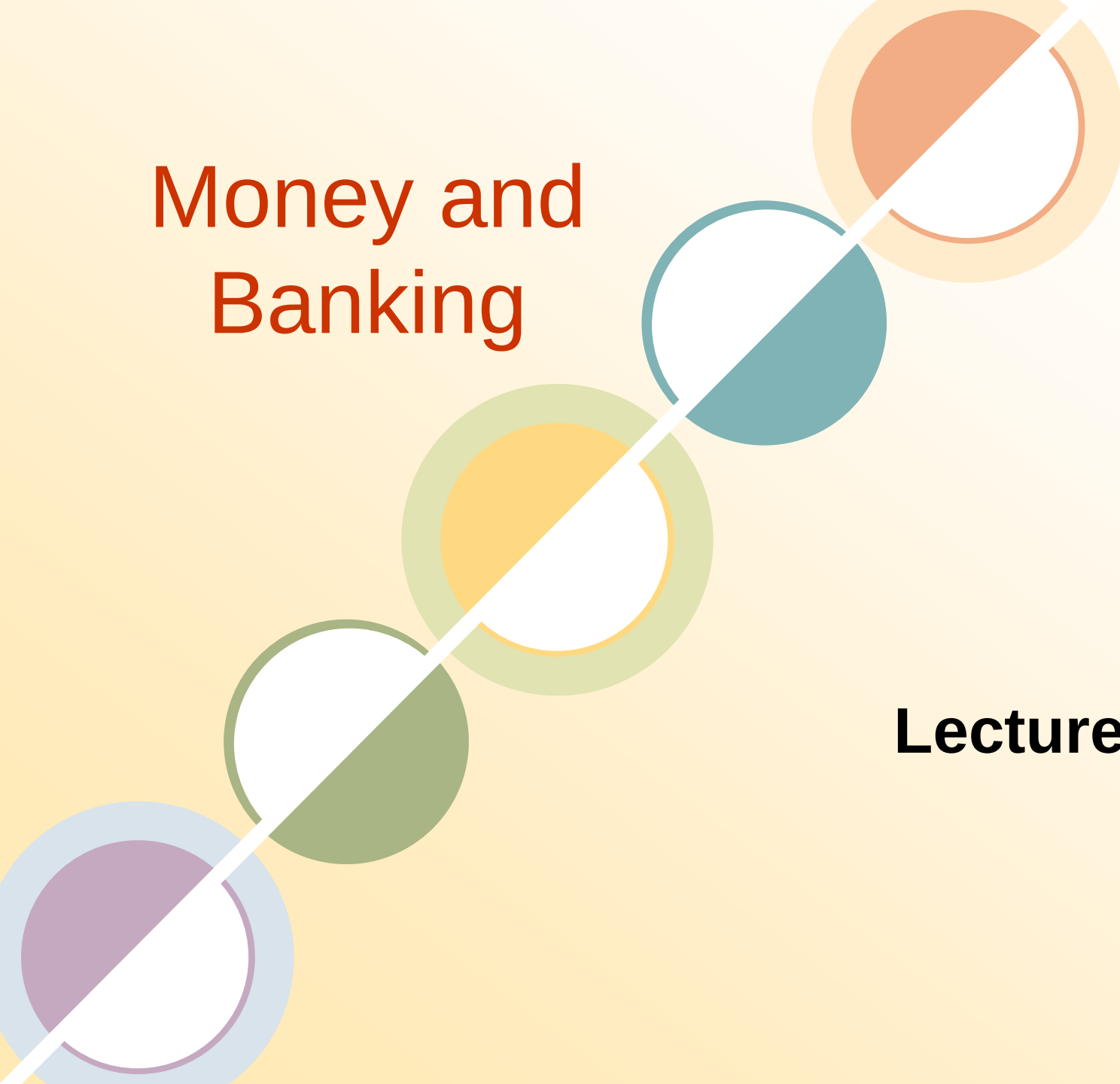
- Financial intermediaries perform five functions:
 1. They pool the resources of small savers;
 2. They provide safekeeping and accounting services as well as access to the payments system;
 3. They supply liquidity;
 4. They provide ways to diversify risk; and
 5. They collect and process information in ways that reduce information costs

Summary

- Theory of Efficient Markets
- Investing in Stocks for Long Run
- Stock Markets' Role in the Economy
- Financial Intermediation
 - Role of Financial Intermediaries

Money and Banking

Lecture 21



Review of the Previous Lecture

- Theory of Efficient Markets
- Investing in Stocks for Long Run
- Stock Markets' Role in the Economy
- Financial Intermediation

Topics Under Discussion



• Role of Financial Intermediaries

- Pool Savings
 - Safekeeping, accounting services and access to the payments system
 - Liquidity
 - Risk diversification
 - Information Services
- 
- 

Role of Financial Intermediaries

- As a general rule, indirect finance through financial intermediaries is much more important than direct finance through the stock and bond markets
- In virtually every country for which we have comprehensive data, credit extended by financial intermediaries is larger as a percentage of GDP than stocks and bonds combined

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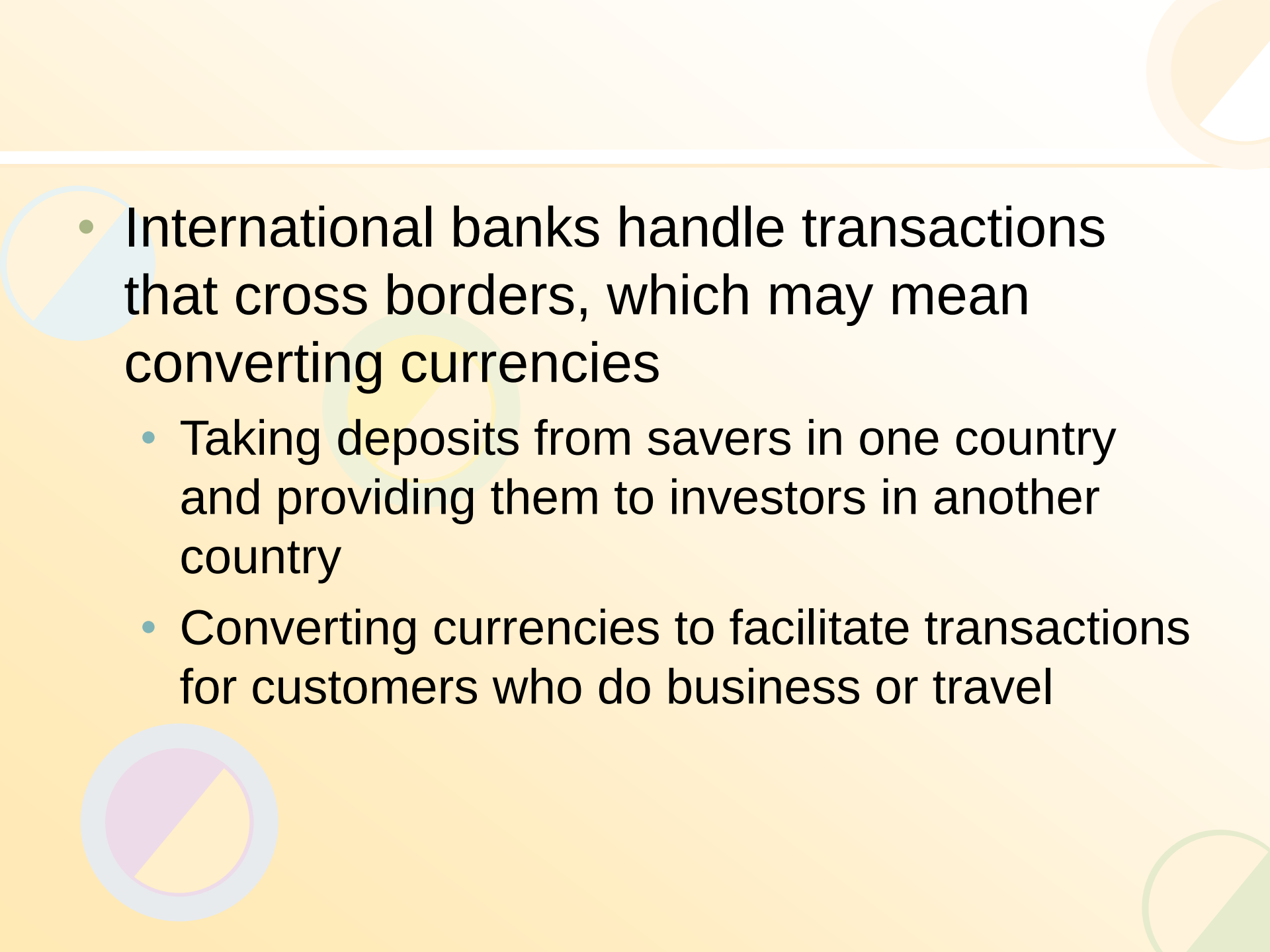
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	Stock Market Capitalization as Percent of GDP (A)	Outstanding Domestic Debt Securities as Percent of GDP (B)	Credit Extended by Banks & Other Financial Institutions as Percent of GDP (C)	Ratio of Indirect to Direct Finance $C/(A+B)$ (D)
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Role of Financial Intermediaries

- Around the world, firms and individuals draw their financing primarily from banks and other financial intermediaries
- The reason for this is information;
 - financial intermediaries exist so that individual lenders don't have to worry about getting answers to all of the important questions concerning a loan and a borrower
- Lending and borrowing involve transactions costs and information costs, and financial intermediaries exist to reduce these costs

Role of Financial Intermediaries

- Financial intermediaries perform five functions:
 1. They pool the resources of small savers;
 2. They provide safekeeping and accounting services as well as access to the payments system;
 3. They supply liquidity;
 4. They provide ways to diversify risk; and
 5. They collect and process information in ways that reduce information costs

- 
- International banks handle transactions that cross borders, which may mean converting currencies
 - Taking deposits from savers in one country and providing them to investors in another country
 - Converting currencies to facilitate transactions for customers who do business or travel

Pooling Savings

- The most straightforward economic function of a financial intermediary is to pool the resources of many small savers
- To succeed in this endeavor the intermediary must attract substantial numbers of savers
- This is the essence of indirect finance, and it means convincing potential depositors of the soundness of the institution
- Banks rely on their reputations and government guarantees like deposit insurance to make sure customers feel that their funds will be safe

Safekeeping, Payments System Access, and Accounting

- Goldsmiths were the original bankers;
- people asked the goldsmiths to store gold in their vaults in return for a receipt to prove it was there
- People soon realized that trading the receipts was easier than trading the gold itself.
- Eventually the goldsmiths noticed that there was gold left in the vaults at the end of the day, so it could safely be lent to others

Safekeeping, Payments System Access, and Accounting

- Today, banks are the places where we put things for safekeeping;
 - we deposit our paychecks and entrust our savings to a bank or other financial institution because we believe it will keep our resources safe until we need them
- Banks also provide other services, like ATMs, checkbooks, and monthly statements, giving people access to the payments system

Safekeeping, Payments System Access, and Accounting

- Financial intermediaries also reduce the cost of transactions and so promote specialization and trade, helping the economy to function more efficiently.
- According to the principle of comparative advantage, people and companies concentrate on the activities
 - at which they are the best and
 - for which their opportunity cost is lower
- This leads to specialization in a particular activity
- More specialization => more trading => more financial transaction => calls for low cost of transaction

Safekeeping, Payments System Access, and Accounting

- The bookkeeping and accounting services that financial intermediaries provide help us to manage our finances
 - Pay-cheques
 - House-rents
 - Utility bills
 - Loan payments
 - Food clothing and other expenses
 - Savings and retirement plans

Safekeeping, Payments System Access, and Accounting

- Providing safekeeping and accounting services as well as access to the payments system forces financial intermediaries to write legal contracts, which are standardized
- Much of what financial intermediaries do takes advantage of economies of scale,
 - The average cost of producing a good or service falls as the quantity produced increases
 - Information is also subject to economies of scale

Summary

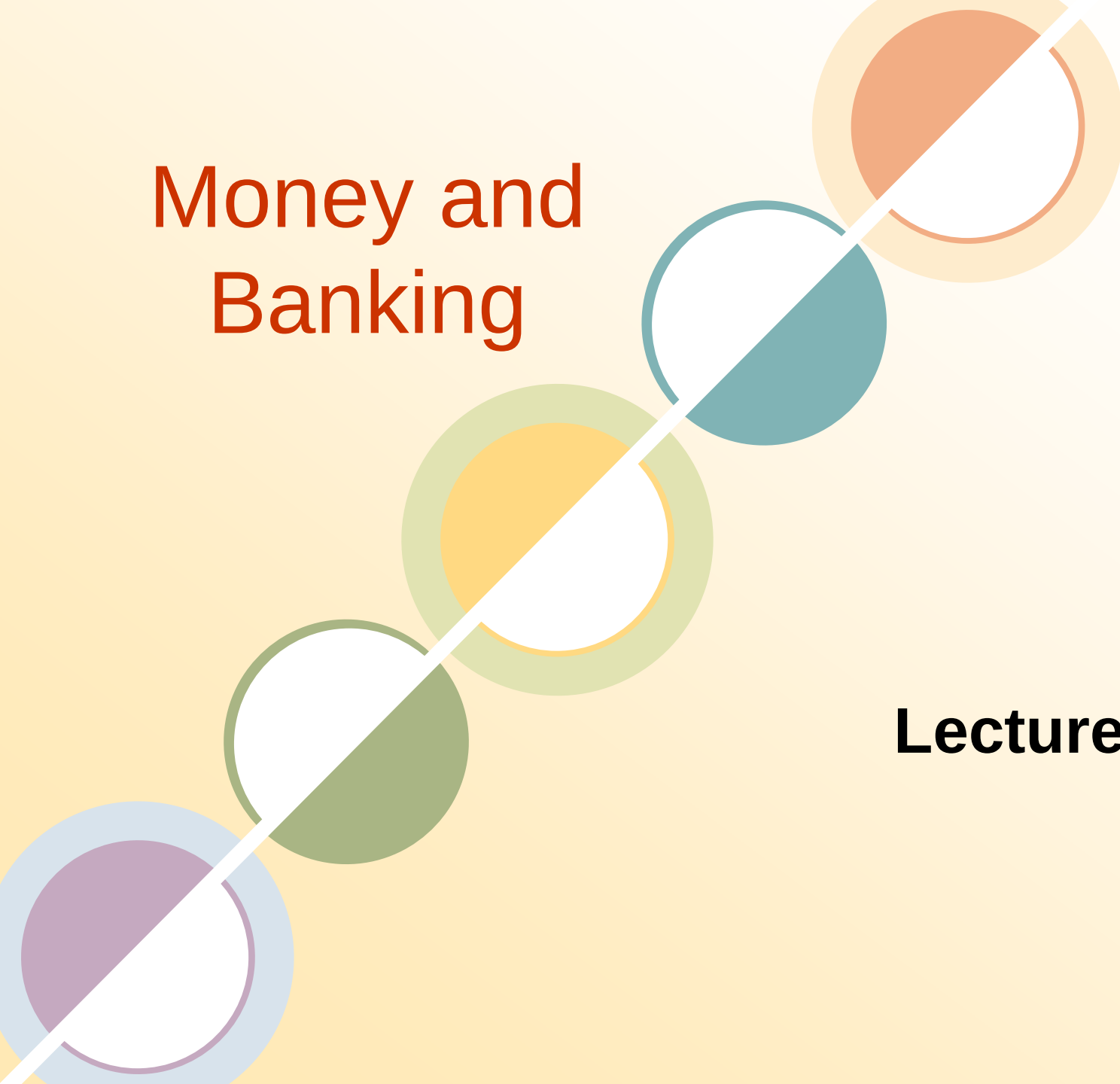


• Role of Financial Intermediaries

- Pool Savings
 - Safekeeping, accounting services and access to the payments system;
 - Liquidity;
 - Risk diversification
 - Information Services
- 
- 

Money and Banking

Lecture 22



Review of the Previous Lecture

- Role of Financial Intermediaries
 - Pool Savings
 - Safekeeping, accounting services and access to the payments system;

Topics under Discussion

- Role of Financial Intermediaries (cont.)
 - Liquidity;
 - Risk diversification
 - Information Services
- Information Asymmetry and Information Costs
 - Adverse Selection
 - Moral Hazards

Providing Liquidity

- Liquidity is a measure of the ease and cost with which an asset can be turned into a means of payment
- Financial intermediaries offer us the ability to transform assets into money at relatively low cost (ATMs are an example)

Providing Liquidity

- Financial intermediaries provide liquidity in a way that is efficient and beneficial to all of us
 - By collecting funds from a large number of small investors, a bank can reduce the cost of their combined investment, offering the individual investor both liquidity and high rates of return
 - Financial intermediaries offer depositors something they can't get from financial markets on their own
 - Financial intermediaries offer both individuals and businesses lines of credit, which are pre-approved loans that can be drawn on whenever a customer needs funds

Diversifying Risk

- Financial intermediaries enable us to diversify our investments and reduce risk
 - While investing, **don't put all your eggs in one basket**
 - Putting \$1 in 100 stocks is better than investing \$100 in just one stock
 - Financial institutions enable us to diversify our investment and reduce risk.

Diversifying Risk

- Banks mitigate risk by taking deposits from a large number of individuals and make thousands of loans with them, thus giving each depositor a small stake in each of the loans
 - Bank may collect \$1,000 from each of one million depositors and then use \$1 billion to make 10,000 loans of \$100,000 each
 - Thus each has a $1/1,000,000$ share in each of the 10,000 loans. This is diversification!
 - And since banks are expert at this game, it can minimize the cost of all such transactions

Diversifying Risk

- All financial intermediaries provide a low-cost way for individuals to diversify their investments
 - Mutual funds

Information Services

- One of the biggest problems individual savers face is figuring out which potential borrowers are trustworthy and which are not
- There is an information asymmetry because the borrower knows whether or not he or she is trustworthy, but the lender faces substantial costs to obtain the same information

Information Services

- Financial intermediaries reduce the problems created by information asymmetries by collecting and processing standardized information
 - Screen loan applications to guarantee the creditworthiness
 - Monitor loan recipients to ensure proper usage of funds

A Summary of the Role of Financial Intermediaries

- Pooling Savings:
 - Accepting resources from a large number of small savers/lenders in order to provide large loans to borrowers.
- Safekeeping and Accounting:
 - Keeping depositors' savings safe, giving them access to the payments system, and providing them with accounting statements that help them to track their income and expenditures.

A Summary of the Role of Financial Intermediaries

- Providing Liquidity:
 - Allowing depositors to transform their financial assets into money quickly, easily, and at low cost.
- Risk sharing:
 - Providing investors with the ability to diversify even small investments.
- Information Services:
 - Collecting and processing large amounts of standardized financial information.

Information Asymmetries and Information Costs

- Information plays a central role in the structure of financial markets and financial institutions
- Markets require sophisticated information in order to work well, and when the cost of obtaining information is too high, markets cease to function

Information Asymmetries and Information Costs

- Asymmetric information
 - issuers of financial instruments – borrowers who want to issue bonds and firms that want to issue stock – know much more about their business prospects and their willingness to work than potential lenders or investors
 - solving this problem is one key to making our financial system work as well as it does

Information Asymmetries and Information Costs

- Lets take up our online store example
 - Buyers must believe that item has been described accurately and they must be sure that the seller will send the item in exchange for their payment
 - Here sellers have much more information than buyers have, creating an information asymmetry
 - To resolve this issue,
 - Induct an insurance system
 - Devise an information system collecting data of purchases and delivery

Information Asymmetries and Information Costs

- Asymmetric information poses two obstacles to the smooth flow of funds from savers to investors:
 - Adverse selection, - involves being able to distinguish good credit risks from bad before the transaction;
 - Moral hazard, - arises after the transaction and involves finding out whether borrowers will use the proceeds of a loan as they claim they will

Adverse Selection

- Potential borrowers know more about the projects they wish to finance than prospective lenders
 - Used Cars and the Market for Lemons:
 - In a market in which there are good cars (“peaches”) and bad cars (“lemons”) for sale, buyers are willing to pay only the average value of all the cars in the market.
 - This is less than the sellers of the “peaches” want, so those cars disappear from the markets and only the “lemons” are left

Adverse Selection

- To solve this problem caused by asymmetric information, companies like Consumer Reports provide information about the reliability and safety of different models, and car dealers will certify the used cars they sell

Adverse Selection

• Adverse Selection in Financial Markets:

- Information asymmetries can drive good stocks and bonds out of the financial market
- If you can't tell the difference between a firm with a good prospects and a firm with bad prospects, you will be willing to pay a price based only on their average qualities
- The stocks of the good company will be undervalued so the managers of these companies will keep the stocks away from the market
- This leaves only the firms with bad prospects in the market

Adverse Selection

- Adverse Selection in Financial Markets:

- The same happens in the bond market
- If a lender can not tell whether a borrower is a good or a bad credit risk, the demand for a risk premium will be based on the average risk
- Borrowers having good credit risk will not pay higher risk premiums and would withdraw from the market
- Only bad credit risk bonds are left in the market

Adverse Selection

- Solving the Adverse Selection Problem
 - The adverse selection problem resulting in good investments not to be undertaken, the economy will not grow as rapidly as it could.
 - So there must be some way of distinguishing good firms from the bad ones
 - Disclosure of Information
 - Collateral and Net Worth

Adverse Selection

- Solving the Adverse Selection Problem
 - Disclosure of Information:
 - Generating more information is one obvious way to solve the problem created by asymmetric information
 - This can be done through government required disclosure and the private collection and production of information
 - e.g. Securities and Exchange Commission regulations
 - Reports from private sources such as rating agencies, brokerage companies and financial analysts
 - The cost and credibility of such information are to be kept in mind

Adverse Selection

- Solving the Adverse Selection Problem
 - Collateral and Net Worth
 - Collateral is something of a value pledged by a borrower to the lender in the event of borrower's default
 - Lenders can be compensated even if borrowers default, and if the loan is so insured then the borrower is not a bad credit risk

Adverse Selection

- Solving the Adverse Selection Problem
 - Collateral and Net Worth
 - Net worth is the owner's stake in the firm, the value of the firm minus the value of its liabilities
 - If a firm defaults on loan, the lender can make a claim against the firm's net worth
 - The same is true for home loans
 - The importance of net worth in reducing adverse selection is the reason owners of new businesses have so much difficulty borrowing money

Moral Hazards

- Moral hazard arises when we cannot observe people's actions, and so cannot judge whether a poor outcome was intentional or just a result of bad luck

Moral Hazards

- Moral Hazard in Equity Finance
 - While purchasing stocks of a company, are you sure that it will use the funds in a way that is best for you?
 - principal-agent problem
 - The separation of ownership from control.
 - When the managers of a company are the owners, the problem of moral hazard in equity financing disappears.

Moral Hazards

- Moral Hazard in Debt Finance
 - Because debt contracts allow owners to keep all the profits in excess of the loan payments, they encourage risk taking
 - a good legal contract can solve the moral hazard problem that is inherent in debt finance.
 - Bonds and loans often carry restrictive covenants

The Negative Consequences of Information Costs

1. Adverse Selection:

- Lenders can't distinguish good from bad credit risks, which discourages transactions from taking place.

Solutions include

- Government-required information disclosure
- Private collection of information
- The pledging of collateral to insure lenders against the borrower's default
- Requiring borrowers to invest substantial resources of their own

The Negative Consequences of Information Costs

2. Moral Hazard:

- Lenders can't tell whether borrowers will do what they claim they will do with the borrowed resources; borrowers may take too many risks.

Solutions include

- Forced reporting of managers to owners
- Requiring managers to invest substantial resources of their own
- Covenants that restrict what borrowers can do with borrowed funds

Financial Intermediaries and Information Costs

- The problems of adverse selection and moral hazard make direct finance expensive and difficult to get.
- These drawbacks lead us immediately to indirect finance and the role of financial institutions.
- Much of the information that financial intermediaries collect is used to reduce information costs and minimize the effects of adverse selection and moral hazard

Financial Intermediaries and Information Costs

- Screening and Certifying to Reduce Adverse Selection
- Monitoring to Reduce Moral Hazard